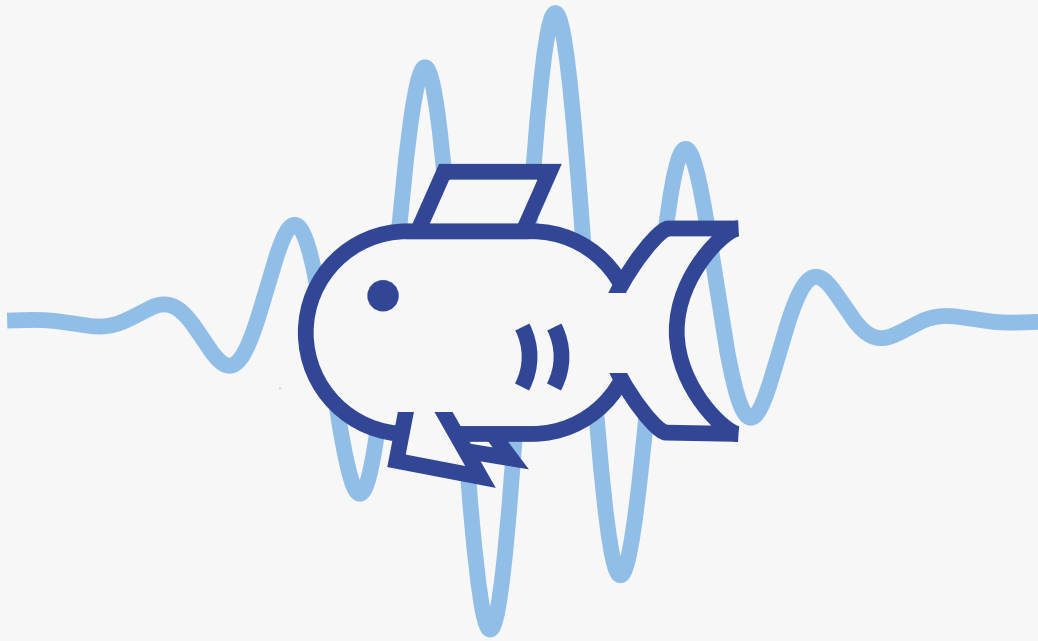




TUNA

Version 0.11.1

August 2026

Harry Brough

Foreword

Welcome to TUNA!

This is a project to deepen my understanding of the theory and implementation of quantum chemistry. A simple to use Python program with a command-line interface, TUNA can calculate energies and properties of diatomic molecules with a range of quantum chemical methods.

Unlike other Python-based programs for quantum chemistry such as Psi4 [1, 2] and PySCF [3], TUNA is not primarily a collection of modules to incorporate into a workflow, but is intended to be a program run directly from the command line. Parts of the syntax are inspired by the ORCA quantum chemistry package [4, 5] (although the tuna is the more majestic fish), but no input files are needed as the geometry of a diatomic molecule is easily specified in a single input line.

The key strength of TUNA is its low-friction, user-friendly interface, hopefully offering an intuitive experience that could serve as a teaching aid. Considerable effort has been taken to document all the code, and extensive examples of how the program works are provided in this manual. The program has grown into quite a large collection of quantum chemistry methods, showing their implementation in the simple yet instructive context of diatomic molecules.

On the technical side, TUNA utilises numerical energy derivatives, so any implemented electronic structure method can be used to calculate geometries, vibrational frequencies, response properties and molecular dynamics trajectories. Because these derivatives are fast for diatomics, almost nothing is written to the disk, and extensive use is made of vectorised operations in NumPy [6], TUNA is very fast for a Python program.

While TUNA is currently fairly modest in scope, its capabilities are set to expand. The long term goal is to develop it into a simple and efficient testing ground for the performance of density functional approximations on diatomics as well as a capable tool for quantum chemistry education.

Who knows? Maybe one day, TUNA could make waves!

Harry Brough

Contents

1	Getting Started	6
1.1	Installation	6
1.2	Simple Input Line	6
1.3	Units and Constants	7
1.4	Philosophy of TUNA	8
1.5	Program Components	8
2	Simple Input Line Examples	10
3	Calculation Types	12
3.1	Single Point Energy	13
3.1.1	Koopmans' Theorem	15
3.1.2	Rotational Constant	15
3.1.3	Multipole Moments	16
3.1.4	Polarisability and Hyperpolarisability	18
3.1.5	Population Analysis	20
3.1.6	Natural Orbitals	21
3.1.7	Plotting Orbitals and Densities	22
3.1.8	Dispersion Correction	23
3.2	Coordinate Scan	26
3.2.1	Plotting Coordinate Scans	27
3.3	Geometry Optimisation	29
3.3.1	Optimisation Convergence	31
3.3.2	Force Calculations	31
3.4	Harmonic Frequency	31
3.4.1	Transition Intensity	33

3.4.2	Thermochemistry	34
3.4.3	Vibrational Perturbation Theory	36
3.5	Anharmonic Frequency	38
3.5.1	Transition Intensity	40
3.5.2	Plotting Nuclear Wavefunctions	41
3.6	Bond Dissociation Energy	42
3.7	Ionisation Potential and Electron Affinity	43
3.8	<i>Ab Initio</i> Molecular Dynamics	44
4	Electronic Structure Methods	46
4.1	Hartree–Fock Theory	48
4.1.1	Initial Guess	50
4.1.2	Symmetry Breaking	51
4.1.3	Stability Analysis	52
4.1.4	Self-consistent Field Convergence	53
4.1.5	Hartree Theory	55
4.2	Møller–Plesset Perturbation Theory	55
4.2.1	Conventional MP2, MP3 and MP4	56
4.2.2	Freezing Core Electrons	57
4.2.3	Spin-component Scaling	58
4.2.4	Orbital-optimised MP2	59
4.2.5	Iterative MP2	59
4.2.6	Laplace Transform AO-MP2	60
4.2.7	Unrelaxed and Relaxed Density Matrices	61
4.3	Configuration Interaction	62
4.3.1	Configuration Interaction Singles and Doubles	63
4.3.2	Configuration Interaction Doubles	63
4.3.3	Configuration Interaction Singles, Doubles and Triples	64
4.4	Coupled Cluster Theory	64
4.4.1	Coupled Cluster Doubles	66
4.4.2	Coupled Cluster Singles and Doubles	67
4.4.3	Coupled Cluster Singles, Doubles and Triples	67

4.4.4	Coupled Cluster Singles, Doubles, Triples and Quadruples	68
4.4.5	Coupled Cluster with Perturbative Triples	68
4.4.6	Coupled Cluster with Perturbative Quadruples	68
4.4.7	Linearised Coupled Cluster	69
4.4.8	Quadratic Configuration Interaction	69
4.4.9	Approximate Coupled Cluster	70
4.4.10	Linearised Density Matrix	70
4.5	Density Functional Theory	70
4.5.1	Local Spin Density Approximation	74
4.5.2	Generalised Gradient Approximation	75
4.5.3	<i>meta</i> -Generalised Gradient Approximation	76
4.5.4	Hybrid Functionals	78
4.5.5	Double-hybrid Functionals	79
4.5.6	Efficient Grid Integration	80
4.6	Excited States	82
4.6.1	Time-dependent Hartree–Fock	83
4.6.2	Configuration Interaction Singles	84
4.6.3	Perturbative Doubles Correction	84
4.6.4	Time-dependent Density Functional Theory	85
4.6.5	Plotting Absorbance Spectra	86
5	Basis Sets	88
5.1	Basis Set Extrapolation	88
5.2	Custom Basis Sets	89
5.3	List of Basis Sets	91
6	Molecular Integrals	97
6.1	Types of Integrals	98
6.2	Spherical and Cartesian Harmonics	98
7	Performance of TUNA	100
7.1	Speed and Efficiency	100
7.2	Memory Requirements	101

7.3 Accuracy and Precision	102
8 Acknowledgements	103
9 List of Keywords	104
Bibliography	117

1 Getting Started

1.1 Installation

First ensure Python 3.12 or higher is installed, then run the following command from a terminal:

```
pip install quantumtuna
```

It is also possible to install TUNA by cloning the GitHub repository, but this is not recommended as you will have to compile the Cython integrals module yourself. Installing with pip will also install TUNA's four dependencies — NumPy [6], SciPy [7], Matplotlib [8] and TermColor [9].

Next locate the folder where TUNA was installed within the Python site packages, `/*/TUNA/`, where `*` is the system path, and add it to the PATH environment variable on Windows.

On MacOS or Linux run the following commands to allow TUNA to be used from anywhere:

```
echo "alias tuna='noglob python3 */TUNA/tuna.py'" >> ~/.zshrc
echo "alias TUNA='noglob python3 */TUNA/tuna.py'" >> ~/.zshrc
source ~/.zshrc
```

Once the folder containing the TUNA Python files is added to PATH, open a new terminal and test that everything is working correctly by running:

```
TUNA --version
```

If the version of TUNA you installed prints to the terminal, you're good to go!

1.2 Simple Input Line

All calculations in TUNA are requested with a single line of commands in the terminal. The input line is not case sensitive and the spaces around colons are optional. It has the structure:

```
TUNA [Calculation] : [Atom A] [Atom B] [Distance] : [Method] [Basis]
```

A single atom can be provided, in which case [Atom B] and [Distance] are omitted. If two

atoms are provided, a [Distance] must be given too. Available calculation types in TUNA are detailed in [chapter 3](#), electronic structure methods in [chapter 4](#) and basis sets in [chapter 5](#).

For example, a [single point energy](#) calculation on an H_2 molecule with a bond length of 1.0 Å, with restricted [Hartree–Fock theory](#) in the STO-3G basis set is requested by:

```
TUNA SPE : H H 1.0 : HF STO-3G
```

Additional parameters can be requested by adding another colon at the end and appending keywords. For example, a [B3LYP/6-31G*](#) [optimisation](#) calculation using [D2 semi-empirical dispersion](#) correction on a H-He^+ molecule starting at a bond length of 0.8 Å is:

```
TUNA OPT : H He 0.8 : B3LYP 6-31G* : CHARGE +1 D2
```

As many keywords as you like can be written after the third colon. Some keywords such as CHARGE require a value (like +1 here) — just write this directly after the keyword, and then more keywords can be added. Many keywords can be used together in this way, and TUNA will figure out what you want. Further input examples to test out TUNA can be found in [chapter 2](#).

1.3 Units and Constants

In the TUNA output, distances are printed in angströms, times in femtoseconds and everything else in atomic units unless otherwise stated. Units are set by the CODATA 2022 recommendations [10], in [Table 1.1](#). All other constants are derived from these and the atomic masses in [Table 3.3](#).

Table 1.1 Fundamental constants used in TUNA

Constant	Symbol	Value
Planck constant	h	$6.62607015 \times 10^{-34} \text{ J s}$
Elementary charge	e	$1.602176634 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.1093837139 \times 10^{-31} \text{ kg}$
Vacuum permittivity	ϵ_0	$8.8541878188 \times 10^{-12} \text{ F m}^{-1}$
Speed of light	c	$2.99792458 \times 10^8 \text{ m s}^{-1}$
Boltzmann constant	k_B	$1.380649 \times 10^{-23} \text{ J K}^{-1}$
Avogadro constant	N_A	$6.02214076 \times 10^{23} \text{ mol}^{-1}$

1.4 Philosophy of TUNA

Numerical differentiation of the molecular energy underpins property calculations in TUNA. For atoms and diatomics it is very efficient, and it is also completely general, so:

any electronic structure method can be combined with **any** calculation type

This makes unusual calculations straightforward and consistent. For example, a coupled cluster [hyperpolarisability](#) calculation can, in principle, be treated with the same precision as a Hartree–Fock calculation. It also means that once a new method is implemented, it automatically becomes available across the program, rapidly expanding the capabilities of TUNA. For numerical first derivatives, for instance used in [geometry optimisations](#) for the force and dipole moment derivatives for [harmonic vibrational intensities](#), the central differences formula is used,

$$f'(x) \approx \frac{1}{2h} [f(x+h) - f(x-h)] .$$

For numerical second derivatives, used in [harmonic frequency](#) or [polarisability](#) calculations, a five point stencil formula is used,

$$f''(x) \approx \frac{1}{12h^2} [-f(x-2h) + 16f(x-h) - 30f(x) + 16f(x+h) - f(x+2h)] .$$

A drawback of numerical derivatives is their sensitivity to the finite difference step size, h . This value has been optimised for each derivative type in TUNA, and can be found in [section 7.3](#).

Numerical derivatives also require well converged energies. For calculations involving first derivatives, TIGHT energy [convergence criteria](#) are used by default, while higher derivatives use EXTREME [convergence criteria](#). Even the standard thresholds in TUNA are tighter than in most programs.

A major advantage of TUNA's numerical philosophy is that these derivative formulae can be combined with [basis set extrapolation](#), allowing approximate complete basis set [geometries](#), [vibrational frequencies](#), [electric properties](#) and [molecular dynamics](#) trajectories to be calculated.

1.5 Program Components

When TUNA is called by the terminal, the main program, `tuna.py`, begins a calculation in which it calls various modules, listed in [Table 1.2](#). The most “fundamental” module is `tuna_util`, which contains all the defining constants for the program. The `tuna_integral` is unique as it is written in Cython [11], which is a compiled extension language to Python. This is because of the difficulty in vectorising the [McMurchie–Davidson molecular integrals over basis functions](#).

In addition to these modules, some external modules are also imported. All these should automatically be installed when `pip install quantumtuna` is run. The most important are NumPy [6] and SciPy [7], which speed up the code enormously compared to pure Python. TermColor [9] makes the output more vibrant, while Matplotlib [8] enables the creation of two-dimensional plots.

Table 1.2 List of Python modules that constitute TUNA, in alphabetical order

Module	Description
tuna	Main program, input parsing
tuna_basis	Storing and setting up basis sets
tuna_calc	Processing keywords and storing calculation information
tuna_cc	Coupled cluster
tuna_ci	Configuration interaction and excited states
tuna_dft	Density functional theory
tuna_energy	Calculating energies, building molecules, coordinate scans
tuna_freq	Harmonic and anharmonic frequencies
tuna_guess	Initial guess
tuna_integral	Evaluating one- and two-electron integrals
tuna_kernel	Low level functions for energy evaluation
tuna_md	<i>Ab initio</i> molecular dynamics
tuna_molecule	Storing and parsing molecular information
tuna_mp	Møller–Plesset perturbation theory
tuna_opt	Calculating gradients, Hessians and optimisation algorithm
tuna_out	Graphical plots and trajectories
tuna_props	Calculating and printing molecular properties
tuna_scf	Main self-consistent field loop, convergence acceleration
tuna_thermo	Thermochemistry after frequency calculations
tuna_util	General utility, units, frequently used functions
tuna_xc	Exchange–correlation functionals

2 Simple Input Line Examples

Several examples of simple input lines are shown here. These can be simply copied and pasted into a terminal window to see TUNA in action. This should give a clear idea of the syntax to run a calculation, and then you can experiment!

Firstly, a single point energy calculation on a hydrogen atom:

```
TUNA SPE : H : HF aug-cc-pVTZ
```

Next, an MP2 geometry optimisation of lithium hydride, with D2 dispersion correction:

```
TUNA OPT : Li H 1.7 : MP2 6-31+G* : D2
```

A single point DFT calculation on dinitrogen, with a decontracted basis and additional print:

```
TUNA SPE : N N 1.10 : B3LYP STO-3G : DECONTRACT P
```

An MP4 frequency calculation on carbon monoxide, with 50% damping and no DIIS:

```
TUNA FREQ : C O 1.230 : MP4 3-21G : DAMP 0.5 NODIIS
```

An optimisation and subsequent frequency calculation on dihydrogen, using coupled cluster theory and extreme SCF and geometry convergence criteria:

```
TUNA OPTFREQ : H H 0.74 : CCSD def2-TZVP : EXTREME EXTREMEOPT
```

A CCSDT calculation on a lithium atom, with reduced maximum coupled cluster iterations:

```
TUNA SPE : Li : CCSDT cc-pVDZ : CORRMXITER 10
```

An orbital-optimised MP2 molecular dynamics simulation of dihydrogen, with a timestep of 0.1 fs and an initial temperature of 298.15 K, printing a trajectory to "tunatraj.xyz":

```
TUNA MD : H H 1.0 : OMP2 6-31G[d] : STEP 0.1 TEMP 298.15 TRAJ TUNATRAJ.XYZ
```

A single point calculation on a hydrogen atom using a ghost lithium basis:

```
TUNA SPE : H XLi 0.50 : HF 6-311++G :
```

A full configuration interaction polarisability calculation on dihydrogen, using a diffuse basis set, plotting the second molecular orbital:

```
TUNA SPE : H H 0.74 : CISD t-aug-cc-pVDZ : POLAR PLOTMO 2
```

An optimisation on triplet oxygen in a finite electric field:

```
TUNA OPT : O O 1.0 : SCS-MP3 3-21G : ML 3 EX 0.05 EY 0.06 EZ 0.10
```

A configuration interaction singles calculation that plots an excited state difference density:

```
TUNA SPE : C H 1.0 : CIS cc-pVDZ : DIFFDENS PLOT ROOT 3
```

A coupled cluster optimisation with single, double, triple and perturbative quadruple excitations on lithium hydride, freezing the core orbitals.

```
TUNA OPT : Li H 1 : CCSDT[Q] STO-3G : FREEZECORE
```

An anharmonic frequency calculation on dihydrogen, where the vibrational wavefunctions are plotted along with the potential energy surface when the calculation has converged:

```
TUNA ANHARM : H H 0.74 : CCSD def2-SVP : VIBPLOT
```

An MP2 atomic calculation on the magnesium atom extrapolated to the complete basis set limit:

```
TUNA SPE : Mg : MP2 def2-TZVPP : EXTRAPOLATE
```

An excited state optimisation and frequency calculation with a perturbative doubles correction:

```
TUNA OPTFREQ : H He 1.0 : CIS[D] 3-21G : ROOT 6
```

Finally, a full configuration interaction polarisability calculation on the helium dimer, with damping for cluster amplitude convergence:

```
TUNA SPE : He He 2.0 : CCSDTQ cc-pVDZ : POLAR CORRDAMP 0.1
```

3 Calculation Types

The calculation types available in TUNA are listed in [Table 3.1](#) with their respective keywords.

Table 3.1 Calculation types available in TUNA

Keyword	Calculation type
SPE	Single point energy
SCAN	Coordinate scan
OPT	Optimisation
FORCE	Force
FREQ	Harmonic frequency
OPTFREQ	Optimisation and subsequent harmonic frequency
ANHARM	Anharmonic frequency
BDE	Bond dissociation energy
IP	Ionisation potential
EA	Electron affinity
MD	<i>Ab initio</i> molecular dynamics

All calculation types run on an initial molecule, defined by its atoms, charge and multiplicity. Element types from hydrogen to argon are implemented, and are requested by their atomic symbols.

“Ghost” atoms are also available, which act as additional basis functions for a single atom and can be used to correct basis set superposition error [12]. The ghost atoms are accessible by typing an “X” before the atomic symbol, like “XH” or “XCl”. For example, the following command will calculate the energy of a hydrogen atom in the presence of a ghost lithium atom’s basis functions, lowering the energy compared to the calculation on a lone atom:

```
TUNA SPE : H XLi 0.735 : HF 6-311++G
```

Trying to run a calculation for a molecule (eg. OPT, FREQ, MD) with a ghost atom present will result in TUNA crashing politely with an error message.

Atoms or molecules can be given a charge and multiplicity with the CHARGE and MULTIPLICITY keywords. For example, triplet H₂ can be requested by:

```
TUNA SPE : H H 1.3 : UHF 6-311G : CHARGE 0 MULTIPLICITY 3
```

The shorter parameters for charge and multiplicity CH and ML can also be used:

```
TUNA SPE : H H 1.3 : UHF 6-311G : CH 0 ML 3
```

If no charge or multiplicity is specified, TUNA defaults to a neutral molecule, which is assumed to be either a singlet or a doublet depending on whether the number of electrons is even or odd.

All calculations start by printing the TUNA logo to the terminal, importing libraries and printing the requested calculation type, electronic structure method and basis set. The molecule is then set up as requested, and the molecular structure, number of basis functions, number of primitive Gaussians, charge, multiplicity, number of (alpha and beta) electrons, number of occupied and virtual orbitals, point group and bond length are printed. The specified calculation then begins.

At the end of any calculation — if it terminates successfully — TUNA prints the total time taken for the calculation. This time is counted from when the Python modules have been imported. More information about the times for each part of a TUNA calculation (molecular integrals, SCF iterations, correlation, etc.) is shown when the additional print keyword, P, is used.

3.1 Single Point Energy

A single point energy calculation in TUNA can be requested with the SPE calculation type keyword. For instance, a single point calculation on H–He⁺ at a distance of 2.0 Å with [restricted Hartree–Fock theory](#) in the 6-311G basis can be requested by:

```
TUNA SPE : H He 2.0 : RHF 6-311G : CH 1
```

The molecule is first set up, and information printed to the console:

```
~~~~~  
Molecule and Basis Information  
~~~~~  
Molecular structure: H ----- He  
  
Number of basis functions: 6
```

Number of primitive Gaussians: 10

Charge: 1

Multiplicity: 1

Number of electrons: 2

Number of alpha electrons: 1

Number of beta electrons: 1

Number of occupied orbitals: 1

Number of virtual orbitals: 5

Point group: C₁

Bond length: 2.0000

~~~~~

The nuclear repulsion energy is then calculated, which for diatomics is simply

$$V_{nn} = \frac{Z_A Z_B}{R}, \quad (3.1)$$

where  $Z$  is the relative nuclear charge and  $R$  is the bond length. Next the optional [dispersion energy](#) is calculated, before the [one- and two-electron molecular integrals](#) are evaluated. The eigenvalues of the overlap matrix,  $\mathbf{S}$ , are then checked to make sure they're not too small and avoid a linearly dependent basis set. The default threshold is  $10^{-7}$  and can be adjusted with the `STHRESH` keyword. The Fock orthogonalisation matrix,  $\mathbf{X} = \mathbf{S}^{-\frac{1}{2}}$ , is formed, then the [guess density matrix](#) is calculated.

If a multi-electron calculation is requested, the convergence accelerators are printed and the [self-consistent field cycle](#) begins. If this converges, the components of the energy are printed in addition to the Virial ratio. This is the ratio between the molecular potential and kinetic energy, and should approach 2.0 as the molecular structure approaches an equilibrium geometry or for an atom. The spin contamination for spin-unrestricted calculations is printed. If a [correlated wavefunction](#) or [excited state](#) calculation has been requested, this then begins.

When the final energy has been calculated, the one-electron reduced density matrix,  $\mathbf{P}$ , is used in property calculations. In the current version, the (un)restricted Hartree–Fock, unrelaxed CIS, (un)relaxed (spin-component scaled) MP2 and linearised coupled cluster density matrices are available in TUNA. The [orbital-optimised MP2](#) density matrix is also implemented, for which there is no difference between an unrelaxed and a relaxed density. If a higher order [perturbation theory](#) calculation is requested, properties will be calculated at the MP2 level. Calculations of

rotational constants, multipole moments and population analysis are disabled for single atoms.

Finally, the energies are printed to the console again. If the additional print keyword, P, is used, molecular orbitals and their eigenvalues will also be printed, separated into  $\alpha$  and  $\beta$  orbitals for spin-unrestricted calculations. Only the first ten virtual orbitals are printed by default, which can be overridden with the PRINTMOS keyword. If the reduced print keyword, T, is used, only the molecular information, SCF cycles and final energies are printed. If the DEBUG keyword is used, loads of information is printed throughout a calculation.

### 3.1.1 Koopmans' Theorem

Koopmans' theorem states that in closed-shell Hartree–Fock theory, the ionisation potential is equal to the negative of the orbital energy of the highest occupied molecular orbital (HOMO),  $-\epsilon_{\text{HOMO}}$ , and that the electron affinity is equal to the negative of the orbital energy of the lowest unoccupied molecular orbital (LUMO),  $-\epsilon_{\text{LUMO}}$  [13]. This is exact within restricted Hartree–Fock (RHF) theory if the ionic orbitals are assumed to be identical to the orbitals of the atom.

The first properties printed in a TUNA calculation on a RHF reference are the Koopmans' parameters  $-\epsilon_{\text{HOMO}}$ ,  $-\epsilon_{\text{LUMO}}$  and the HOMO–LUMO gap,  $\epsilon_{\text{LUMO}} - \epsilon_{\text{HOMO}}$ . These values are not printed for calculations with unrestricted (UHF) references. They are also printed for restricted Kohn–Sham density functional theory (DFT) calculations, although the results are theoretically less meaningful due to the absence of an exact exchange–correlation functional.

The “exact” ionisation potential or electron affinity also accounts for orbital relaxation, and can be calculated from two converged self-consistent field calculations (for vertical ionisations) or from two converged geometry optimisations (for adiabatic ionisations). These calculations can be performed easily using the IP or EA calculation types, which are discussed in detail in [section 3.7](#).

### 3.1.2 Rotational Constant

The next property printed, in units of GHz and  $\text{cm}^{-1}$ , is the rotational constant, calculated by

$$B = \frac{1}{4\pi\mu R^2}, \quad (3.2)$$

where  $\mu$  is the reduced mass of the molecule, using the masses in [section 3.4](#).



### 3.1.3 Multipole Moments

In the presence of a perturbative electric field,  $\mathbf{F}$ , the energy can be expanded as

$$E(\mathbf{F}) = E - \sum_i \mu_i F_i - \frac{1}{2} \sum_{ij} \alpha_{ij} F_i F_j - \frac{1}{6} \sum_{ijk} \beta_{ijk} F_i F_j F_k + \dots, \quad (3.3)$$

where the dipole moment can be identified with the first derivative of the energy with respect to the field, polarisability with the second derivative and hyperpolarisability with the third.

The origin for dipole moment calculations (which is printed) is the centre of mass,  $R_0$ , calculated with the masses in [section 3.4](#). The nuclear dipole moment,

$$\mu_{\text{nuc}} = Z_A(R_A - R_0) + Z_B(R_B - R_0), \quad (3.4)$$

and the analytical electronic dipole moment,

$$\mu_{\text{ele}} = - \sum_{\mu\nu} P_{\mu\nu} D_{\mu\nu}^z, \quad (3.5)$$

are calculated, where  $D_{\mu\nu}^\alpha$  are the dipole moment integrals,  $\langle \mu | \mathbf{r} - \mathbf{R}_0 | \nu \rangle$ , evaluated with the rest of the [one-electron integrals](#) at the start of a calculation. Note that these dipole moments are one-dimensional by the symmetry of a diatomic molecule.

In addition to these values and the total dipole moment,  $\mu = \mu_{\text{nuc}} + \mu_{\text{ele}}$ , a diagram of the molecular structure is printed with an arrow pointing in the direction of the negative charge. For example, a RHF/6-311G calculation on H-He<sup>+</sup> at 2 Å gives this output, meaning the hydrogen atom is more negatively charged than the helium atom:

```

~~~~~
 Dipole Moment
~~~~~
Nuclear: -1.4987929      Electronic: -1.4774574

Total: -2.9762504      H --- He  <---+
~~~~~

```

This can be reproduced with the following input line:

```
TUNA SPE : H He 2.0 : HF 6-311G : CH 1
```

The analytical dipole moment as described above is implemented for [Hartree–Fock](#), pure and hybrid [density functional theory](#), [\(un\)relaxed MP2](#), [unrelaxed CIS](#) and [linearised coupled cluster density matrices](#). The numerical dipole moment can be calculated for all methods by taking the derivative of the energy with respect to the  $z$  component of an applied electric field,

$$\mu_{\text{ele}} = \frac{\partial E}{\partial F} . \quad (3.6)$$

This is requested with the `DIPOLE` keyword. For methods with a one-electron density matrix implemented, this will give the same result as the version automatically printed after the SCF calculation. However this dipole moment will be correct for all methods, where the analytical one may not be — for example the analytical MP2 dipole moment with the unrelaxed density matrix. The numerical dipole moment (and relaxed analytical dipole moment) will be correct:

`TUNA SPE : C O 1.0 : MP2 cc-pVDZ : DIPOLE`

The quadrupole moment is also calculated analytically by default, and numerically with the `QUADRUPOLE` keyword. Just as for the dipole moment, these results are identical if an exact analytical density matrix is implemented in TUNA, but otherwise different, with the numerical results being correct.

The nuclear quadrupole moment is

$$Q_{\text{nuc}} = Z_{\text{A}}(R_{\text{A}} - R_0)^2 + Z_{\text{B}}(R_{\text{B}} - R_0)^2 \quad (3.7)$$

and the analytical electronic quadrupole moment is a rank-two tensor, with just two independent components for a diatomic aligned on the  $z$  axis,

$$Q_{\text{ele}}^{\alpha} = - \sum_{\mu\nu} P_{\mu\nu} Q_{\mu\nu}^{\alpha} , \quad (3.8)$$

where  $\alpha$  is either  $xx$  or  $zz$  and  $Q_{\mu\nu}$  are the [quadrupole moment integrals](#) over basis functions,  $\langle \mu | (\mathbf{r} - \mathbf{R}_0)^2 | \nu \rangle$ . The total  $zz$  component of the quadrupole tensor is then

$$Q^{zz} = Q_{\text{nuc}} + Q_{\text{ele}}^{zz} . \quad (3.9)$$

The isotropic quadrupole is

$$Q_{\text{iso}} = \frac{1}{3}(2Q^{xx} + Q^{zz}) \quad (3.10)$$

and the anisotropic quadrupole is

$$Q_{\text{aniso}} = Q^{zz} - Q^{xx} . \quad (3.11)$$

Similarly to the dipole moment, the nuclear, isotropic and anisotropic components of the analytical quadrupole moment are printed at the energy of a calculation by default, with a diagram showing the orientation of the quadrupole, here that the outsides of the molecule are more positive:

```

~~~~~
                          Quadrupole Moment
~~~~~
Nuclear: 10.2716330 Anisotropic: 10.2716330

Isotropic: 2.6323036 H --- He +-> <-+
~~~~~

```

When the QUADRUPOLE keyword is used, the quadrupole components are calculated as derivatives of the energy with respect to an applied electric field gradient,

$$Q_{\text{ele}}^{\alpha} = \frac{\partial E}{\partial(\partial_{\alpha} F_{\alpha})} . \quad (3.12)$$

This requires four additional energy evaluations, but recalculation of the molecular integrals can be skipped, speeding things up. The  $xx$  and  $zz$  electronic components of the quadrupole, as well as the total isotropic and anisotropic quadrupole, are printed.

```
TUNA SPE : C O 1.0 : MP2 cc-pVDZ : QUADRUPOLE
```

### 3.1.4 Polarisability and Hyperpolarisability

Another electric property calculated by TUNA is the polarisability, requested with the POLAR keyword. In general, there are nine components of the polarisability matrix which are

$$\alpha_{\alpha\beta} = \frac{\partial^2 E}{\partial F_{\alpha} \partial F_{\beta}} . \quad (3.13)$$

Due to the symmetry of a diatomic molecule, there are only two unique components of the dipole–dipole polarisability matrix so only two second derivatives are necessary — eight additional energy calculations in total. These give the parallel and perpendicular components of polarisability, then

the isotropic polarisability is

$$\alpha_{\text{iso}} = \frac{1}{3}(\alpha_{\perp} + \alpha_{\perp} + \alpha_{\parallel}) \quad (3.14)$$

and the anisotropic polarisability is

$$\alpha_{\text{aniso}} = \alpha_{\parallel} - \alpha_{\perp} . \quad (3.15)$$

These are both printed (along with the parallel and perpendicular component with P) with the dipole moment during a polarisability calculation. The following input line calculates polarisability on the hydrogen molecule, giving a value of  $5.2 a_0^3$ , close to the experimental value of  $5.4 a_0^3$  [14].

```
TUNA SPE : H H 0.741 : CCSD aug-cc-pVTZ : POLAR
```

The dipole–dipole–dipole hyperpolarisability can also be calculated in TUNA with the HYPER keyword, and in general has  $3 \times 3 \times 3 = 27$  components since

$$\beta_{\alpha\beta\gamma} = \frac{\partial^3 E}{\partial F_{\alpha} \partial F_{\beta} \partial F_{\gamma}} . \quad (3.16)$$

For diatomics, like the polarisability there are only two independent components, the parallel and perpendicular components,  $\beta_{zzz}$  and  $\beta_{xxz}$ . This means twelve energy evaluations are necessary so the calculation isn't too much slower than for polarisability.

The hyperpolarisability of a  $D_{\infty h}$  molecule is necessarily zero due to symmetry, but can be calculated anyway. A  $C_{\infty v}$  molecule has two independent components. For example:

```
TUNA SPE : H He 1.0 : CCSD aug-cc-pVTZ : HYPER CH 1
```

Calculations can also be requested where an external electric field is applied to the molecule. This can be requested with the EX, EY and EZ for the Cartesian electric field components, in atomic units. For example an optimisation in the field

$$\mathbf{F} = 0.05 \hat{\mathbf{x}} + 0.06 \hat{\mathbf{y}} + 0.10 \hat{\mathbf{z}} \quad (3.17)$$

is requested by:

```
TUNA OPT : H H 0.74 : SCS-MP3 cc-pVTZ : EX 0.05 EY 0.06 EZ 0.10
```

Similarly, calculations in an applied electric field gradient are requested with EGX, EGY and EGZ.

### 3.1.5 Population Analysis

Towards the end of a calculation, TUNA prints Mulliken [15], Löwdin [16] and Mayer [17] population analysis information. Note that all the equations in this section are true for diatomics, not necessarily molecules in general. These calculations can be disabled with the T keyword, or requested after every geometry iteration in an optimisation with P.

The Mulliken analysis [15] is very widely used — despite its considerable weaknesses [18] — and partitions the electron density by the basis functions assigned to each atom. The number of electrons,  $N$ , is the integral of the density and can be calculated by

$$N = \sum_{\mu\nu} P_{\mu\nu} S_{\mu\nu} . \quad (3.18)$$

When the density is partitioned equally between atoms, the number of electrons on atom A is

$$N_A = \sum_{\mu \in A} \sum_{\nu \in A} P_{\mu\nu} S_{\mu\nu} + \frac{1}{2} B_{AB} , \quad (3.19)$$

where  $B_{AB}$  is the Mulliken bond order, given by

$$B_{AB} = 2 \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu} S_{\mu\nu} . \quad (3.20)$$

The partial charge of an atom in a molecule is then given by  $Q_A = Z_A - N_A$ . In a population analysis, TUNA prints these charges and the Mulliken bond order, as well as the sum of the charges which should be equal to the requested molecular charge.

Because the molecular orbitals that generate a specific energy and density are not unique, population analysis can be performed with non-localised orthogonal basis functions in Löwdin analysis [16] where the density matrix is rotated by

$$\mathbf{P}^L = \mathbf{S}^{\frac{1}{2}} \mathbf{P} \mathbf{S}^{\frac{1}{2}} . \quad (3.21)$$

Using this density matrix, atomic populations are calculated by

$$N_A = \sum_{\mu \in A} P_{\mu\mu}^L , \quad (3.22)$$

and the bond order is calculated by

$$B_{AB} = \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu}^L P_{\nu\mu}^L . \quad (3.23)$$

Similarly to in Mulliken analysis, the Löwdin charges and bond order are printed.

The results of a Mayer analysis [17] are also printed. The Mayer charges are the same as the Mulliken charges, but the bond order (and bonded valence) is given by

$$B_{AB} = \sum_{\mu \in A} \sum_{\nu \in B} (\mathbf{PS})_{\mu\nu} (\mathbf{PS})_{\nu\mu} + (\mathbf{RS})_{\mu\nu} (\mathbf{RS})_{\nu\mu} , \quad (3.24)$$

where  $\mathbf{R}$  is the spin density matrix,  $\mathbf{P}_\alpha - \mathbf{P}_\beta$ . In Mayer analysis the valence of each atom is calculated. The total valence, which is similar to the expected valence of an atom (ie. one for hydrogen, three for boron), is

$$V_A = 2N_A - \sum_{\mu \in A} \sum_{\nu \in B} (\mathbf{PS})_{\mu\nu} (\mathbf{PS})_{\nu\mu} \quad (3.25)$$

and the free valence, which is a measure of the ability to form further bonds, is calculated by  $F_A = V_A - B_{AB}$ . These valences are printed under the Mayer analysis heading.

### 3.1.6 Natural Orbitals

For UHF, second-order perturbation theory and coupled cluster methods, natural orbitals can be calculated. These are the eigenfunctions of the one-electron reduced density matrix

$$\mathbf{P}\mathbf{\Lambda}_i = \lambda_i \mathbf{\Lambda}_i , \quad (3.26)$$

where  $\lambda_i$  represents the occupancy of the  $i$ th natural orbital, a value between 0.0 and 2.0. Natural orbitals provide a basis that converges most efficiently to the exact wavefunction [19], and the occupancies are useful for determining systems which might benefit from a multireference treatment, beyond the single determinant model of Hartree–Fock theory. Occupancies that deviate strongly from 2.0 (occupied) or 0.0 (virtual) indicate natural orbitals with strong correlation.

The calculation of natural orbitals is disabled by default, but can be activated with the NATORBS keyword. A coupled cluster doubles calculation with natural orbitals is requested by:

```
TUNA SPE : H H 2.0 : CCD def2-SVPD : NATORBS
```

The sum of the natural orbital occupancies is also printed along with the orbital occupancies, along with the density matrix trace. These values should be equal to the total number of electrons. Natural orbital calculations are possible with both unrelaxed and relaxed density matrices for MP2.

Using the PRINTMOS keyword, the occupancies and coefficients of each natural orbital can be seen:

```
TUNA SPE : H H 3.0 : UHF cc-pVDZ : PRINTMOS 5 NATORBS
```

### 3.1.7 Plotting Orbitals and Densities

The infrastructure that supports [density functional theory](#) calculations enables the expression of orbitals and electron densities onto a grid, which can be plotted by TUNA. While the grids for DFT integration are more dense near the nuclei and more sparse further away, the plotting grids are two-dimensional Cartesian grids — with size defined by the bond length. These two-dimensional grids are much more efficient to compute than three-dimensional “cube” files, and contain all the information necessary to describe a diatomic, due to these molecules’ symmetry.

The simplest plot is the electron density, requested via the DENSPLIT keyword, which shows a magenta blob of the converged electron density:

```
TUNA SPE : H H 0.74 : HF STO-3G : DENSPLIT
```

For unrestricted calculations, the SPINDENSPLIT keyword plots the spin density. For example, the following calculation on the carbyne radical, shown in [Figure 3.1](#), shows the majority of the spin density is localised on the carbon atom. Red indicates  $\alpha$  spin density and blue shows  $\beta$  density.

```
TUNA SPE : C H 1.0 : HF cc-pVDZ : SPINDENSPLIT COREGUESS
```

Difference densities of the  $i$ th excited state from [excited state calculations](#), calculated by

$$\mathbf{P}_{\text{diff}}^i = \mathbf{P}_{\text{excited}}^i - \mathbf{P}_{\text{HF}} \quad (3.27)$$

can be plotted with the DIFFDENSPLIT keyword. A similar equation calculates the difference spin density, given by the DIFFSPINDENSPLIT keyword. For example, the third excited state difference density of the carbyne radical, shown in [Figure 3.2](#), is plotted by:

```
TUNA SPE : C H 1.0 : CIS cc-pVDZ : DIFFDENSPLIT ROOT 3
```

Areas of positive difference density, which have gained electrons in the excited state, are shown in red, and areas which have lost electrons are shown in blue.

The converged molecular orbitals from a [Hartree–Fock](#) or [DFT](#) calculation can also be plotted. The PLOTMO keyword is used to do this, along with the number of molecular orbital that is desired to be plotted. The orbital numbers can be checked with the additional print keyword, P or PRINTMOS. For unrestricted references, the  $\alpha$  and  $\beta$  orbitals are interleaved. [Figure 3.3](#) shows a virtual 4d

Spin density from UHF/CC-PVDZ calculation on C—H molecule

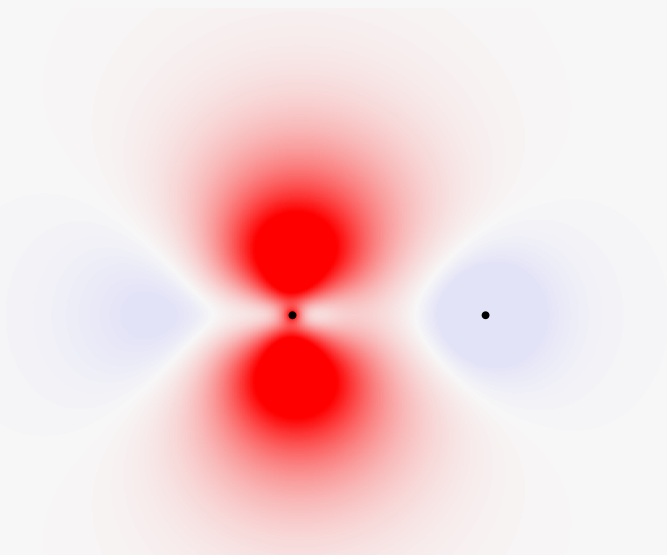


Figure 3.1: The spin density of carbyne, showing the vast majority of the spin density is found on the carbon atom, in the shape of a p orbital.

orbital of a hydrogen atom, plotted by:

```
TUNA SPE : H : PBE cc-pVQZ : PLOTMO 29
```

The HOMO and LUMO can also be plotted with the PLOTHOMO and PLOTLUMO keywords, respectively. The following calculation shows the  $\pi^*$  orbital of the triplet oxygen molecule.

```
TUNA SPE : O O 1.1 : B3LYP 6-311++G** : ML 3 PLOTHOMO
```

If [natural orbitals](#) have been calculated with NATORBS they can be plotted similarly, using the PLOTNO keyword. If no natural orbitals are found, the plot will just be blank.

```
TUNA SPE : C O 1.1 : CCSD cc-pVDZ : NATORBS PLOTNO 7
```

### 3.1.8 Dispersion Correction

The dispersion energy is long range and poorly described by most exchange–correlation functionals, so semi-empirical corrections are often used to improve the description of non-covalent interactions. There are two forms of dispersion correction available in TUNA: the atomic pairwise D2 scheme



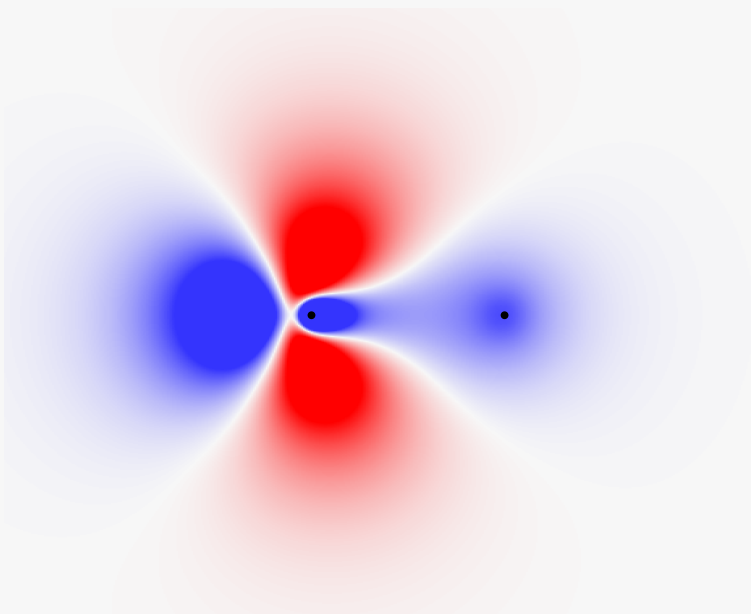


Figure 3.2: The difference density from an excited state of carbyne. This shows that the electron density has moved from around the lone pair on the carbon atom (deep blue), to the bond area.

by Grimme [20], and the non-local VV10 method based on integrating the electron density [21].

The D2 correction is requested with the D2 keyword:

TUNA SPE : H H 1.0 : PBE 6-311++G : D2

In this method, the dispersion energy is calculated by

$$E_{\text{disp}} = -\frac{s_6}{f_{\text{damp}}} \frac{C_6^{AB}}{R^6}, \quad (3.28)$$

where the damping function is given by

$$f_{\text{damp}} = 1 + \exp \left[ -d \left( \frac{R}{R_{\text{VdW}}^{AB}} - 1 \right) \right], \quad (3.29)$$

and  $C_6^{AB} = \sqrt{C_6^A C_6^B}$  while  $R_{\text{VdW}}^{AB} = R_{\text{VdW}}^A + R_{\text{VdW}}^B$ .

The default value of the damping factor  $d$  is 20, and the  $s_6$  value is 1.2 in line with the parameterisation of Hartree–Fock in ORCA [4, 5]. The values of the Van der Waals radii,  $R_{\text{VdW}}$ , and  $C_6$

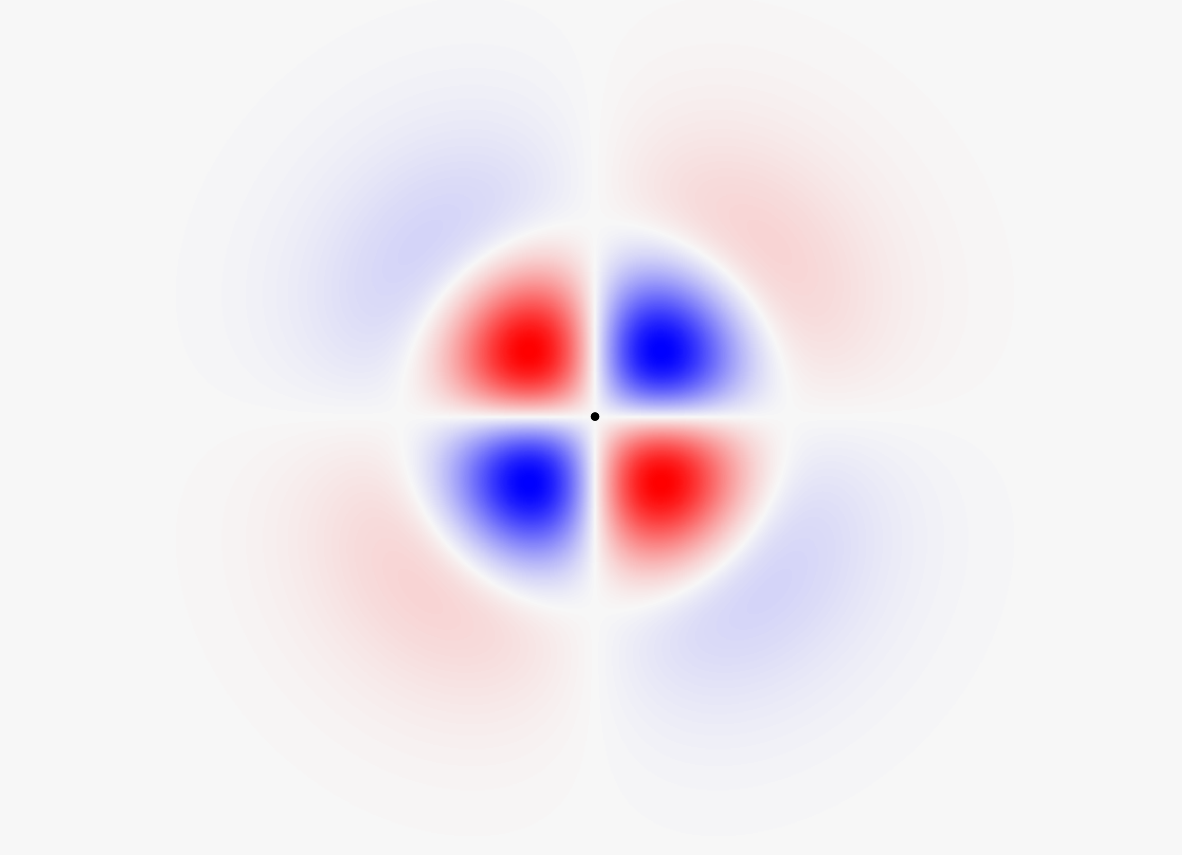


Figure 3.3: Using a large basis set like cc-pVQZ allows high-lying unoccupied orbitals to be visualised. This is a virtual 4d orbital for the hydrogen atom.

coefficients are element-specific. The optimised values of  $s_6$  are implemented for various [density functional theory](#) methods [20]. These are PBE, BLYP, B3LYP, B2PLYP, mPW2PLYP, BP86 and TPSS. The  $s_6$  value will print out to the console whenever the P keyword is used.

The non-local VV10 dispersion correction is requested with the NL keyword:

```
TUNA SPE : H H 1.0 : PBE 6-311++G : NL
```

In VV10, the dispersion energy is a non-local double integral,

$$E_{\text{disp}} = \int n(\mathbf{r}) \left[ \beta + \frac{1}{2} \int d\mathbf{r}' n(\mathbf{r}') \Phi(\mathbf{r}, \mathbf{r}') \right] d\mathbf{r}, \quad (3.30)$$

where  $n(\mathbf{r})$  is the electron density,  $\beta$  is an exchange–correlation functional-dependent parameter,

and  $\Phi(\mathbf{r}, \mathbf{r}')$  is known as the VV10 kernel.

In TUNA, the VV10 energy is computed at the end of the SCF loop, not self-consistently every iteration — this is not likely to make a meaningful difference to the final energy. The double integral is calculated with the same [numerical quadrature](#) used for [density functional theory](#) calculations, but is still very slow as the full non-local grid cannot be stored simultaneously in memory.

This method is parameterised for [Hartree–Fock](#), and a number of [density functional theory](#) methods. These are PBE, RPBE, revPBE, PBE0, revPBE0, revPBE38, BLYP, B3LYP, B2PLYP, DSD-BLYP, BP86, TPSS, SCAN, rSCAN, r<sup>2</sup>SCAN, TPSSH, TPSS0, r<sup>2</sup>SCANh, r<sup>2</sup>SCAN0, r<sup>2</sup>SCAN50, PW1PW, B3PW91, B3P86, Pr<sup>2</sup>SCAN50, Pr<sup>2</sup>SCAN69. The B97M-V functional is also parameterised for VV10, and in that case the NL keyword does not need to be explicitly used:

```
TUNA SPE : H H 1.0 : B97M-V 6-311++G :
```

## 3.2 Coordinate Scan

A coordinate scan can be requested in TUNA with the SCAN calculation type. This calculates the potential energy surface between two points, increasing the interatomic distance at each step, using the chosen electronic structure method, basis set and miscellaneous keywords.

There are two mandatory parameters for a coordinate scan. The distance between steps must be specified by the STEP keyword, and the number of scan steps is specified by the NUM keyword. There are no default parameters here; these must be specified when a SCAN calculation is requested.

For instance, an MP2 scan over the potential energy surface of H<sub>2</sub>, starting at a bond length of 0.3 Å and increasing with 20 steps of 0.1 Å can be requested by:

```
TUNA SCAN : H H 0.3 : MP2 6-31G : NUM 20 STEP 0.1
```

By default, the previous step's SCF density matrix is read in at each scan step to accelerate convergence; this can be disabled using the NOMOREAD keyword. For UHF calculations on singlet  $D_{\infty h}$  molecules, the molecular orbitals are rotated after being read, unless the NOROTATE keyword is used. See [subsection 4.1.1](#) for a discussion of the initial guess and ROTATE keywords.

Any of the keywords that can be used with a [single point energy](#) calculation (eg. D2, SLOWCONV, DIPOLE, EXTRAPOLATE) can also be combined with the coordinate scan calculation. At the end, a table of bond lengths and final energies is printed, which can be easily copied into a spreadsheet.

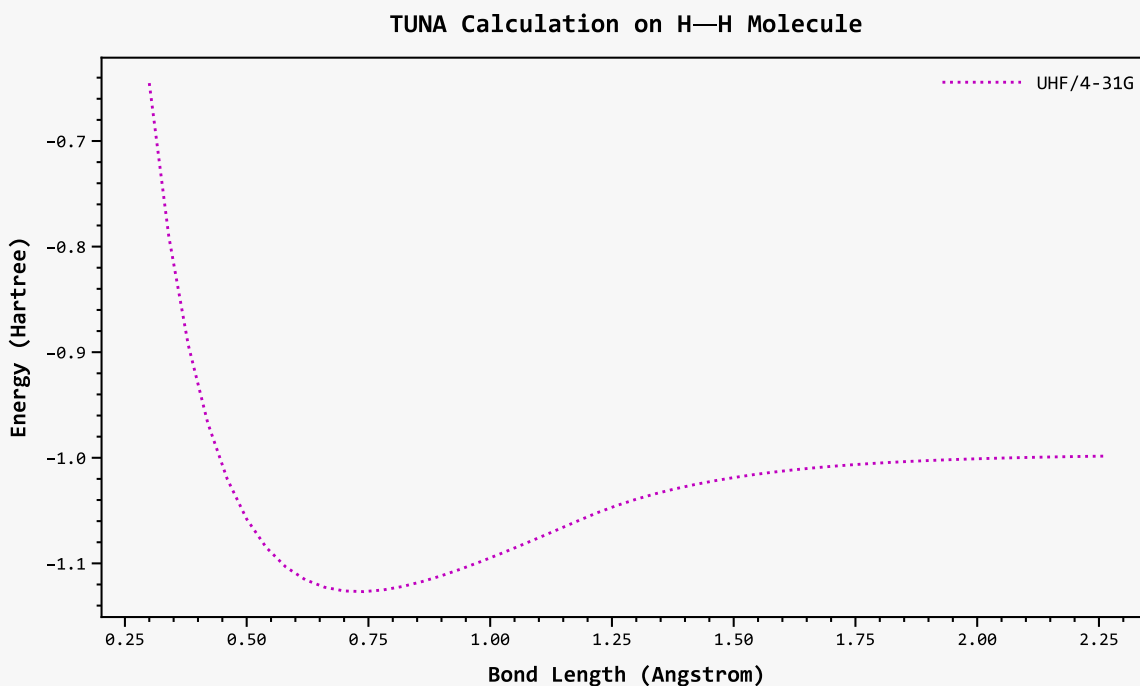


Figure 3.4: Potential energy surface of  $\text{H}_2$  calculated in TUNA with UHF/4-31G

### 3.2.1 Plotting Coordinate Scans

Due to the Matplotlib [8] integration in TUNA, two-dimensional plots can easily be generated from coordinate scan calculations with the SCANPLOT keyword. The line is straight and blue by default, but can be changed to dashed using the keyword DASH, dotted using DOT, and the colour can be changed to any of BLACK, WHITE, RED, BLUE, YELLOW, GREEN, CYAN, and MAGENTA.

For example, a plot with a magenta dotted line over the UHF  $\text{H}_2$  potential energy surface is:

```
TUNA SCAN : H H 0.3 : UHF 4-31G : NUM 40 STEP 0.04 SCANPLOT DOT MAGENTA
```

This produces the plot in Figure 3.4, with the legend showing the electronic structure method and basis set. Plots can be saved by the keyword SAVEPLOT [filepath], for example SAVEPLOT "MAGENTAPLOT.PDF". Standard raster and vector image extensions are supported.

Any plot colour can be chosen with the COLOUR keyword, followed by a hexadecimal hash:

```
TUNA SCAN : Li H 0.6 : HF def2-SVP : COLOUR #00FF00 NUM 40 STEP 0.1 SCANPLOT
```

Multiple calculations can be plotted on the same axes for easy comparison. To do this, use the

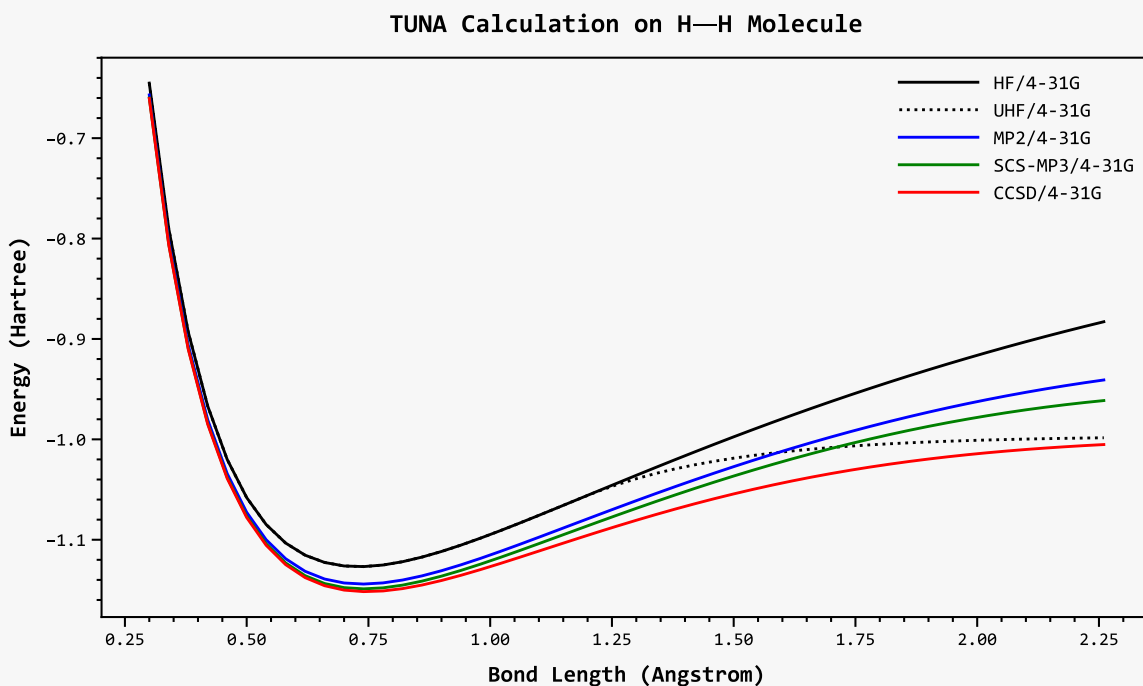


Figure 3.5: Potential energy surfaces of  $\text{H}_2$  calculated in TUNA with various electronic structure methods in 4-31G basis set, overlaid using the ADDPLOT keyword and shown with SCANPLOT

keyword ADDPLOT with every calculation you want overlaid on the same axes. Then just run calculations one after the other. To clear the axes, use the DELPLOT keyword once. The DELPLOT keyword is parsed before ADDPLOT, so they can be used together to start from a blank canvas. The following calculations, run one after the other, produced [Figure 3.5](#).

```
TUNA SCAN : H H 0.3 : HF 4-31G : NUM 50 STEP 0.04 SCANPLOT BLACK ADDPLOT
```

```
TUNA SCAN : H H 0.3 : UHF 4-31G : NUM 50 STEP 0.04 SCANPLOT BLACK DOT ADDPLOT
```

```
TUNA SCAN : H H 0.3 : MP2 4-31G : NUM 50 STEP 0.04 SCANPLOT BLUE ADDPLOT
```

```
TUNA SCAN : H H 0.3 : SCS-MP3 4-31G : NUM 50 STEP 0.04 SCANPLOT GREEN ADDPLOT
```

```
TUNA SCAN : H H 0.3 : CCSD 4-31G : NUM 50 STEP 0.04 SCANPLOT RED ADDPLOT
```

This kind of plotting can also be combined with [excited state calculations](#) to easily see the different potential energy surfaces. Note that TUNA does not currently have a root-following algorithm, so it's easy for potential energy surfaces to get mixed up part way through a coordinate scan.

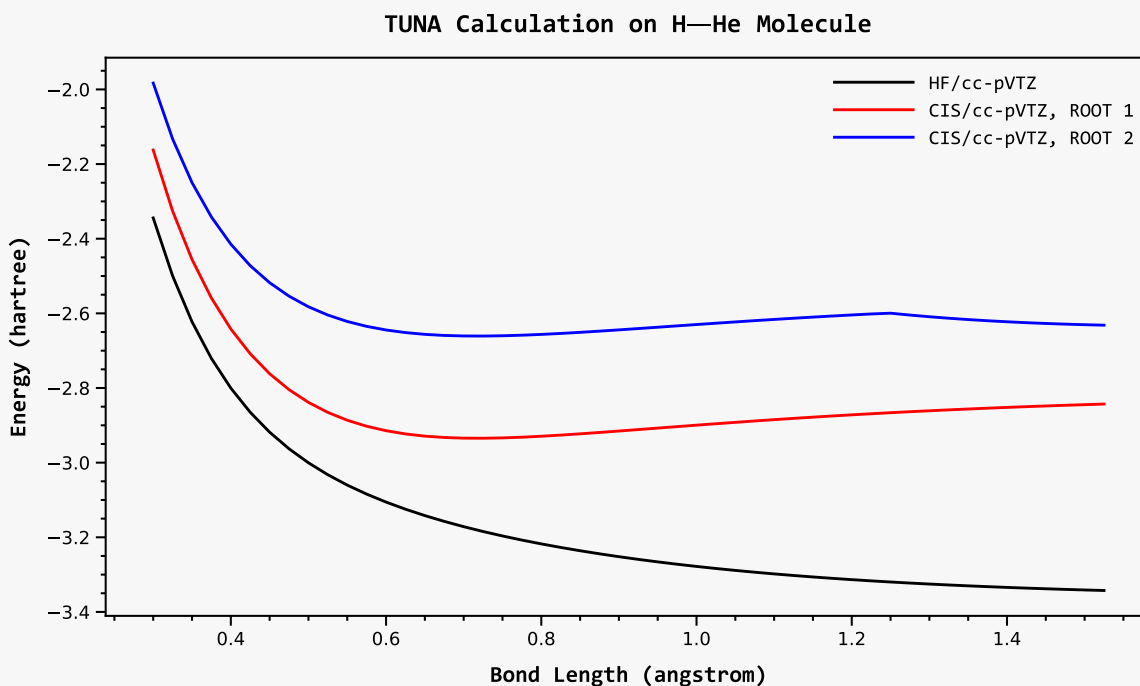


Figure 3.6: Potential energy surfaces of the H—He ground state and first two excited states calculated in TUNA with HF/cc-pVTZ and CIS/cc-pVTZ

The following calculations on the H—He molecule produced [Figure 3.6](#). The ground state calculation shows a bond does not form, and the atoms dissociate. However absorbing a photon does allow a metastable molecule to form, with local minima found in the excited state surfaces. The sharp turn in the second excited surface indicates a crossing between two potential energy surfaces.

```
TUNA SCAN : H He 0.3 : HF cc-pVTZ : NUM 50 STEP 0.025 SCANPLOT BLACK ADDPLOT
```

```
TUNA SCAN : H He 0.3 : CIS cc-pVTZ : NUM 50 STEP 0.025 SCANPLOT RED ADDPLOT
```

### 3.3 Geometry Optimisation

A geometry optimisation finds a stationary point on the potential energy surface, where

$$\frac{\partial E}{\partial R} = 0, \quad (3.31)$$

and is requested by the OPT calculation type keyword:

```
TUNA OPT : H He 1.0 : RHF 6-311G : CH 1
```

At the start of an optimisation, the convergence criteria, maximum number of iterations and maximum allowed step are printed, before the calculation sets off. In each geometry iteration, the molecule is set up and the energy is calculated with the chosen method. Tight [energy convergence criteria](#) are used by default for optimisations and the SCF density matrix is read in from the previous optimisation step — this guess strategy can be disabled using the NOMOREAD keyword.

Next, the gradient is calculated using the central differences method, by

$$\nabla E_n = \frac{\partial E}{\partial R_n} = \frac{E(R_n + h) - E(R_n - h)}{2h}, \quad (3.32)$$

where  $R$  is the bond length and  $h$  is defined in [section 7.3](#). Next, an approximation to the geometric Hessian is calculated by

$$H_n = \frac{\nabla E_n - \nabla E_{n-1}}{R_n - R_{n-1}}. \quad (3.33)$$

Alternatively, the exact Hessian can be calculated using the CALCHESS keyword. This increases calculation time at each step significantly, but tends to reduce the number of geometry iterations needed. For diatomics, the exact Hessian calculation is probably pointless unless the potential energy surface is very flat. For optimisations to a minimum, either the exact or approximate Hessian is used in the convex region of the potential energy surface, whereas for optimisations to maxima (using OPTMAX), these Hessians are used in the concave region. Beyond these regions, a default Hessian of 1/4 is used. This value can be changed using the DEFAULTHESS keyword.

After the Hessian is determined, the bond length is updated by

$$R_{n+1} = R_n - \frac{\nabla E_n}{H_n}. \quad (3.34)$$

Note that for a one-dimensional potential energy surface there is no choice necessary about the update — this is the only method that uses derivatives up to second order.

The maximum value for the change in bond length is 0.2 Å by default, to prevent exploding optimisations from a bad starting guess geometry. This can be changed with the MAXSTEP keyword, giving the maximum allowed step in angstroms. The trajectory of a geometry optimisation can be printed to an .xyz file using the TRAJ keyword. The structures and energies at each point are printed to a file called "tuna-trajectory.xyz" in the directory of the terminal. An example of an optimisation with an exact Hessian, reduced maximum step and printed trajectory is:

TUNA OPT : H F 0.9 : HF 4-31G : CALCHESS MAXSTEP 0.1 TRAJ

**Table 3.2** Optimisation convergence criteria in TUNA

| Convergence | Gradient / hartree bohr <sup>-1</sup> | Step / Å         |
|-------------|---------------------------------------|------------------|
| LOOSEOPT    | 10 <sup>-3</sup>                      | 10 <sup>-2</sup> |
| MEDIUMOPT   | 10 <sup>-4</sup>                      | 10 <sup>-4</sup> |
| TIGHTOPT    | 10 <sup>-6</sup>                      | 10 <sup>-5</sup> |
| EXTREMEOPT  | 10 <sup>-8</sup>                      | 10 <sup>-7</sup> |

### 3.3.1 Optimisation Convergence

Four optimisation convergence criteria keywords are available: LOOSEOPT, MEDIUMOPT, TIGHTOPT and EXTREMEOPT. Medium convergence is the default for standard optimisation calculations, while [optimisation and frequency](#) and [anharmonic frequency](#) calculations use tight default criteria.

The gradient and step convergence criteria of these are listed in [Table 3.2](#). If the geometry has not converged to the chosen criteria within the maximum number of iterations, that maximum can be increased with the MAXGEOMITER keyword, which has a value of 30 by default.

When the gradient and step fall below the convergence criteria, the optimisation will terminate, and the final density will be used to calculate [analytical properties](#). These can be calculated at every step with the additional print keyword, P. Finally, TUNA prints the bond length of the converged geometry and the optimised energy. If a [harmonic frequency](#) calculation is desired immediately after the optimisation terminates, the OPTFREQ calculation type should be used instead of OPT.

### 3.3.2 Force Calculations

The FORCE keyword calculates the force on the current molecular geometry, without optimisation:

```
TUNA FORCE : H C 1.1 : mPW1LYP def2-SVP : CH 1
```

This calculates the energy and the gradient at the bond length and prints them to the terminal.

## 3.4 Harmonic Frequency

A harmonic frequency calculation can be requested on a molecule with the FREQ keyword:

```
TUNA FREQ : H H 0.7375 : RHF 6-311++G :
```



This can — and often should — be combined with a prior [geometry optimisation](#), where the frequency calculation is run on the optimised structure, using the OPTFREQ calculation type. All the keywords available for the OPT calculation type are available for OPTFREQ.

```
TUNA OPTFREQ : H O 1.0 : CCSD[T] 6-31G* : CH -1
```

Harmonic frequency calculations in TUNA determine the numerical second derivative, which is the bond's force constant,  $k$ , (and the Hessian,  $H$ ) by

$$k = \frac{\partial^2 E}{\partial R^2}. \quad (3.35)$$

The use of five [single point calculations](#) makes harmonic frequency calculations slow, but ensures the frequencies are numerically stable. For additional reliability, EXTREME SCF [convergence criteria](#) are used for frequency calculations by default, but by optimising the numerical step length, the frequencies have been found to be converged to  $0.01 \text{ cm}^{-1}$  even with LOOSE energy convergence.

After the second derivatives are determined, the vibrational frequency is calculated by

$$\omega = \sqrt{\frac{k}{\mu}}, \quad (3.36)$$

where  $\mu$  is the reduced mass, calculated using the atomic masses in [Table 3.3](#). These are the masses of the most abundant isotope. Custom masses can be used via the M1 and M2 keywords, where the first and second atom's mass can be chosen. For example, this calculation gives the second atom — chlorine — a mass of 36.0. The atomic masses can be printed using P.

```
TUNA OPTFREQ : H Cl 1.30 : RHF 6-311++G : M2 36.0
```

The vibrational frequency, reduced mass and force constant are all printed to the console. This printed frequency has units of  $\text{cm}^{-1}$ . If  $k$  is negative, such as at a maximum on the potential energy surface,  $\omega$  will be imaginary and TUNA will output the frequency prepended with an “i”.

The harmonic frequencies of  $\text{H}_2$  calculated from some different methods implemented in TUNA are shown in [Table 3.4](#), where [coupled cluster](#) methods perform much better than [perturbation theory](#) and B3LYP does very well. These values were calculated using [two-point basis set extrapolation](#) with triple- and quadruple- $\zeta$  bases. For example:

```
TUNA OPTFREQ : H H 1.0 : CCSD cc-pVTZ : EXTRAPOLATE
```

**Table 3.3** Default atomic masses in TUNA, using data from [22, 23].

| Atom       | Major Isotope    | Mass / amu |
|------------|------------------|------------|
| Hydrogen   | <sup>1</sup> H   | 1.007825   |
| Helium     | <sup>4</sup> He  | 4.002603   |
| Lithium    | <sup>7</sup> Li  | 7.016004   |
| Beryllium  | <sup>9</sup> Be  | 9.012182   |
| Boron      | <sup>11</sup> B  | 11.009305  |
| Carbon     | <sup>12</sup> C  | 12.000000  |
| Nitrogen   | <sup>14</sup> N  | 14.003074  |
| Oxygen     | <sup>16</sup> O  | 15.994915  |
| Fluorine   | <sup>19</sup> F  | 18.998403  |
| Neon       | <sup>20</sup> Ne | 19.992440  |
| Sodium     | <sup>23</sup> Na | 22.989770  |
| Magnesium  | <sup>24</sup> Mg | 23.985042  |
| Aluminium  | <sup>27</sup> Al | 26.981538  |
| Silicon    | <sup>28</sup> Si | 27.976927  |
| Phosphorus | <sup>31</sup> P  | 30.973762  |
| Sulfur     | <sup>32</sup> S  | 31.972071  |
| Chlorine   | <sup>35</sup> Cl | 34.968853  |
| Argon      | <sup>40</sup> Ar | 39.962383  |

### 3.4.1 Transition Intensity

A commonly used measure of fundamental absorption intensity is the integral absorption coefficient,  $\mathcal{A}$ , which has units  $\text{km mol}^{-1}$ . This coefficient is calculated by

$$\mathcal{A} = \frac{\pi N_{\text{A}}}{3c^2} \left( \frac{\partial \boldsymbol{\mu}}{\partial \mathbf{q}} \right)^2. \quad (3.37)$$

This absorption coefficient is calculated as it is independent of the experimental setup — including the incident light intensity, molar concentration and length of sample cell [25]. This is printed under the "transition intensity" header, along with the dipole moment derivative and its square.

This intensity is proportional to the derivative of the dipole moment at the equilibrium geometry

**Table 3.4** Harmonic frequencies of H<sub>2</sub> calculated with [cc-pVTZ, cc-pVQZ] extrapolation

| Method     | Harmonic Frequency / cm <sup>-1</sup> |
|------------|---------------------------------------|
| HF         | 4580.4                                |
| LSDA       | 4179.1                                |
| PBE        | 4310.5                                |
| B3LYP      | 4409.7                                |
| MP2        | 4517.3                                |
| MP3        | 4466.1                                |
| MP4        | 4431.7                                |
| CCD        | 4412.8                                |
| CCSD       | 4401.4                                |
| Exact [24] | 4401.2                                |

in mass-weighted normal coordinates,

$$\mathcal{A} \propto \frac{\partial \mu}{\partial \mathbf{q}}, \quad (3.38)$$

which is determined by [numerical differentiation](#) while the Hessian is determined.

The [dipole moment](#) at each geometry is calculated analytically by default, but can be calculated numerically with the DIPOLE keyword. This will give the correct intensity for electronic structure methods without an implemented relaxed density matrix — see [subsection 3.1.3](#). For example, this input line calculates the [full configuration interaction](#) harmonic intensity for lithium hydride:

TUNA FREQ : Li H 1.54747 : CCSDTQ cc-pVDZ : DIPOLE

### 3.4.2 Thermochemistry

After the vibrational frequency calculation is finished TUNA begins a quick calculation of thermochemical parameters. The contributions to internal energy,

$$U = E + E_{\text{zero-point}} + E_{\text{translational}} + E_{\text{vibrational}} + E_{\text{rotational}}, \quad (3.39)$$

and entropy,

$$S = S_{\text{electronic}} + S_{\text{translational}} + S_{\text{vibrational}} + S_{\text{rotational}} , \quad (3.40)$$

are first calculated, before the enthalpy is determined,

$$H = U + k_{\text{B}}T , \quad (3.41)$$

where  $k_{\text{B}}$  is the Boltzmann constant and  $T$  is the absolute temperature. The Gibbs free energy,

$$G = H - TS , \quad (3.42)$$

is evaluated and printed. All these values, including entropies, are expressed in hartree.

The zero-point energy is given by  $E_{\text{zero-point}} = \frac{1}{2}\omega$ , and is excluded if the vibrational frequency is imaginary. The translational contribution to energy is  $E_{\text{translational}} = \frac{3}{2}k_{\text{B}}T$  and the rotational contribution is  $E_{\text{rotational}} = k_{\text{B}}T$ . Finally, the vibrational contribution is calculated by

$$E_{\text{vibrational}} = \frac{\omega}{\exp(\omega/k_{\text{B}}T) - 1} , \quad (3.43)$$

assuming the molecule behaves as a harmonic oscillator, which is a bad approximation for diatomics, particularly at high temperatures. This entropy is also excluded for imaginary frequencies.

The electronic contribution to entropy is calculated from the multiplicity (related to the total spin  $S$ ) assuming only the ground state is populated, by

$$S_{\text{electronic}} = k_{\text{B}} \ln(2S + 1) . \quad (3.44)$$

The translational, vibrational and rotational contributions are calculated by three fairly horrible formulas, where  $p$  is the pressure,  $B$  is the [rotational constant](#) and  $M$  is the molecular mass:

$$S_{\text{translational}} = k_{\text{B}} \left[ \frac{5}{2} + \ln \left( \frac{M k_{\text{B}} T}{h} \right)^{\frac{3}{2}} + \ln \left( \frac{k_{\text{B}} T}{p} \right) \right] \quad (3.45)$$

$$S_{\text{vibrational}} = k_{\text{B}} \left[ \frac{\omega}{k_{\text{B}} T (\exp(\omega/k_{\text{B}}T) - 1)} - \ln \left( 1 - \exp \left( -\frac{\omega}{k_{\text{B}} T} \right) \right) \right] \quad (3.46)$$

$$S_{\text{rotational}} = k_B + k_B \ln \left( \frac{k_B T}{2\pi\sigma Bc} \right) \quad (3.47)$$

For the rotational entropy, the symmetry number,  $\sigma$ , is determined from the molecular point group. Symmetric,  $D_{\infty h}$  molecules have  $\sigma = 2$  while asymmetric,  $C_{\infty v}$  molecules have  $\sigma = 1$ .

The enthalpy, entropy, and Gibbs free energy depend on the temperature and pressure. Using the TEMP and PRES keywords, these quantities can be specified. For instance, a thermochemistry calculation at 340 K and 101500 Pa can be requested by:

TUNA FREQ : H H 0.6750 : RHF 6-311++G : TEMP 340 PRES 101500

### 3.4.3 Vibrational Perturbation Theory

The Taylor expansion of the potential energy surface around a stationary point is

$$E(R) = E + \frac{\partial E}{\partial R} R + \frac{1}{2} \frac{\partial^2 E}{\partial R^2} R^2 + \frac{1}{6} \frac{\partial^3 E}{\partial R^3} R^3 + \frac{1}{24} \frac{\partial^4 E}{\partial R^4} R^4 + \dots \quad (3.48)$$

and since the gradient is zero and we can choose the energy gauge, this simplifies to

$$E(R) = \frac{1}{2} k R^2 + \frac{1}{6} \frac{\partial^3 E}{\partial R^3} R^3 + \frac{1}{24} \frac{\partial^4 E}{\partial R^4} R^4 + \dots \quad (3.49)$$

where the first term is the harmonic potential, used to solve the nuclear Schrödinger equation in the [quantum harmonic oscillator](#) model of vibration. Vibrational perturbation theory provides an improved treatment by considering the next terms. Because first- and second-order vibrational perturbation theory share the same computational cost, the second-order treatment, VPT2, is used in practice although the first-order VPT1 is also implemented in TUNA.

This requires third and fourth derivatives of the energy, which are calculated in TUNA with eight- and nine-point stencils respectively, discussed in [section 7.3](#). These higher derivatives are used to calculate an approximate anharmonic fundamental frequency, overtone frequency and zero-point energy, which can then feed into a [thermochemistry](#) calculation.

Two intermediate quantities, based on third and fourth derivatives, are calculated for diatomic molecules. These are the terms from second and first order perturbation theory respectively. The

VPT approach uses isoinertial normal coordinates involving the reduced mass [26].

$$g_3 = -\frac{1}{\mu^3 \omega^4} \left( \frac{\partial^3 E}{\partial R^3} \right)^2 \quad (3.50)$$

$$g_4 = \frac{1}{\mu^2 \omega^2} \frac{\partial^4 E}{\partial R^4} \quad (3.51)$$

The fundamental frequency then becomes

$$\omega_{\text{fund}} = \omega + \frac{5}{24}g_3 + \frac{1}{8}g_4, \quad (3.52)$$

and the zero-point energy is

$$E_{\text{zero-point}} = \frac{1}{2}\omega + \frac{11}{288}g_3 + \frac{1}{32}g_4. \quad (3.53)$$

The VPT2 keyword is used with a frequency (or optimisation and frequency) calculation:

```
TUNA OPTFREQ : H H 1.0 : CCSD def2-TZVP : VPT2
```

During the execution, the potential energy surface is scanned symmetrically to calculate the third and fourth derivatives. Since several energy evaluations are already completed to calculate the harmonic frequency, only four additional calculations are necessary to achieve sufficient precision. The VPT2 results are often within a couple of  $\text{cm}^{-1}$  of the full [anharmonic vibrational treatment](#). Naturally, the more “quadratic-like” the potential, and the lower the anharmonicity constant, the better results VPT2 will yield. The VPT1 keyword is also implemented and provides a first-order treatment, but this saves no computational cost and loses a lot of accuracy.

After the four additional energies are evaluated, the new zero-point energy and equilibrium energy is printed, along with the fundamental frequency and first overtone in  $\text{cm}^{-1}$ . The anharmonicity constant,  $\chi$ , is also printed. Running a VPT2 calculation on a non-stationary point is meaningless, so TIGHTOPT [convergence criteria](#) are used by default when VPT2 is requested.

Table 3.5 shows how the VPT2 frequencies hold up against our implementation of the [full anharmonic treatment](#), for various molecules calculated with CCSD(T) and def2-TZVP. These complex calculations were performed with the notably simple input line:

```
TUNA ANHARM : H H 1.0 : CCSD[T] def2-TZVP : VPT2
```

**Table 3.5** Fundamental frequencies in  $\text{cm}^{-1}$  calculated with the [quantum harmonic oscillator](#) model, second-order [vibrational perturbation theory](#) and [full nuclear Hamiltonian diagonalisation](#) in TUNA. Results are from CCSD(T) in the def2-TZVP basis set for various molecules

| Molecule | FREQ   | VPT2   | ANHARM |
|----------|--------|--------|--------|
| H—H      | 4430.4 | 4182.9 | 4184.6 |
| H—F      | 4132.9 | 3954.0 | 3956.9 |
| Li—H     | 1413.7 | 1360.4 | 1360.3 |
| C—O      | 2468.9 | 2447.1 | 2447.1 |

### 3.5 Anharmonic Frequency

Since the potential energy surface of a diatomic molecule is one dimensional, it is feasible to “exactly” solve the nuclear Schrödinger equation using the surface generated by TUNA, using any electronic structure method and basis set combination with the ANHARM calculation type.

TUNA ANHARM : C O 1.0 : MP2 6-31G :

The time taken to generate the potential energy surface is the rate limiting step in these calculations, and the building and diagonalisation of the nuclear Hamiltonian is rapid. Assuming full configuration interaction and a complete one-electron basis set, this allows the calculation of full anharmonic vibrational frequencies to  $\text{sub-cm}^{-1}$  accuracy for many molecules.

The fundamental transition frequency is

$$\omega_{\text{fund}} = E_1 - E_0 , \quad (3.54)$$

where  $\{E_i\}$  are the eigenvalues of the nuclear Hamiltonian for the  $i$ th wavefunction,

$$\hat{H}_{\text{nuc}}\Psi_i(R) = E_i\Psi_i(R) . \quad (3.55)$$

Anharmonic frequency calculations are performed by TUNA as follows:

- The molecule is optimised to find its equilibrium bond length
- The harmonic vibrational frequency is calculated at this bond length
- A small 0.4 Å section of the potential energy surface is scanned, forwards and backwards
- This potential energy surface is interpolated

- The nuclear Hamiltonian is formed and diagonalised
- A further 0.1 Å is scanned on either side of the potential energy surface

This process of further scanning repeats until diagonalisation of the nuclear Hamiltonian yields a fundamental transition frequency that is converged to within 0.01 cm<sup>-1</sup>. This default convergence criteria can be changed with the ANHARMCONV keyword. A looser convergence is obtained as:

TUNA ANHARM : N N 1.0 : QCISD 6-31G : ANHARMCONV 0.1

The step length for the coordinate scans is 0.05 Å by default, which is almost always sufficiently converged, but can be changed with the STEP keyword. Each iteration, the fundamental transition frequency, harmonic frequency, anharmonicity constant and bond length range of the potential energy surface are printed into a table.

This iterative approach ensures that a minimal number of energy evaluations are performed, without any loss of accuracy. But bear in mind that anharmonic frequency calculations are still very intensive and require around 30 energy evaluations in total. All ANHARM calculations use EXTREME SCF convergence criteria by default. To avoid issues with linear independence, the bond length is not allowed to get too small — this doesn't affect the quality of results as at short bond length the wavefunctions become tiny. However, you'll probably want to avoid basis sets with diffuse functions, and may want to reduce the overlap matrix eigenvalue threshold with STHRESH.

The Hamiltonian of nuclei moving on a potential energy surface (PES) calculated by TUNA is

$$\hat{H}_{\text{nuc}} = -\frac{1}{2\mu} \frac{\partial^2}{\partial R^2} + V_{\text{PES}} . \quad (3.56)$$

The nuclear Schrödinger equation is solved on a grid, using a basis set of delta functions. This gives a matrix on the order of 1000 × 1000, but it can be diagonalised for the first few states efficiently since it is tridiagonal — only consecutive points are coupled due to the kinetic energy term so the matrix is very sparse. The second derivative is expressed as finite differences as

$$\hat{H}_{\text{nuc}}\Psi(R_n) = -\frac{1}{2\mu} \frac{\Psi(R_{n+1}) - 2\Psi(R_n) + \Psi(R_{n-1}))}{(R_n - R_{n-1})^2} + V_{\text{PES}}(R_n)\Psi(R_n) . \quad (3.57)$$

This can be split into a diagonal and off diagonal part, for a uniform grid where  $\Delta R = R_n - R_{n-1}$

$$\hat{H}_{\text{nuc}}\Psi(R_n) = \left[ \frac{1}{\mu\Delta R^2} + V_{\text{PES}}(R_n) \right] \Psi(R_n) - \frac{\Psi(R_{n+1}) + \Psi(R_{n-1}))}{2\mu\Delta R^2} . \quad (3.58)$$



This produces a Hamiltonian matrix with the diagonal part

$$\hat{H}_n^n = \frac{1}{\mu\Delta R^2} + V_{\text{PES}}(R_n), \quad (3.59)$$

and the off-diagonal part, where  $n \neq m$ ,

$$\hat{H}_n^m = -\frac{\delta_{n+1,m} + \delta_{n-1,m}}{2\mu\Delta R^2} \quad (3.60)$$

so the two closest off diagonal matrix elements are, where  $m = n \pm 1$

$$\hat{H}_n^m = -\frac{1}{2\mu\Delta R^2} \quad (3.61)$$

and all the other elements are zero — this is the case for a “tridiagonal” matrix. This matrix is formed by TUNA after interpolation of the potential energy surface using cubic splines.

The lowest eigenvalue of the nuclear Hamiltonian,  $E_0$ , is the anharmonic zero-point energy. The energy gap between this and the first excited vibrational state is the fundamental transition frequency, and comparison to the harmonic frequency defines the anharmonicity constant,

$$\chi = \frac{\omega_0^1 - \omega_1^2}{2\omega_{\text{harm}}}, \quad (3.62)$$

where  $\{\omega_i^j\}$  are the transition matrix elements from the  $i$ th to  $j$ th nuclear state.

### 3.5.1 Transition Intensity

After convergence is reached, the infrared spectrum is printed for various transitions between nuclear states  $i \rightarrow j$ , and their intensities are calculated. The transitions are printed as energies in hartree, wavenumbers in  $\text{cm}^{-1}$ , and wavelengths in nm. Intensities are printed in  $\text{km mol}^{-1}$ .

The intensity of a vibrational transition depends on the transition dipole moment operator

$$\mathcal{A}_{ij} \propto \langle \Psi_i | \hat{\mu} | \Psi_j \rangle = \int \Psi_i^*(R) \mu(R) \Psi_j(R) dR, \quad (3.63)$$

which is calculated in TUNA by simple numerical integration with the trapezium rule, interpolating the dipole moment at each bond length along the coordinate scans with cubic splines.

By default, the intensities use the analytical dipole moment to calculate the transition dipole matrix elements. This is correct for any method with an [implemented one-electron response density matrix](#), but not exact for methods with an approximate “unrelaxed” density matrix. However,

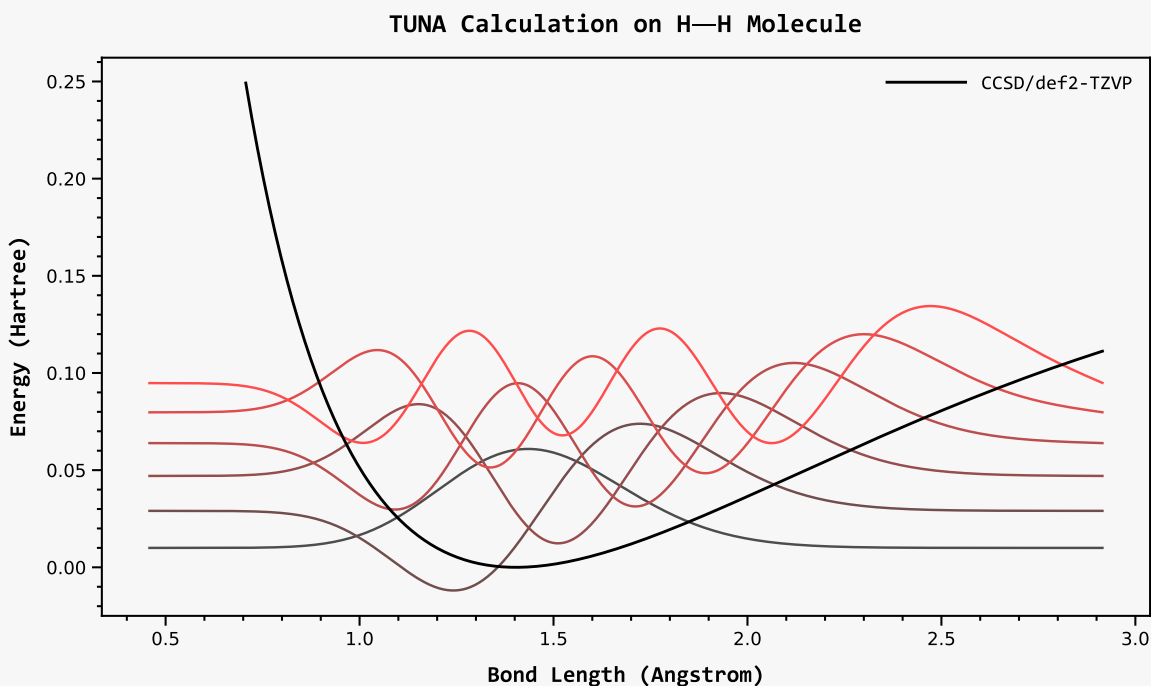


Figure 3.7: Potential energy surface (black) and nuclear wavefunctions (increasingly bright red colour with energy) calculated in TUNA and shown with the VIBPLOT keyword

the [numerical dipole moment](#) is correct for any electronic structure method, and can be requested with the DIPOLE keyword. The following calculation gives [full configuration interaction](#) anharmonic intensities — within the tiny one-electron basis set.

```
TUNA ANHARM : He H 1.0 : CISD 3-21G : CH 1 DIPOLE
```

The calculated intensities differ somewhat from (and are more accurate than) a calculation using analytical dipole moments from the [linearised CCSD density matrix](#):

```
TUNA ANHARM : He H 1.0 : CISD 3-21G : CH 1
```

After the infrared spectrum is printed, so is the anharmonic zero-point energy, which can be used to calculate anharmonic thermochemical parameters with the additional print keyword, P.

### 3.5.2 Plotting Nuclear Wavefunctions

At the end of an ANHARM calculation, the nuclear wavefunctions and potential energy surface can be plotted with the VIBPLOT keyword. The SAVEPLOT keyword can be used in conjunction with this. The vibrational wavefunctions are plotted at their corresponding eigenstate energies, with more intense colouring the higher the energy. Only the first six nuclear eigenstates are plotted.

The following calculation will determine the quadruple- $\zeta$  basis set, [full configuration interaction](#) anharmonic vibrational transition frequencies for the hydrogen molecule. Running the (fairly long) calculation gives a result of  $4161\text{ cm}^{-1}$ , within  $\text{sub-cm}^{-1}$  spectroscopic accuracy [27].

```
TUNA ANHARM : H H 0.74176 : CISD def2-QZVPP : VIBPLOT
```

## 3.6 Bond Dissociation Energy

Commonly calculated quantities for diatomics have been automated in TUNA, such as the bond dissociation energy which can be calculated by the simple BDE calculation type, which determines

$$E_{\text{BDE}} = E(\text{A}) + E(\text{B}) - E(\text{A-B}) \quad (3.64)$$

between atoms A and B. In these calculations, the first step is a [geometry optimisation](#). [Single point energies](#) of the constituent atoms are then determined, in the presence of “ghost” basis functions to counteract basis set superposition error, by default [12]. This can be disabled with the NOCP keyword. In BDE calculations, TIGHT [energy convergence](#) is used by default.

To avoid redundant calculations, TUNA will only perform [full configuration interaction](#) if an even higher order calculation is requested. For example, a CCSDTQ bond dissociation energy calculation on lithium hydride will calculate the energy of the molecule with CCSDTQ, the lithium atom with CISDT and the hydrogen atom with [Hartree-Fock](#) — these are all the exact energies within the one-electron basis set. This approach can speed up the atomic calculations a lot.

```
TUNA BDE : Li H 1.5 : CCSDTQ 3-21G : NOCP
```

To improve agreement with experiment, the zero-point energy of the molecule can be calculated using the ZPE keyword. [Second-order vibrational perturbation theory](#) with VPT2 can also be requested, to provide a good approximation to anharmonic effects and improve the energetics.

This bond dissociation energy calculation on dihydrogen gives an energy of  $0.16387 E_{\text{h}}$ , in pretty good agreement with the experimental value of  $0.16457 E_{\text{h}}$  [28].

```
TUNA BDE : H H 1.0 : CISD cc-pVQZ : ZPE
```

The discrepancy is partly due to the harmonic zero-point energy. The [anharmonic zero-point energy](#) could be calculated exactly with an ANHARM calculation, but it is simpler to just request second-order [vibrational perturbation theory](#). The following calculation gives  $0.16397 E_{\text{h}}$ .

```
TUNA BDE : H H 1.0 : CISD cc-pVQZ : ZPE VPT2
```

An even better result of  $0.16475 E_h$  comes by [extrapolation](#) from triple- $\zeta$  to quadruple- $\zeta$ :

```
TUNA BDE : H H 1.0 : CISD cc-pVTZ : ZPE VPT2 EXTRAPOLATE
```

Finally, a very intensive [basis set extrapolation](#) from quadruple- $\zeta$  to quintuple- $\zeta$  gives a bond dissociation energy of  $0.16460 E_h$  which is only  $0.02 \text{ kcal mol}^{-1}$  from experiment — bang on!

```
TUNA BDE : H H 1.0 : CISD cc-pVQZ : ZPE VPT2 EXTRAPOLATE
```

## 3.7 Ionisation Potential and Electron Affinity

There are also inbuilt calculation types to determine the ionisation potential,

$$\text{IP} = E(\text{X}^+) - E(\text{X}) , \quad (3.65)$$

and the electron affinity

$$\text{EA} = E(\text{X}) - E(\text{X}^-) . \quad (3.66)$$

These quantities can be calculated with the calculation types IP and EA respectively. Unlike the [ionisation potential from Koopmans' theorem](#), the orbitals are allowed to fully relax. If a correlated calculation is requested, it will be performed on these converged orbitals. An ionisation potential calculation on the carbon monoxide molecule is:

```
TUNA IP : C O 1.0 : HF 3-21G :
```

The analogous electron affinity calculation is:

```
TUNA EA : C O 1.0 : HF 3-21G :
```

By default, an IP or EA calculation on a molecule will be adiabatic. First, the molecule is [geometry optimised](#) in its native charge state (normally neutral). The final energy of this molecule is saved as  $E(\text{X})$  and the final molecular structure is used as a starting guess for the ionised (or electron attached) calculation. With the new charge assigned, TUNA then optimises the geometry again to return the final charged energy such that the adiabatic IP or EA can be calculated.

For atoms, which can't be optimised, the vertical IP or EA is calculated instead. The vertical IP

or EA can be calculated on molecules using the VERTICAL keyword:

```
TUNA EA : H H 0.7414 : PBE0 ano-pVDZ : VERTICAL
```

By default, the IP calculation will remove one electron and the EA calculation will add one electron to whatever charge and multiplicity state the calculation is initially called with. To calculate the IP or EA with more than one removed or added electron, the NELEC keyword can be used. For example, this calculation will determine the ionisation potential,

$$\text{IP} = E(\text{Be}^{2+}) - E(\text{Be}) . \quad (3.67)$$

```
TUNA IP : Be : CCSDT cc-pVTZ : NELEC 2
```

The output for IP or EA calculations first shows the charges of the two states, then prints the energy of the reference and charged system. For adiabatic calculations, the optimised bond lengths of the two systems are also printed, then for all calculations the final IP or EA is presented.

If one of the calculations to obtain the IP or EA will be on a zero-electron system, or produces any other weird state, the calculation will exit with an error message as it normally would.

### 3.8 *Ab Initio* Molecular Dynamics

The implementation in TUNA of *ab initio* molecular dynamics (AIMD) is Born–Oppenheimer molecular dynamics [29], where nuclear positions are updated by the classical equations of motion,  $\mathbf{F} = m\mathbf{a}$ , over the quantum chemical potential energy surface. Therefore, unlike other AIMD methods such as Car–Parrinello molecular dynamics where the electronic degrees of freedom are also dynamic [30], the *ab initio* energy and gradient must be recalculated at every timestep.

To run an AIMD calculation in TUNA, use the MD calculation type with any electronic structure method and basis, and set the timestep (fs), initial temperature (K) and number of steps with the keywords STEP, TEMP and NUM respectively. By default, STEP is 0.1 fs, TEMP is 0 K and NUM is 50.

For instance, a 200 step AIMD simulation using MP2/6-31G with a timestep of 0.2 fs at an initial temperature of 350 K can be requested by:

```
TUNA MD : H H 1.2 : MP2 6-31G : STEP 0.2 NUM 200 TEMP 350
```

As positions and velocities are continually updated, a trajectory is mapped out. By default, TUNA prints this trajectory to a file called "tuna-trajectory.xyz" in the terminal's directory — this

logging can be disabled using the NOTRAJ keyword. The TRAJ [filepath] keyword can also be used to change the name of the file — just include the name in quotation marks.

The output prints the step number, time, bond length and temperature. The classical kinetic energy of the nuclei, as well as their potential energy (the total electronic energy) is also printed, with the total molecular energy. The timestep should be chosen to conserve the total system energy. At each timestep, all energy and gradient information can be printed using the additional print keyword, P, although this makes a big mess.

Because the Velocity Verlet algorithm [31] yields velocities and positions at the time point and is numerically stable, it is implemented in TUNA. Positions are updated in three dimensions by

$$\mathbf{R}(t + \Delta t) = \mathbf{R}(t) + \mathbf{v}(t)\Delta t + \frac{1}{2}\mathbf{a}(t)\Delta t^2, \quad (3.68)$$

and velocities are updated by

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{\mathbf{a}(t) + \mathbf{a}(t + \Delta t)}{2}\Delta t. \quad (3.69)$$

By default, the AIMD calculation begins with the atoms at rest. However, by using the TEMP keyword an initial temperature can be specified from which the total classical kinetic energy of the molecule is calculated by

$$K = 3k_{\text{B}}T, \quad (3.70)$$

where

$$K = \frac{1}{2}m_{\text{A}}\mathbf{v}_{\text{A}}^2 + \frac{1}{2}m_{\text{B}}\mathbf{v}_{\text{B}}^2. \quad (3.71)$$

From this kinetic energy, velocities are randomly determined to satisfy the Maxwell–Boltzmann distribution. Next, net molecular translations are removed before the velocities are rescaled to produce the specified temperature exactly. Net rotations are not removed, so to handle this three-dimensional data — which is stressful for TUNA — the coordinates are rotated onto one dimension to calculate the energy and gradient before forces are back-transformed into three dimensions.

The initial specified temperature will not be conserved, as thermostats don't really make sense for diatomics, so all AIMD calculations are run in the *NVE* ensemble. The shorter the timestep, the more faithful the energy convergence. Like the rest of TUNA, in principle any [electronic structure method](#), [basis set](#) and keyword combination can be used for a molecular dynamics simulation.

## 4 Electronic Structure Methods

The wavefunction theory electronic structure methods available in TUNA are listed in [Table 4.1](#). A plethora of [density functional approximations](#) are also available, which are presented in [section 4.5](#).

Most of these methods are available for restricted and unrestricted references — the unrestricted reference can be requested for a singlet molecule by adding the letter "U" to the start of a method.

**Table 4.1** Wavefunction-based electronic structure methods implemented in TUNA

| Keyword  | Electronic Structure Method                                  |
|----------|--------------------------------------------------------------|
| HF       | Hartree–Fock theory                                          |
| UHF      | Unrestricted Hartree–Fock theory                             |
| H        | Hartree theory                                               |
| UH       | Unrestricted Hartree theory                                  |
| MP2      | Second-order Møller–Plesset perturbation theory              |
| UMP2     | Unrestricted second-order Møller–Plesset perturbation theory |
| SCS-MP2  | Spin-component-scaled MP2 perturbation theory                |
| USCS-MP2 | Unrestricted spin-component-scaled MP2 theory                |
| OMP2     | Orbital-optimised MP2 theory                                 |
| UOMP2    | Unrestricted orbital-optimised MP2 theory                    |
| IMP2     | Iterative MP2 theory                                         |
| AO-MP2   | Laplace transform AO-MP2 theory                              |
| MP3      | Third-order Møller–Plesset perturbation theory               |
| UMP3     | Unrestricted third-order Møller–Plesset perturbation theory  |
| SCS-MP3  | Spin-component-scaled MP3 perturbation theory                |
| USCS-MP3 | Unrestricted spin-component-scaled MP3 theory                |

| Keyword  | Electronic Structure Method                                         |
|----------|---------------------------------------------------------------------|
| MP4      | Fourth-order Møller–Plesset perturbation theory                     |
| MP4[DQ]  | Restricted MP4 theory with no singles or triples                    |
| MP4[SDQ] | Restricted MP4 theory with no triples                               |
| TDHF     | Time-dependent Hartree–Fock theory                                  |
| UTDHF    | Unrestricted time-dependent Hartree–Fock theory                     |
| CIS      | Configuration interaction singles                                   |
| UCIS     | Unrestricted configuration interaction singles                      |
| CIS[D]   | CIS with perturbative doubles                                       |
| UCIS[D]  | Unrestricted CIS with perturbative doubles                          |
| CID      | Configuration interaction doubles                                   |
| UCID     | Unrestricted configuration interaction doubles                      |
| CISD     | Configuration interaction singles and doubles                       |
| UCISD    | Unrestricted configuration interaction singles and doubles          |
| CISDT    | Configuration interaction singles, doubles and triples              |
| UCISDT   | Unrestricted configuration interaction singles, doubles and triples |
| CEPA0    | Coupled electron pair approximation                                 |
| UCEPA0   | Unrestricted coupled electron pair approximation                    |
| LCCD     | Linearised coupled cluster doubles                                  |
| ULCCD    | Unrestricted linearised coupled cluster doubles                     |
| LCCSD    | Linearised coupled cluster singles and doubles                      |
| ULCCSD   | Unrestricted linearised coupled cluster singles and doubles         |
| CC2      | Approximate coupled cluster singles and doubles                     |
| CC3      | Approximate coupled cluster singles, doubles and triples            |
| CCD      | Coupled cluster doubles                                             |
| UCCD     | Unrestricted coupled cluster doubles                                |



| Keyword   | Electronic Structure Method                                            |
|-----------|------------------------------------------------------------------------|
| CCSD      | Coupled cluster singles and doubles                                    |
| UCCSD     | Unrestricted coupled cluster singles and doubles                       |
| CCSD[T]   | Coupled cluster singles, doubles and perturbative triples              |
| UCCSD[T]  | Unrestricted coupled cluster singles, doubles and perturbative triples |
| CCSDT     | Coupled cluster singles, doubles and triples                           |
| UCCSDT    | Unrestricted coupled cluster singles, doubles and triples              |
| CCSDT[Q]  | Coupled cluster singles, doubles, triples and perturbative quadruples  |
| CCSDTQ    | Coupled cluster singles, doubles, triples and quadruples               |
| QCISD     | Quadratic configuration interaction singles and doubles                |
| UQCISD    | Unrestricted quadratic CI singles and doubles                          |
| QCISD[T]  | Quadratic CI singles, doubles and perturbative triples                 |
| UQCISD[T] | Unrestricted quadratic CI singles, doubles and perturbative triples    |

## 4.1 Hartree–Fock Theory

The simplest *ab initio* quantum chemistry method is Hartree–Fock theory [32].

Rather than calculating instantaneous electron–electron interactions, Hartree–Fock assumes each electron experiences an average field generated by all the others, described with one-electron wavefunctions known as molecular orbitals. Because the orbitals depend on the field and the field depends on the orbitals, the problem is solved iteratively: an initial guess for the orbitals is used to construct the one-electron Hamiltonian (the Fock matrix), which then generates new, refined orbitals. This cycle repeats until the orbitals stop changing, resulting in a [self-consistent field \(SCF\)](#).

Both restricted and unrestricted Hartree–Fock calculations are available in TUNA. The HF method keyword defaults to RHF for singlet molecules and UHF for non-singlets. Unrestricted Hartree–Fock can also be requested on a singlet molecule using the UHF method, such as:

```
TUNA SPE : H H 1.85 : UHF 6-311++G
```

The Roothaan–Hall equations [33, 34] turn the variational differential restricted Hartree–Fock equations [32] into a simpler eigenvalue problem,

$$\mathbf{F}\mathbf{M} = \mathbf{S}\mathbf{M}\epsilon , \quad (4.1)$$

where  $\mathbf{F}$  is the Fock matrix (the one-electron Hamiltonian),  $\mathbf{M}$  is the molecular orbital matrix in the basis of atomic orbitals,  $\mathbf{S}$  is the overlap matrix and  $\epsilon$  is a diagonal matrix of orbital energies. The Fock matrix consists of contributions from the kinetic energy matrix,  $\mathbf{T}$ , nuclear–electron attraction matrix,  $\mathbf{V}$ , Coulomb matrix,  $\mathbf{J}$ , and exchange matrix,  $\mathbf{K}$ , by

$$\mathbf{F} = \mathbf{T} + \mathbf{V} + 2\mathbf{J} - \mathbf{K} . \quad (4.2)$$

These matrices are determined from [one- and two-electron integrals of atomic orbital basis functions](#). To solve the Roothaan–Hall equations, they need to be converted into a conventional eigenvalue problem; the overlap matrix needs to become the identity matrix. To do this, the basis functions can be rotated to an orthonormal basis. One method to achieve this defines  $\mathbf{X} = \mathbf{S}^{-\frac{1}{2}}$ , such that when the Fock matrix is rotated by

$$\mathbf{F}' = \mathbf{X}^\dagger \mathbf{F} \mathbf{X} , \quad (4.3)$$

the Roothaan–Hall equations become a conventional eigenvalue problem,

$$\mathbf{F}'\mathbf{M}' = \mathbf{M}'\epsilon . \quad (4.4)$$

Now the rotated Fock matrix,  $\mathbf{F}'$ , is diagonalised for its eigenvectors,  $\mathbf{M}'$ , and eigenvalues,  $\epsilon$ . The eigenvectors are then rotated back to the original basis, where they represent molecular orbitals,

$$\mathbf{M} = \mathbf{X}\mathbf{M}' . \quad (4.5)$$

Now the molecular orbitals have been determined the restricted Hartree–Fock density matrix,  $\mathbf{P}$ , is constructed by summing over the lowest energy occupied orbitals,

$$P_{\mu\nu} = 2 \sum_i^{\text{occ}} M_{\mu i} M_{\nu i}^* . \quad (4.6)$$

Finally, the energy can be evaluated by contracting this new density matrix with the Fock and one-electron matrices, and adding on the constant nuclear repulsion energy,

$$E = V_{\text{nn}} + \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} (F_{\mu\nu} + H_{\mu\nu}^{\text{core}}) . \quad (4.7)$$

Here,  $\mathbf{H}^{\text{core}} = \mathbf{T} + \mathbf{V}$ . The density matrix is contracted with the two-electron integrals to determine the new two-electron contribution to the Fock matrix,

$$2\mathbf{J} - \mathbf{K} = \sum_{\kappa\lambda} P_{\kappa\lambda} [(\mu\nu|\kappa\lambda) - \frac{1}{2}(\mu\kappa|\nu\lambda)] . \quad (4.8)$$

The Fock matrix is then constructed again, from which new molecular orbitals are determined as this procedure repeats. When the energy and density matrix change from the last iteration less than the [convergence criteria](#), the mean field has become self-consistent and a stationary point on the surface of orbital rotations has been found — the Hartree–Fock wavefunction.

### 4.1.1 Initial Guess

In TUNA, the simplest choice of initial guess for a [single point energy](#) calculation is the density matrix from a one-electron calculation, where contributions from  $\mathbf{J}$  and  $\mathbf{K}$  are ignored and the one-electron Hamiltonian to diagonalise is simply

$$\mathbf{F} = \mathbf{H}^{\text{core}} . \quad (4.9)$$

This works well for diatomics of hydrogen and helium, but fails catastrophically even for relatively light diatomics in the p block. The guess density is so contracted in these molecules that the SCF can converge to a [metastable state](#) or a [saddle point](#). The core Hamiltonian guess method can be requested with the COREGUESS keyword. Doing this for the hydrogen molecule will make the calculation slightly faster, without losing any accuracy — probably.

TUNA SPE : H H 1.0 : HF pc-1 : COREGUESS

Another widely used guess strategy is the superposition of atomic densities guess. Here the guess density matrix is evaluated as a block diagonal matrix of atomic density matrices, assuming zero interaction between the atoms (ie. a simple superposition of the densities),

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_B \end{bmatrix} . \quad (4.10)$$

This produces a much less contracted density that tends to reduce the number of SCF iterations necessary to converge. The initial atomic density matrices are stored in TUNA as spherically averaged STO-3G density matrices. These are projected onto the requested basis set by

$$\mathbf{P}_2 = \mathbf{S}_2^{-1} \mathbf{S}_{12} \mathbf{P}_1 \mathbf{S}_{12} \mathbf{S}_2^{-1} , \quad (4.11)$$

where  $\mathbf{P}_1$  is the minimal basis set and  $\mathbf{P}_2$  is the requested basis set. For this, the cross overlap

matrix between the two basis sets is needed, which is happily provided by the [molecular integral engine](#). A downside is that simple densities are not as well represented, so the SCF may take longer to converge in some cases. This guess strategy can be requested with the SADGUESS keyword.

```
TUNA SPE : N F 1.0 : HF pc-1 : SADGUESS
```

The default guess strategy in TUNA embodies the best of both worlds, and can be accessed with the SCFGUESS keyword. Here, a STO-3G minimal basis set SCF calculation is first fully converged, using the superposition of atomic densities guess for *that* calculation. The converged density matrix is then projected using [Equation 4.11](#) onto the larger basis, and the true calculation begins.

We have found this to be the most robust guess strategy, and the one which minimises the number of needed SCF iterations on average, but it is naturally a fair bit slower than the other methods. This is felt most during [DFT calculations](#) as the basis functions need to be expressed on a grid, but considering the benefit in reducing the necessary SCF iterations it is probably still worth it.

```
TUNA SPE : N F 1.0 : HF pc-1 : SCFGUESS
```

In all cases, the initial guess energy is contracted from the guess density matrix contracted with the core Hamiltonian matrix,

$$E_{\text{guess}} = \sum_{\mu\nu} H_{\mu\nu}^{\text{core}} P_{\mu\nu} . \quad (4.12)$$

### 4.1.2 Symmetry Breaking

If an unrestricted Hartree–Fock calculation is run on a singlet molecule with equal  $\alpha$  and  $\beta$  guess densities the SCF will likely get caught in a local minimum or saddle point on the surface of orbital rotations, the restricted Hartree–Fock solution. This happens for diatomics at long bond lengths.

To handle this, TUNA rotates the HOMO with the LUMO after the guess density is made, by

$$\begin{bmatrix} \psi'_{\text{HOMO}} \\ \psi'_{\text{LUMO}} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \psi_{\text{HOMO}} \\ \psi_{\text{LUMO}} \end{bmatrix} . \quad (4.13)$$

By default,  $\theta = 45^\circ$ , which corresponds to a half-and-half mixing. This can cause some problems with convergence when the RHF and UHF solutions are (near-)degenerate, so the rotation can be disabled with the NOROTATE keyword. Orbital rotation can be activated with the ROTATE keyword. If the default value of  $\theta$  is not converging well, use the ROTATE keyword with the desired rotation angle in degrees to change it. The value of THETA is printed when orbital rotation is used.

```
TUNA SPE : H He 1.2 : UHF 6-31G : ROTATE 45
```

A well-considered guess rotation may allow a metastable energy state to be converged. Whether an initial guess rotation is necessary to converge the lowest energy Hartree–Fock state can be checked by performing a [stability analysis](#).

### 4.1.3 Stability Analysis

The [self-consistent field](#) process optimises the [Hartree–Fock](#) or [Kohn–Sham](#) wavefunction to a stationary point in the space of orbital rotations. However, this may be a saddle point rather than a local minimum. To determine whether there is an instability in the wavefunction with respect to an orbital rotation, a stability calculation can be requested with the STAB keyword:

```
TUNA SPE : H H 2.0 : RHF STO-3G : STAB
```

The above calculation shows that there is an instability with respect to unrestricted Hartree–Fock. Switching to UHF, we see that this instability has gone, and the energy has lowered:

```
TUNA SPE : H H 2.0 : UHF STO-3G : STAB
```

This only works because of the [automatic symmetry-breaking](#) in TUNA. Turning this off with NOROTATE still results in an instability due to the symmetric guess leading to the RHF solution:

```
TUNA SPE : H H 2.0 : UHF STO-3G : STAB NOROTATE
```

This works by forming and diagonalising the orbital Hessian. Similarly to diagonalising the geometric Hessian to check if an optimised structure is a local minimum, a negative eigenvalue of the orbital Hessian indicates an instability with respect to a molecular orbital rotation.

For a spin-restricted calculation, both the “singlet” and “triplet” Hessian are calculated. The former checks for RHF/RHF instabilities, and the latter checks for RHF/UHF instabilities.

The singlet orbital Hessian is calculated by

$$\mathbf{H} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{bmatrix}, \quad (4.14)$$

where

$$A_{ia,jb} = (\varepsilon_a - \varepsilon_i)\delta_{ij}\delta_{ab} + 2(ia|jb) - \beta(ij|ab) + (1 - \beta)K_{ijab}^{\text{XC}} \quad (4.15)$$

**Table 4.2** Self-consistent field convergence criteria in TUNA

| Convergence | $\Delta E$ | MAX( $\Delta \mathbf{P}$ ) | RMS( $\Delta \mathbf{P}$ ) | RMS( $[\mathbf{F}, \mathbf{PS}]$ ) |
|-------------|------------|----------------------------|----------------------------|------------------------------------|
| LOOSE       | $10^{-6}$  | $10^{-5}$                  | $10^{-6}$                  | $10^{-4}$                          |
| MEDIUM      | $10^{-7}$  | $10^{-6}$                  | $10^{-7}$                  | $10^{-5}$                          |
| TIGHT       | $10^{-9}$  | $10^{-8}$                  | $10^{-9}$                  | $10^{-7}$                          |
| EXTREME     | $10^{-11}$ | $10^{-10}$                 | $10^{-11}$                 | $10^{-9}$                          |

and

$$B_{ia,jb} = 2(ia|bj) - \beta(ib|aj) + (1 - \beta)K_{ijab}^{\text{XC}}. \quad (4.16)$$

Here the density functional kernel matrix,  $K_{ijab}^{\text{XC}}$ , is the same as in [time-dependent DFT](#). Therefore stability analysis is only available for exchange–correlation functionals compatible with TD-DFT.

#### 4.1.4 Self-consistent Field Convergence

A number of SCF convergence criteria are available in TUNA, shown in [Table 4.2](#). These can be activated by the keywords LOOSE, MEDIUM (default for [single point](#) calculations), TIGHT (default for first derivative calculations) and EXTREME (default for higher derivative calculations).

The number of SCF iterations before TUNA gives up can be picked with the keyword MAXITER, which is 100 by default. Convergence is measured by the change in energy, maximum and root-mean-square change in the density matrix, and the DIIS error (labelled Error in the TUNA output), which is the root-mean-square of the commutator between  $\mathbf{F}$  and  $\mathbf{PS}$ . More precise control over each of these can be found with the ECONV, MAXDP, RMSDP and DIISERR keywords, respectively. Note that ECONV also determines the energy convergence for post-SCF iterative methods.

There are several methods available to accelerate SCF convergence. The most powerful is probably Fock matrix extrapolation, known as direct inversion of the iterative subspace (DIIS) [\[35\]](#).

At SCF convergence, the density matrix commutes with the Fock matrix,

$$\mathbf{SPF} - \mathbf{FPS} = \mathbf{0}. \quad (4.17)$$

An error matrix can therefore be defined at the  $i$ th iteration, whose root-mean-square indicates

how far the Hartree–Fock solution is from self-consistency,

$$\mathbf{S}\mathbf{P}_i\mathbf{F}_i - \mathbf{F}_i\mathbf{P}_i\mathbf{S} = \mathbf{e}_i. \quad (4.18)$$

In DIIS, an error vector is built, where each error matrix is associated with a Fock matrix. This error vector is then optimised in a least-squares minimisation to yield the proportion of each Fock matrix,  $c_j$ , which can be used to make a new Fock matrix with a lower error matrix,

$$\mathbf{F}' = \sum_j c_j \mathbf{F}_j. \quad (4.19)$$

DIIS tends to be very robust. TUNA stores 6 previous Fock matrices at each SCF step by default — which can be changed using DIIS [Num. Matrices] — and will reset this stored array if the equations approach linear dependency and break, and a message will be printed to relay this at the end of the SCF iteration. The DIIS procedure begins after the second SCF step, to prevent extrapolation before convergence has properly begun, and can be deactivated with NODIIS.

Another convergence accelerator, which may be the most conceptually simple, is damping, where the density matrix,  $\mathbf{P}$ , is mixed with a fraction of the density from the previous SCF cycle, by

$$\mathbf{P}' = a\mathbf{P}_{\text{old}} + (1 - a)\mathbf{P}. \quad (4.20)$$

Static damping, where the value of  $a$  is fixed, can be invoked with the SLOWCONV or VERYSLOWCONV keywords, where  $a = 0.5$  and  $a = 0.85$  respectively. These can be very helpful to force a difficult case to converge, although should be used with a very large value of MAXITER. A calculation with a maximum of 200 iterations and  $a = 0.5$  can be called by:

```
TUNA SPE : H He 0.6 : HF 6-31+G : SLOWCONV MAXITER 200
```

The value of  $a$  can be chosen specifically with the DAMP keyword. For example, a static damping factor of  $a = 0.3$  can be requested by:

```
TUNA SPE : H He 0.6 : HF 6-31+G : DAMP 0.3
```

However, the default method is dynamic damping, where the damping factor,  $a$ , is some function of the distance from convergence. The implemented scheme for dynamic damping is that from Zerner and Hehenberger [36]. In this method, Mulliken gross atomic populations [15] are calculated in each cycle and extrapolated for an optimal damping factor, the average of the calculated atomic damping factors weighted by the proportion of basis functions on each atom.

This dynamic approach gives a damping factor that more rapidly converges, and avoids unnecessary damping in the static case. The dynamical damping method is enabled by default, can be disabled with the `NODAMP` keyword and is compatible with DIIS. The maximum damping factor is 0.7 by default, and can be controlled with the `MAXDAMP` keyword.

#### 4.1.5 Hartree Theory

To isolate how the Fock exchange contribution affects energies and properties, calculations can be performed with Hartree theory with the `H` method, which sets the proportion of exchange to zero.

```
TUNA OPT : H H 1.0 : H ano-pVDZ : DAMP 0.3
```

The Hartree bond length, 1.06 Å, for dihydrogen is much longer (and less accurate) compared to the Hartree–Fock bond length of 0.74 Å.

The HFX parameter enables more precise investigations of Fock exchange. The following calculation represents 50% of the exact exchange being used, and gives an improved bond length of 0.87 Å.

```
TUNA OPT : H H 1.0 : HF ano-pVDZ : HFX 0.5
```

## 4.2 Møller–Plesset Perturbation Theory

The instantaneous electron–electron interaction neglected in Hartree–Fock theory makes up only a small proportion of the overall molecular energy,  $E$ , but is necessary to describe phenomena including [dispersion forces](#) and entanglement and to provide quantitatively accurate energetics.

Various approaches to calculate this molecular “correlation” energy,

$$E_{\text{corr}} = E - E_{\text{HF}} , \quad (4.21)$$

have been developed.

The simplest post-Hartree–Fock electronic structure method to introduce electron–electron correlation is found in perturbation theory, where the exact molecular Hamiltonian is split into the solvable one-electron part, the Fock operator, and a perturbative part,

$$\hat{H} = \hat{F} + \hat{V} , \quad (4.22)$$

then the standard equations of perturbation theory are applied for working formulae [\[37\]](#). This



approach defines the total molecular energy as a series of energy contributions,

$$E = E_{\text{MP0}} + E_{\text{MP1}} + E_{\text{MP2}} + E_{\text{MP3}} + E_{\text{MP4}} \cdots \quad (4.23)$$

The zeroeth and first order terms sum to the Hartree–Fock energy, so the first and most important contribution to electron correlation comes at second-order, and is known as MP2 [38].

### 4.2.1 Conventional MP2, MP3 and MP4

Conventional Møller–Plesset perturbation theory [38] to second (MP2) and third (MP3) order are available for both RHF and UHF references, and MP4 is implemented for restricted calculations.

An MP2 calculation can be requested by:

TUNA SPE : H H 1.0 : MP2 6-311G :

If your molecule has an even number of electrons but an unrestricted reference is desired, the method keywords UMP2 and UMP3 can be used:

TUNA SPE : H H 1.0 : UMP3 6-311G :

All correlated wavefunction methods require an initial  $\mathcal{O}(N^5)$  transformation of the [two-electron integrals](#) into the molecular orbital basis. Calculations with MP3 and MP4 also require slower contractions to calculate the final energies. To speed these calculations up, [core electrons can be excluded](#) from the correlation treatment which only loses a small proportion of the accuracy.

Mathematically, this initial transformation into the spatial orbital basis involves matrix multiplications with the molecular orbital coefficients,

$$(ia|jb) = \sum_{\mu\kappa\nu\lambda} M_{\mu}^i M_{\kappa}^a M_{\nu}^j M_{\lambda}^b (\mu\kappa|\nu\lambda) . \quad (4.24)$$

This is the rate-limiting step of an MP2 calculation, and necessary for all correlated wavefunction methods. The spatial orbital MP2 energy can then be evaluated as

$$E_{\text{MP2}} = \sum_{ijab} \frac{(ia|jb)[2(ia|jb) - (ib|ja)]}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} . \quad (4.25)$$

For unrestricted calculations, the [two-electron integrals](#) are instead transformed into the spin orbital

basis, where  $\alpha$  and  $\beta$  spins are treated separately by spin-blocked orbitals,

$$\langle ij|ab\rangle = \sum_{\mu\kappa\nu\lambda} M_{\mu}^i M_{\kappa}^a M_{\nu}^j M_{\lambda}^b \langle \mu\nu|\kappa\lambda\rangle. \quad (4.26)$$

These integrals are then antisymmetrised by

$$\langle ij||ab\rangle = \langle ij|ab\rangle - \langle ij|ba\rangle, \quad (4.27)$$

and the MP2 energy is evaluated as

$$E_{\text{MP2}} = \frac{1}{4} \sum_{ijab} \frac{|\langle ij||ab\rangle|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}. \quad (4.28)$$

The same-spin and opposite-spin contributions to the MP2 energy are calculated by TUNA. For unrestricted calculations, this is decomposed into the energy due to  $\alpha\alpha$ ,  $\beta\beta$  and  $\alpha\beta$  electron pairs.

The MP4 energy can be decomposed as

$$E_{\text{MP4}} = E_{\text{MP4}}^{\text{singles}} + E_{\text{MP4}}^{\text{doubles}} + E_{\text{MP4}}^{\text{triples}} + E_{\text{MP4}}^{\text{quadruples}}. \quad (4.29)$$

The calculation of the triples is the steepest scaling step, at  $\mathcal{O}(N^7)$ , while other steps scale at most at  $\mathcal{O}(N^6)$ . Therefore, a speedup can be achieved by simply neglecting the triples, yielding the MP4(SDQ) method. This can be calculated in TUNA using the method type MP4[SDQ]:

```
TUNA SPE : H F 1.2 : MP4[SDQ] 6-31G :
```

Similarly, the calculation of singles can be skipped either with the MP4[DQ] method.

```
TUNA SPE : C O 1.2 : MP4[DQ] 6-31G :
```

Naturally, conventional MP4 can be requested by the MP4 method type:

```
TUNA SPE : H H 0.7 : MP4 6-31G :
```

In all cases, TUNA prints the energy from the single, double, triple and quadruple excitations.

### 4.2.2 Freezing Core Electrons

Core electrons, like 1s electrons in second period elements, have a fairly consistent correlation energy between molecular geometries. This core correlation can often be safely ignored with the “frozen core” approximation. This is requested in TUNA with the FREEZECORE keyword.

Using FREEZECORE on its own will freeze the default number of core electrons depending on the elements used in the calculation. This number is 0 for H–Be, 2 for B–Mg and 10 for Al–Ar. Alternatively, if a specific number of *orbitals* (not electrons) is wanted to be frozen, this number can be given after the FREEZECORE keyword. For example, the following calculation will freeze the three lowest energy spatial orbitals of nitric oxide:

```
TUNA SPE : N O 1.0 : MP2 6-311G : FREEZECORE 3
```

Note that for unrestricted calculations there are twice the number of orbitals to freeze as in restricted calculations. During the transformation to a molecular orbital basis, TUNA will print the number of frozen orbitals — which will be half the number of frozen electrons for RHF references, or the same as the number of frozen electrons for UHF references.

### 4.2.3 Spin-component Scaling

Some Møller–Plesset perturbation theory methods can be “spin-component-scaled” (SCS), which has been reported to increase their accuracy [39, 40]. The total MP2 energy can be written in terms of same-spin and opposite-spin components,

$$E_{\text{MP2}} = E_{\text{MP2}}^{\text{SS}} + E_{\text{MP2}}^{\text{OS}}. \quad (4.30)$$

The SCS-MP2 approximation applies different scaling factors to the same spin,  $E_{\text{MP2}}^{\text{SS}}$ , and opposite spin,  $E_{\text{MP2}}^{\text{OS}}$ , contributions to correlation. These calculations can be called in TUNA by:

```
TUNA SPE : H He 1.2 : SCS-MP2 6-31G
```

In SCS-MP2, the correlation energy is given by

$$E_{\text{SCS-MP2}} = c_{\text{S}} E_{\text{MP2}}^{\text{SS}} + c_{\text{O}} E_{\text{MP2}}^{\text{OS}}, \quad (4.31)$$

where by default,  $c_{\text{S}} = 1/3$  and  $c_{\text{O}} = 6/5$ . These values can be changed by the SSS and OSS keywords respectively. The following calculation removes the same-spin scaling entirely, and increases the opposite spin scaling:

```
TUNA SPE : H H 0.74 : SCS-MP2 6-31G : SSS 0 OSS 1.5
```

In SCS-MP3, which often improves on conventional MP3, the total energy is given by

$$E_{\text{SCS-MP3}} = E_{\text{SCS-MP2}} + c_{\text{MP3}} E_{\text{MP3}}, \quad (4.32)$$

where  $c_{\text{MP3}} = 1/4$  by default. The value of  $c_{\text{MP3}}$  can be changed with the MP3S keyword. For

example a value of  $c_{\text{MP3}} = 0.5$  is requested by:

```
TUNA SPE : H He 1.2 : SCS-MP3 6-31G : MP3S 0.5
```

#### 4.2.4 Orbital-optimised MP2

While the conventional theory is applied to the Hartree–Fock state, the MP2 energy can be minimised with respect to orbital rotations, in orbital-optimised MP2 [41, 42]. This can be requested with the OMP2 method keyword, or UOMP2 to force an unrestricted calculation:

```
TUNA SPE : H H 0.6 : OMP2 6-311++G :
```

The correlation energy is determined using the one- and two-particle reduced density matrices at each OMP2 iteration, where the molecular orbitals are updated using a rotation matrix,  $\Theta$ , by

$$\mathbf{M}_{n+1} = \mathbf{M}_n \exp(\Theta) . \quad (4.33)$$

This cycle continues until the change in energy falls below a convergence criteria of  $10^{-8}$  by default. This value can be changed by using the ECONV keyword. The maximum number of OMP2 iterations is 30 by default, mutable with the CORRMAXITER keyword. The density matrix from orbital-optimised MP2 is [fully relaxed](#), and is used to calculate [molecular properties](#).

#### 4.2.5 Iterative MP2

Conventional MP2 requires a Hartree–Fock reference. For calculations with non-canonical orbitals (such as those using localised molecular orbitals), the MP2 energy cannot be calculated in the conventional way, but can be evaluated through minimisation of the Hylleraas functional [43].

In iterative MP2, the following residuals are iteratively minimised

$$R_{ij}^{ab} = (ia|jb) + F_{ac} t_{ij}^{cd} S_{db} + S_{ac} t_{ij}^{cd} F_{db} - S_{ac} F_{ik} t_{kj}^{cd} S_{db} - S_{ac} F_{kj} t_{ik}^{cd} S_{db} . \quad (4.34)$$

Where the overlap matrix elements allow for the possibility of a non-orthogonal basis and Fock matrix elements allow for the possibility of non-canonical molecular orbitals. The doubles amplitudes,

$$t_{ij}^{ab} = \frac{(ia|jb)}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} , \quad (4.35)$$

are updated as

$$t_{ij}^{ab} \leftarrow t_{ij}^{ab} + \frac{R_{ij}^{ab}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} . \quad (4.36)$$

The pair energy for each  $ij$  occupied orbital pair can then be calculated as

$$e_{ij} = \sum_{ab} [(ia|jb) + R_{ij}^{ab}] (4t_{ij}^{ab} - 2t_{ji}^{ab}) , \quad (4.37)$$

and then the total MP2 energy is a sum of these pair energies

$$E_{\text{MP2}} = \frac{1}{2} \sum_{ij} e_{ij} . \quad (4.38)$$

This process repeats until the difference in energy falls below a convergence criterion, changed with ECONN. The maximum number of iterations can be changed with the CORRMAXITER keyword.

An iterative MP2 calculation on CO, with a reduced number of maximum iterations is:

TUNA SPE : C O 1.6 : IMP2 def2-SVP : ECONN 1E-10 CORRMAXITER 8

Iterative MP2 is only available, as a pilot implementation, for restricted references. The [unrelaxed MP2 density matrix](#) is also calculated and used for [molecular property calculations](#).

#### 4.2.6 Laplace Transform AO-MP2

Conventional MP2 requires molecular orbitals and eigenvalues from diagonalisation of the Fock matrix. However, as matrix diagonalisation has formal scaling  $\mathcal{O}(N^3)$ , this cannot be used for large systems. One route towards linear scaling MP2 is to exploit the Laplace transform for the energy denominators and express the MP2 energy as a functional of the HF density matrix [44].

Using the Laplace identity for energy denominators,

$$\frac{1}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} = \int_0^\infty e^{-s(\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j)} ds , \quad (4.39)$$

the MP2 correlation energy can be written as an integral over the parameter  $s$  without explicit orbital energies. The implementation in TUNA forms energy-weighted density matrices

$$\mathbf{X}(s) = \exp(s\mathbf{PF}) \mathbf{P} \quad \mathbf{Y}(s) = \exp(-s\mathbf{QF}) \mathbf{Q} , \quad (4.40)$$

where  $\mathbf{Q}$  is the “hole” density matrix, containing unoccupied orbital information, and contracts

them with atomic orbital basis electron repulsion integrals to obtain an integrand

$$e(s) = \sum_{mnl\sigma} \sum_{gdke} X_{mg}(s) Y_{nd}(s) X_{kl}(s) Y_{eo}(s) (gd|ke) L_{mnl\sigma}, \quad (4.41)$$

which can be integrated by numerical quadrature

$$E_{\text{MP2}} = - \int_0^\infty e(s) ds. \quad (4.42)$$

The  $[0, \infty)$  integral is mapped from  $r \in (0, 1)$  via  $s = g(r)$  and evaluated by a  $\tau$ -point quadrature,

$$E_{\text{MP2}} \approx - \frac{1}{\tau + 1} \sum_{k=1}^{\tau} e(s_k) g'(r_k), \quad r_k = \frac{k}{\tau + 1}, \quad s_k = g(r_k),$$

where  $g'(r_k) = ds/dr$  is the analytic weight used in the TUNA. The Euler–Maclaurin quadrature method is used which needs very few quadrature points to reach microhartree precision [45].

The number of quadrature points can be changed with the MPGRID keyword which is 10 by default. An example of a calculation using AO-MP2 with 5 grid points is:

TUNA SPE : H H 0.6 : AO-MP2 6-311++G : MPGRID 5

This method is a work in progress and is currently only available for restricted references. There is no current implementation of a Laplace transform AO-MP2 [density matrix](#).

## 4.2.7 Unrelaxed and Relaxed Density Matrices

The approximate doubles amplitudes are used to update the occupied–occupied and virtual–virtual blocks of the density matrix to form the “unrelaxed” density, which is calculated automatically during an MP2 calculation. Specifically, the occupied–occupied and virtual–virtual blocks are

$$P_{ij} = -\frac{1}{2} \sum_{abk} t_{jk}^{ab} t_{ik}^{ab}, \quad (4.43)$$

and

$$P_{ab} = \frac{1}{2} \sum_{ijc} t_{ij}^{ac} t_{ij}^{bc}. \quad (4.44)$$

This unrelaxed density is evaluated for all MP2-based methods. For MP3 or MP4 calculations, the MP2 density is evaluated instead. The unrelaxed density is used for [population analysis](#) and evaluating [multipole moments](#). If the additional print keyword P is used, this density is also used to compute the expectation value of the  $\hat{S}^2$  operator to evaluate the spin contamination.

The relaxed MP2 density matrix, which includes the orbital response contributions in the occupied–virtual blocks, is also available and can be requested using the RELAXED keyword. This response density yields the correct analytic properties (such as [molecular dipole moments](#)) that precisely match the results obtained from [numerical differentiation](#) of the MP2 energy.

Computing the relaxed density is more demanding than the unrelaxed density because it requires solving the coupled-perturbed Hartree–Fock equations to account for orbital relaxation,

$$\sum_{jb} H_{ia,jb} Z_{jb} = -L_{ia} , \quad (4.45)$$

where  $H_{ia,jb}$  represents the [orbital Hessian](#) and  $L_{ia}$  is the MP2 Lagrangian. The solutions to this linear system,  $Z_{ia}$ , define the occupied–virtual blocks of the relaxed contribution to the density.

These examples demonstrate the difference. Using RELAXED yields an analytic MP2 [dipole moment](#) that exactly matches numerical energy derivatives with DIPOLE, whereas the unrelaxed example does not. The relaxed analytical (or numerical) dipole moment is the correct way around, with the negative charge on the oxygen whereas the unrelaxed dipole makes carbon more negative.

```
TUNA SPE : C O 1.1 : MP2 6-31G : UNRELAXED DIPOLE
```

```
TUNA SPE : C O 1.1 : MP2 6-31G : RELAXED DIPOLE
```

Both the unrelaxed and relaxed density matrices are compatible with the [frozen core approximation](#) and [spin-component scaling](#). [Natural orbitals](#) from MP2 can be calculated using either density:

```
TUNA SPE : H H 1.4 : MP2 6-31G : RELAXED NATORBS PRINTMOS
```

## 4.3 Configuration Interaction

As the Hartree–Fock state cannot encode electron–electron correlation, configuration interaction (CI) constructs the many-body wavefunction as a superposition of the Hartree–Fock Slater determinant with various excited determinants. These excited determinants are generated by promoting electrons from occupied spin orbitals (denoted by indices  $i, j, \dots$ ) to virtual orbitals ( $a, b, \dots$ ),

$$|\Psi\rangle = c_{\text{HF}}|\Phi_{\text{HF}}\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \sum_{i<j} \sum_{a<b} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \dots \quad (4.46)$$

When this linear combination of determinants is complete such that all electrons have a probability of being found in all molecular orbitals, the exact wavefunction is constructed (within the one-

electron basis set) in “full configuration interaction”. Once  $\{c_{\text{occ}}^{\text{virt}}\}$  are known, the energy is

$$E = E_{\text{HF}} + \frac{1}{4} \sum_{ijab} c_{ij}^{ab} \langle ij || ab \rangle . \quad (4.47)$$

However, because the number of possible determinants scales factorially with system size, full configuration interaction is computationally intractable for all but the smallest systems. Consequently, the expansion is typically truncated at a practical level of excitation, such as CI with singles and doubles (CISD). Systematically retaining higher-order excitations progressively increases the accuracy of the final computed energy. All the same keywords for [coupled cluster](#) calculations to speed up convergence can be used for configuration interaction calculations.

### 4.3.1 Configuration Interaction Singles and Doubles

Full configuration interaction for two-electron systems like the helium atom or hydrogen molecule only requires single and double excitations, where

$$|\Psi\rangle \approx c_{\text{HF}} |\Phi_{\text{HF}}\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \frac{1}{4} \sum_{ijab} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle . \quad (4.48)$$

This is a more efficient alternative to full CI compared to [coupled cluster](#) for these systems, as disconnected terms do not need to be calculated, although the computational scaling remains the same at  $\mathcal{O}(N^6)$ . A configuration interaction singles and doubles (CISD) [optimisation and harmonic frequency](#) calculation on the hydrogen molecule, within the cc-pVTZ basis, is called by:

TUNA OPTFREQ : H H 0.7 : CISD cc-pVTZ :

This gives the same answer as the [CCSD](#) calculation, in about half the time, and is available in TUNA for both [restricted and unrestricted Hartree–Fock](#) references.

Although more efficient for two-electron systems, CISD is rarely used for many-electron molecules due to its lack of size consistency — the energy of two well-separated molecules is not twice the energy of one isolated molecule. The preferred method, [coupled cluster theory](#), is size consistent.

### 4.3.2 Configuration Interaction Doubles

An approximation to CISD is configuration interaction doubles, where

$$|\Psi\rangle \approx c_{\text{HF}} |\Phi_{\text{HF}}\rangle + \frac{1}{4} \sum_{ijab} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle . \quad (4.49)$$



This approach is essentially a non-perturbative version of [MP2](#) — the same space of double excitations is used, but these are refined iteratively. A configuration interaction doubles (CID) [optimisation and frequency](#) calculation on the hydrogen molecule, within cc-pVTZ, is requested:

TUNA OPTFREQ : H H 0.7 : CID cc-pVTZ :

### 4.3.3 Configuration Interaction Singles, Doubles and Triples

The CISDT method is exact for three-electron systems, and is defined by

$$|\Psi\rangle \approx c_{\text{HF}}|\Phi_{\text{HF}}\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \frac{1}{4} \sum_{ijab} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \frac{1}{8} \sum_{ijkabc} c_{ijk}^{abc} |\Phi_{ijk}^{abc}\rangle \quad (4.50)$$

This is available for both [RHF](#) and [UHF](#) references, but CISDT is not spin-adapted in TUNA. Like all truncated CI methods, CISDT is not size consistent so [CCSDT](#) is preferred for molecules.

TUNA OPTFREQ : Li H 1.0 : CISDT cc-pVDZ : CH 1

## 4.4 Coupled Cluster Theory

In coupled cluster theory, the exact wavefunction is given by an exponential ansatz,

$$|\Psi\rangle = \exp(\hat{T})|\Phi_{\text{HF}}\rangle, \quad (4.51)$$

where  $|\Phi_{\text{HF}}\rangle$  is the Hartree–Fock reference determinant, and  $\hat{T}$  is the cluster operator, which produces a linear combination of excited Slater determinants. This is written as

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3 + \cdots, \quad (4.52)$$

where  $\hat{T}_1$  is the operator of all single excitations,  $\hat{T}_2$  is the operator of all double excitations, etc. In the formalism of second quantisation, these operators can be expressed as

$$\hat{T}_1 = \sum_{ia} t_i^a \hat{a}_a^\dagger \hat{a}_i, \quad (4.53)$$

and

$$\hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \hat{a}_b^\dagger \hat{a}_j^\dagger \hat{a}_a \hat{a}_i. \quad (4.54)$$

In these formulae,  $\hat{a}^\dagger$  and  $\hat{a}$  are the second quantisation creation and annihilation operators, respectively, and  $t_i^a$  and  $t_{ij}^{ab}$  are the singles and doubles amplitudes, which are iteratively updated

until convergence in a coupled cluster calculation.

Expanding the exponential operator,  $\exp(\hat{T})$ , as a Taylor series yields

$$\exp(\hat{T}) = 1 + \hat{T} + \frac{1}{2!}\hat{T}^2 + \frac{1}{3!}\hat{T}^3 + \dots \quad (4.55)$$

which, upon substitution of the definition of  $T$ , yields

$$\exp(\hat{T}) = 1 + \hat{T}_1 + \hat{T}_2 + \frac{1}{2}\hat{T}_1^2 + \frac{1}{2}\hat{T}_1\hat{T}_2 + \frac{1}{2}\hat{T}_2\hat{T}_1 + \frac{1}{2}\hat{T}_2^2 + \dots \quad (4.56)$$

Curtailing  $\hat{T}$  yields approximate wavefunctions. For instance curtailing  $\hat{T}$  to  $\hat{T}_1 + \hat{T}_2$  yields the [coupled cluster singles and doubles \(CCSD\)](#) method, which, as demonstrated in [Equation 4.56](#), includes approximate contributions from higher than double excitations, through so-called “disconnected” excitations like  $\hat{T}_1\hat{T}_2$ . This means coupled cluster accounts for more electron correlation than [configuration interaction](#) curtailed to the same level.

Because single excitations do not mix with the Hartree–Fock determinant, and differences of more than two excitations do not either, the expression for the coupled cluster energy only depends on connected and disconnected double excitations, no matter the curtailing. The difference in energy comes from effects from eg. triples on the converged singles and doubles  $t$ -amplitudes.

The unrestricted total coupled cluster energy is calculated by

$$E = E_{\text{HF}} + \frac{1}{4} \sum_{ijab} \langle ij || ab \rangle t_{ij}^{ab} + \frac{1}{2} \sum_{ijab} \langle ij || ab \rangle t_i^a t_j^b, \quad (4.57)$$

and the spin-restricted coupled cluster energy is

$$E = E_{\text{HF}} + \sum_{ijab} [2(ia|jb) - (ib|ja)](t_{ij}^{ab} + t_i^a t_j^b). \quad (4.58)$$

In all coupled cluster methods, an iterative method is used to converge the  $t$ -amplitudes. Similarly to the Hartree–Fock [self-consistent field](#), [DIIS](#) is used by default to dramatically accelerate convergence. The choice of error vector in TUNA to minimise is the difference between subsequent pairs of  $t$ -amplitudes, the residual. This extrapolation is on by default and can be turned off with the `NODIIS` keyword, and the number of amplitudes that DIIS will remember (six by default) can be changed with the keyword `DIIS [Num. Amps]`.

Damping of successive  $t$ -amplitudes can also be requested with the `CORRDAMP` keyword, whose

parameter determines how much old  $t$ -amplitudes should be mixed with new ones,

$$t_{\mu} \leftarrow a t_{\mu}^{\text{old}} + (1 - a) t_{\mu}^{\text{new}} . \quad (4.59)$$

This damping is off by default. A CCSD calculation with damping at 50% and DIIS remembering seven sets of  $t$ -amplitudes is given by:

TUNA SPE : 0 0 0.9 : CCSD 3-21G : DIIS 7 CORRDAMP 0.50

The TUNA output for coupled cluster calculation begins by printing the energy convergence for the coupled cluster iterations, which is mutable with the ECONV keyword — this is the same as the SCF convergence by default. In addition to energy convergence, the amplitudes themselves must converge to within  $10^{-8}$ , which can be adjusted with the AMPCONV keyword. Next the MP2 energy, which is calculated from the MP2 approximate  $t_{ij}^{ab}$  amplitudes, which are used as the guess amplitudes, is printed. Information about the convergence acceleration is then printed, before the iterations begin. Once convergence is achieved, the final correlation energy is shown. A maximum of CORRMAXITER iterations are iterated through, which is 100 by default.

The norm of  $t_i^a$  and the  $\mathcal{T}_1$  diagnostic is then printed, which gives an idea if a single reference calculation is valid [46]. The  $\mathcal{T}_1$  diagnostic is related to the norm of the  $t_i^a$  amplitudes by

$$\mathcal{T}_1 = \frac{||t_i^a||}{\sqrt{N_{\text{occ}}}} , \quad (4.60)$$

where  $N_{\text{occ}}$  is the number of correlated occupied spin orbitals.

Next, the largest singles and doubles amplitudes are printed, showing the most important excitations out of the Hartree–Fock orbitals. The number of amplitudes shown is ten by default, and can be changed with the PRINTAMPS keyword. Finally, the linearised density is calculated and molecular property and population analysis calculations can begin.

All coupled cluster methods — except CC2, CC3, CCSDT(Q) and CCSDTQ which require a restricted Hartree–Fock reference — are available in both spin unrestricted and restricted spin-adapted forms, making calculations on closed-shell singlet molecules much faster.

#### 4.4.1 Coupled Cluster Doubles

In CCD, the cluster operator is curtailed to

$$T = T_2 , \quad (4.61)$$

giving equations where  $t_{ij}^{ab}$  needs to be optimised. In TUNA, CCD can be requested with the CCD method type. A CCD [single point energy](#) calculation on carbon monoxide is requested by:

```
TUNA SPE : C O 0.9 : CCD 6-31+G
```

#### 4.4.2 Coupled Cluster Singles and Doubles

In CCSD, the cluster operator is curtailed to

$$T = T_1 + T_2 , \quad (4.62)$$

giving equations where  $t_i^a$  and  $t_{ij}^{ab}$  need to be optimised together [47]. In TUNA, CCSD can be requested with the CCSD method type. For example, a CCSD [optimisation and harmonic frequency](#) calculation on the hydrogen molecule, in the cc-pVTZ basis set, can be called by:

```
TUNA OPTFREQ : H H 0.7 : CCSD cc-pVTZ
```

For hydrogen, with only two electrons, CCSD corresponds to full [configuration interaction](#) so this is the exact frequency (exact [harmonic frequency](#), within the [basis set](#), within the Born–Oppenheimer approximation, within non-relativistic quantum mechanics). The CCSD method is quite computationally intensive, with formal  $\mathcal{O}(N^6)$  scaling with respect to basis functions.

#### 4.4.3 Coupled Cluster Singles, Doubles and Triples

In CCSDT, the cluster operator is curtailed to

$$T = T_1 + T_2 + T_3 , \quad (4.63)$$

giving equations where  $t_i^a$ ,  $t_{ij}^{ab}$  and  $t_{ijk}^{abc}$  need to be optimised together [48]. This can be requested using CCSDT. For example, a CCSDT [single point energy](#) on lithium with cc-pVDZ is called by:

```
TUNA SPE : Li : CCSDT cc-pVDZ :
```

As lithium has three electrons, the CCSDT energy corresponds to the [full CI](#) energy. The CCSDT method is very slow, with formal  $\mathcal{O}(N^8)$  scaling with respect to [basis functions](#). To improve convergence, CCSDT and [CCSDTQ](#) amplitudes that are not physically meaningful — and build up noise — are projected out each coupled cluster iteration. This can reduce the number of necessary iterations by a factor of 5, at little extra cost per iteration.

#### 4.4.4 Coupled Cluster Singles, Doubles, Triples and Quadruples

In CCSDTQ, the cluster operator is curtailed to

$$T = T_1 + T_2 + T_3 + T_4, \quad (4.64)$$

giving equations where  $t_i^a$ ,  $t_{ij}^{ab}$ ,  $t_{ijk}^{abc}$  and  $t_{ijkl}^{abcd}$  need to be optimised together [49, 50]. The CCSDTQ method can be requested using CCSDTQ. A CCSDTQ [single point](#) calculation on beryllium is:

```
TUNA SPE : Be : CCSDTQ cc-pVDZ
```

As beryllium has four electrons, the CCSDTQ energy corresponds to the [full CI](#) energy. The CCSDTQ method is extremely slow and [memory intensive](#), with formal  $\mathcal{O}(N^{10})$  scaling with respect to [basis functions](#). This is currently the most intensive electronic structure method possible in TUNA, especially because the convergence of the quadruples amplitudes can be slow for difficult systems. All CCSDTQ calculations must be used with spin-restricted references.

#### 4.4.5 Coupled Cluster with Perturbative Triples

In CCSD(T), often called the “gold standard” of quantum chemistry, a perturbative correction is made to the [CCSD](#) energy, based on the [MP4](#) and [MP5](#) triples amplitudes [51, 52]. This provides approximate treatment of triples, which often gives better results than [CCSDT](#), with lower  $\mathcal{O}(N^7)$  scaling, due to the undershooting of triples resembling the inclusion of quadruples [46].

Because TUNA is run entirely from the terminal CCSD(T) is called by the CCSD[T] method keyword. An [optimisation](#) of lithium hydride in the 6-311G\*\* basis is given by:

```
TUNA OPT : Li H 1.0 : CCSD[T] 6-311G**
```

Note that the “CCSD[T]” method in the literature is not the same as the “CCSD(T)” method. Unfortunately, parentheses cannot be used directly in the terminal in Windows, so TUNA changes circular brackets to square brackets. Either will work fine in Linux or MacOS.

#### 4.4.6 Coupled Cluster with Perturbative Quadruples

If the “gold standard” of CCSD(T) is not enough — for example in a multireference situation — the “platinum standard” of CCSDT(Q) can be used, which incorporates approximate connected quadruples excitations through contributions based on MP5 and MP6 [53, 54]. The cost of this method is huge — at  $\mathcal{O}(N^9)$ , but does not rely on the slow convergence of full [CCSDTQ](#).

The CCSDT(Q) method is called by the CCSDT[Q] keyword. A single point calculation on lithium

hydride in the cc-pVTZ basis is given by:

```
TUNA SPE : Li H 1.0 : CCSDT[Q] cc-pVTZ
```

#### 4.4.7 Linearised Coupled Cluster

Linearised coupled cluster makes an approximation to  $\exp(\hat{T})$ , keeping only terms to first order

$$\exp(\hat{T}) \approx 1 + T = 1 + T_1 + T_2 + T_3 + \dots \quad (4.65)$$

This comes at the cost of size consistency and is normally less accurate than conventional coupled cluster, however computation of the equations becomes much faster.

In TUNA, linearised CCD and linearised CCSD are implemented, and can be requested using the LCCD and LCCSD method types, respectively. In linearised CCD,  $T = T_2$  and in linearised CCSD,  $T = T_1 + T_2$ . For instance, an [optimisation](#) on dinitrogen with LCCD can be requested by:

```
TUNA OPT : N N 1.2 : LCCD 6-31G
```

Linearised coupled cluster coincides with the coupled electron pair approximation (CEPA0). By default, the CEPA0 method is the same as LCCSD in TUNA. A calculation using CEPA0, with single excitations (unshifted) on the carbon atom is shown:

```
TUNA SPE : C : CEPA0 3-21G : ML 3
```

#### 4.4.8 Quadratic Configuration Interaction

A modification to [configuration interaction](#) to make it size consistent known as quadratic configuration interaction singles and doubles (QCISD) is implemented in TUNA [46, 55]. This turns out to be the same as a simplified version of [CCSD](#).

The keyword for this method is QCISD, and a spin-adapted implementation is available:

```
TUNA OPT : H Li 1.3 : QCISD def2-SVP :
```

Moreover, the method can be combined with perturbative triples with the QCISD[T] method.

```
TUNA OPTFREQ : H Li 1.6 : QCISD[T] def2-TZVPPD :
```

#### 4.4.9 Approximate Coupled Cluster

In the approximate coupled cluster singles and doubles method, CC2, the [CCSD](#) equations are simplified to only second order in the fluctuation potential, with iterative singles and a perturbative, [MP2](#)-like approach to the doubles amplitudes. This reduces the scaling of [CCSD](#) to  $\mathcal{O}(N^5)$  [56].

Similarly, in the CC3 method the full [CCSD](#) equations are iteratively solved, but an approximation to connected triples with an [MP4](#)-like second-order triples expression. This reduces the scaling of [CCSDT](#) down to  $\mathcal{O}(N^7)$ , the same as [CCSD\(T\)](#).

These methods are implemented in TUNA with a T1-dressed Hamiltonian which reduces the complexity of the working equations, for spin-restricted references only. A [polarisability](#) calculation with CC2 on carbon monoxide is requested by:

```
TUNA SPE : C O 1.0 : CC2 cc-pVDZ : POLAR
```

Similarly, an [optimisation and frequency](#) calculation on lithium hydride with the iterative CC3 method can be requested by:

```
TUNA OPTFREQ : Li H 1.0 : CC3 def2-SVP :
```

#### 4.4.10 Linearised Density Matrix

All implemented coupled cluster methods generate a density matrix after the energy is calculated, which is then used for [property](#) calculations such as [natural orbitals](#), spin contamination, [dipole moment](#) and [population analysis](#). This density can also be used for [plotting](#).

However, bear in mind that this is not the [relaxed response density](#), nor the full unrelaxed density – the implemented density is linearised. This means that only terms of  $t$ -amplitudes up to quadratic are included in the density matrix construction.

### 4.5 Density Functional Theory

Taking a very different approach to the [correlated wavefunction methods](#), density functional theory (DFT) aims to recover electron–electron correlation directly from the electron density rather than the complex many-body wavefunction. Because the electron density depends on only three spatial coordinates — regardless of the number of electrons in the system — DFT offers a highly favourable balance between computational cost and physical accuracy.

In the widely used Kohn–Sham formulation of DFT [57, 58], the interacting many-body problem is mapped onto a mathematically equivalent, fictitious system of non-interacting electrons. The total electronic energy is expressed as a functional of the electron density,  $n(\mathbf{r})$ , by

$$E[n] = T_s[n] + V_{\text{ne}}[n] + J[n] + E_{\text{xc}}[n] . \quad (4.66)$$

Here, the terms represent the non-interacting kinetic energy ( $T_s$ ), the external nuclear–electron attraction ( $V_{\text{ne}}$ ), the classical Coulomb repulsion between electrons ( $J$ ), and the exchange–correlation energy ( $E_{\text{xc}}$ ), which accounts for quantum mechanical effects. To compute the kinetic energy term Kohn and Sham introduced a set of independent-particle orbitals [58]. In TUNA,  $T_s[n]$ ,  $V_{\text{ne}}[n]$ , and  $J[n]$  are calculated from analytical [integrals over these Kohn–Sham orbitals](#).

The ground state electron density is constructed from the occupied Kohn–Sham orbitals,

$$n(\mathbf{r}) = \sum_i^{\text{occ}} |\psi_i(\mathbf{r})|^2 . \quad (4.67)$$

The  $E_{\text{xc}}[n]$  term accounts for all non-classical many-body effects, including quantum mechanical exchange, dynamic electron correlation, and the small correction to the kinetic energy not captured by  $T_s[n]$ . The implementation in TUNA defines all exchange–correlation functionals as

$$E_{\text{xc}}[n] = \int f_{\text{xc}}[n] \, d\mathbf{r} \quad (4.68)$$

where  $f_{\text{xc}}[n] = ne_{\text{xc}}[n]$  with  $e_{\text{xc}}$  being the exchange–correlation energy density per particle. Because the exact mathematical form of  $E_{\text{xc}}$  remains unknown, various approximations are employed to evaluate this term numerically over a grid.

At self-consistency, the Kohn–Sham equations [58],

$$\left( -\frac{1}{2}\nabla^2 + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, d\mathbf{r}' + \frac{\delta E_{\text{xc}}}{\delta n} + v_{\text{ext}}(\mathbf{r}) \right) \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r}) , \quad (4.69)$$

are satisfied, where  $\psi_i(\mathbf{r})$  and  $\varepsilon_i$  are the Kohn–Sham orbitals and energies and  $v_{\text{ext}}(\mathbf{r})$  is the external potential. Practically, the [Roothaan–Hall equations from Hartree Fock](#) are modified so the spin-restricted Fock matrix is calculated using an exchange–correlation matrix,  $\mathbf{K}_{\text{xc}}$

$$\mathbf{F} = \mathbf{T} + \mathbf{V} + 2\mathbf{J} + \mathbf{K}_{\text{xc}} , \quad (4.70)$$

and a proportion of Hartree–Fock exchange can be simply implemented through the exchange matrix,  $\mathbf{K}$ . This Fock matrix can then be used in the standard [Hartree–Fock procedure](#), leveraging the available [convergence accelerators](#) in TUNA.



All density functional methods in TUNA can be extended to open-shell systems by using unrestricted references. This is done simply by prepending a “U” to the functional name, which allows  $\alpha$  and  $\beta$  electrons to occupy different spatial orbitals. For all standard methods, the energy and density are evaluated iteratively until self-consistency is achieved. For more advanced [double-hybrid functionals](#), which mix exact exchange with perturbative correlation, a [second-order Møller–Plesset](#) correction is evaluated and applied only after self-consistency has been reached.

The exchange or correlation contributions can be turned off with the NOX or NOC keywords. For example, this reproduces the Hartree–Fock–Slater [59] method with no DFT correlation:

TUNA SPE : H H 1.0 : LDA 3-21G : NOC

[Table 4.3](#) and [Table 4.4](#) list all the available exchange and correlation functionals in TUNA. Methods are often thought of as belonging to a certain rung of Perdew’s “Jacob’s ladder” of exchange–correlation functionals [78], shown in [Figure 4.1](#).

**Table 4.3** Exchange functionals in TUNA

| Functional               | Description                                                             |
|--------------------------|-------------------------------------------------------------------------|
| S [59]                   | Slater LDA exchange                                                     |
| B88 [60]                 | Becke GGA exchange                                                      |
| PBE [61]                 | Perdew–Burke–Ernzerhof GGA exchange                                     |
| revPBE [62]              | Revised Perdew–Burke–Ernzerhof GGA exchange                             |
| RPBE [63]                | Modified Perdew–Burke–Ernzerhof GGA exchange                            |
| PW91 [64]                | Perdew–Wang GGA exchange                                                |
| mPW [65]                 | Modified Perdew–Wang GGA exchange                                       |
| B97 [66]                 | Becke GGA exchange                                                      |
| B97-D [67]               | Dispersion-corrected Becke GGA exchange                                 |
| TPSS [68]                | Tao–Perdew–Staroverov–Scuseria <i>meta</i> -GGA exchange                |
| revTPSS [69]             | Revised Tao–Perdew–Staroverov–Scuseria <i>meta</i> -GGA exchange        |
| SCAN [70]                | Strongly constrained and appropriately normed <i>meta</i> -GGA exchange |
| rSCAN [71]               | Regularised SCAN <i>meta</i> -GGA exchange                              |
| r <sup>2</sup> SCAN [72] | Regularised and restored SCAN <i>meta</i> -GGA exchange                 |
| B97M-V [73]              | Non-local dispersion-corrected Becke <i>meta</i> -GGA exchange          |

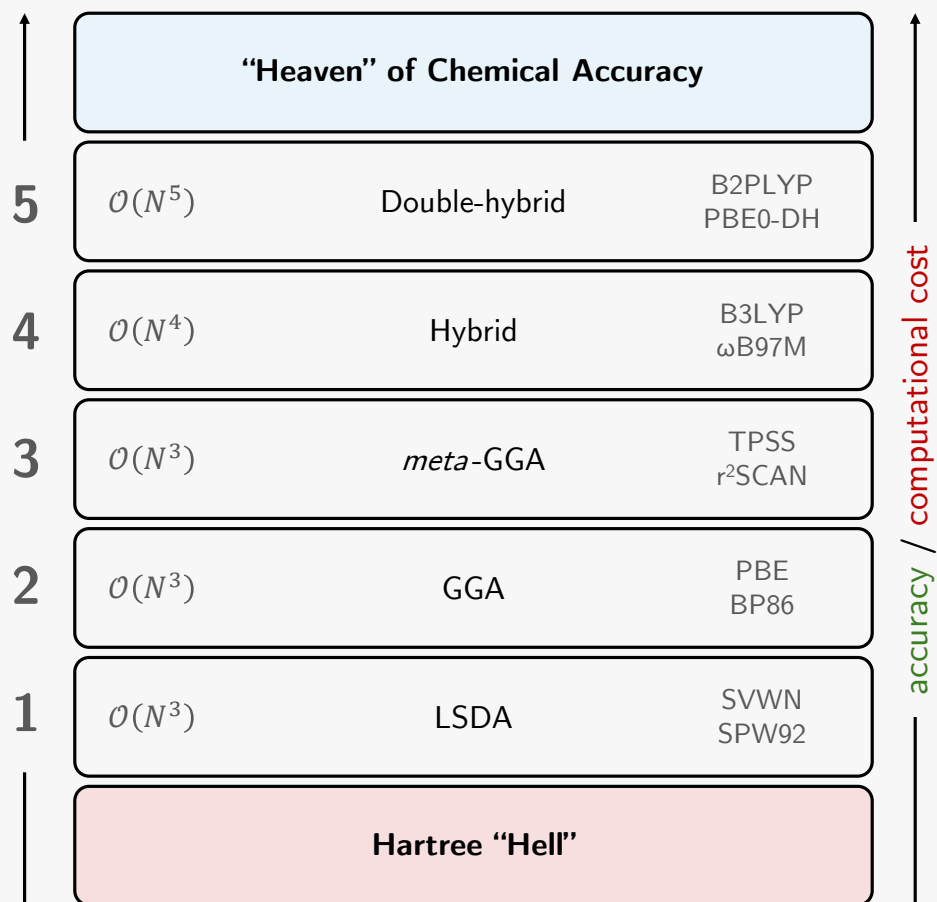


Figure 4.1: The Jacob's ladder of exchange–correlation functionals [78]. Climbing the ladder results in higher computational cost and, on average, higher accuracy

**Table 4.4** Correlation functionals in TUNA

| Functional               | Description                                                                |
|--------------------------|----------------------------------------------------------------------------|
| VWN3 [74]                | Vosko–Wilk–Nusair LDA correlation, parameter set III                       |
| VWN5 [74]                | Vosko–Wilk–Nusair LDA correlation, parameter set V                         |
| PW [75]                  | Perdew–Wang LDA correlation                                                |
| P86 [76]                 | Perdew 1986 GGA correlation                                                |
| PW91 [64]                | Perdew–Wang GGA correlation                                                |
| PBE [61]                 | Perdew–Burke–Ernzerhof GGA correlation                                     |
| LYP [77]                 | Lee–Yang–Parr GGA correlation                                              |
| B97 [66]                 | Becke GGA correlation                                                      |
| B97-D [67]               | Dispersion-corrected Becke GGA correlation                                 |
| TPSS [68]                | Tao–Perdew–Staroverov–Scuseria <i>meta</i> -GGA correlation                |
| revTPSS [69]             | Revised Tao–Perdew–Staroverov–Scuseria <i>meta</i> -GGA correlation        |
| SCAN [70]                | Strongly constrained and appropriately normed <i>meta</i> -GGA correlation |
| rSCAN [71]               | Regularised SCAN <i>meta</i> -GGA correlation                              |
| r <sup>2</sup> SCAN [72] | Regularised and restored SCAN <i>meta</i> -GGA correlation                 |
| B97M-V [73]              | Non-local dispersion-corrected Becke <i>meta</i> -GGA correlation          |

### 4.5.1 Local Spin Density Approximation

The local density contribution to the exchange–correlation matrix is

$$K_{xc}^{\mu\nu} = \int \varphi_{\mu}(\mathbf{r}) \frac{\delta f_{xc}}{\delta n} \varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} , \quad (4.71)$$

and the energy density is a functional of the local spin densities,  $e_{xc} = e_{xc}[n_{\alpha}, n_{\beta}]$ . The implemented local density functionals are listed in Table 4.5. The derivatives with respect to the density needed to form the exchange–correlation matrix have been implemented analytically.

The LSDA functional is equivalent to parameter set V of the VWN exchange–correlation functional, VWN5 [74]. The Perdew–Wang 1992 functional is a simpler and more accurate parameterisation of the correlation energy of the uniform electron gas [75]. Parameter set III for VWN can be requested with the SVWN3 functional [74]. The only local density exchange is that of the uniform electron gas, Slater exchange. This can be used in lieu of Hartree–Fock exchange by the Hartree–Fock–Slater

(HFS) method keyword [59].

The exchange energy of the spin-restricted uniform electron gas is

$$E_x[n] = -\alpha \frac{9}{8} \left( \frac{3}{\pi} \right)^{\frac{1}{3}} \int n(\mathbf{r})^{\frac{4}{3}} d\mathbf{r}, \quad (4.72)$$

where  $\alpha = 2/3$  by default. The XA keyword can be used to change this, in the X- $\alpha$  method. For example, changing the value of  $\alpha$  to 0.7 in a HFS calculation is requested by:

```
TUNA SPE : N N 1.0 : HFS 3-21G : XA 0.7
```

An LSDA [geometry optimisation](#) can be requested in the usual way. Here, a medium [integration grid](#) is requested. The LSDA does worse for dihydrogen's bond length compared to [Hartree–Fock](#).

```
TUNA OPT : H H 0.74 : LSDA cc-pVTZ : MEDIUMGRID
```

#### 4.5.2 Generalised Gradient Approximation

The treatment of electronic structure can be improved by incorporating information about the gradients of the spin densities into the exchange–correlation functional, in “generalised gradient approximation” (GGA) functionals,

$$e_{xc} = e_{xc}[n_\alpha, n_\beta, \nabla n_\alpha, \nabla n_\beta]. \quad (4.73)$$

**Table 4.5** Local spin density approximation exchange–correlation functionals in TUNA

| Functional     | $E_x[n]$ | $E_c[n]$ |
|----------------|----------|----------|
| LSDA [59, 74]  | S        | VWN5     |
| SVWN [59, 74]  | S        | VWN5     |
| SVWN5[59, 74]  | S        | VWN5     |
| SVWN3 [59, 74] | S        | VWN3     |
| SPW [59, 75]   | S        | PW       |
| HFS [59]       | S        | —        |

**Table 4.6** Generalised gradient approximation exchange–correlation functionals in TUNA

| Functional      | $E_x[n]$ | $E_c[n]$ |
|-----------------|----------|----------|
| SLYP [59, 77]   | S        | LYP      |
| HFB [60]        | B88      | —        |
| BVWN [60, 74]   | B88      | VWN5     |
| BVWN3 [60, 74]  | B88      | VWN3     |
| BLYP [60, 77]   | B88      | LYP      |
| PWP [64, 76]    | PW91     | P86      |
| BP86 [60, 76]   | B88      | P86      |
| PBE [61]        | PBE      | PBE      |
| revPBE [79]     | revPBE   | PBE      |
| RPBE [63]       | RPBE     | PBE      |
| mPWLYP [77, 80] | mPW      | LYP      |
| mPWPW [64, 80]  | mPW      | PW91     |
| B97-D [67]      | B97-D    | B97-D    |

The GGA contribution to exchange–correlation depends on the derivative of  $f_{xc}$  with respect to

$$\sigma(\mathbf{r}) = \nabla n \cdot \nabla n . \quad (4.74)$$

It also depends on the [basis function](#) gradients, which are calculated analytically by TUNA and expressed on the [integration grid](#),

$$K_{xc}^{\mu\nu} = \int \varphi_\mu(\mathbf{r}) \frac{\delta f_{xc}}{\delta \sigma} \nabla \varphi_\nu(\mathbf{r}) \nabla n(\mathbf{r}) d\mathbf{r} . \quad (4.75)$$

The implemented GGA functionals are listed in [Table 4.6](#), among which are the popular PBE and BLYP [60, 61, 77], as well as many other combinations of exchange and correlation.

TUNA OPT : Li Li 2.2 : BLYP def2-SVP : TIGHTGRID

TUNA SPE : H H 1.2 : UPBE def2-SVP : PLOTLUMO

### 4.5.3 meta-Generalised Gradient Approximation

On the next step along the Jacob’s ladder of exchange–correlation functionals [78] are *meta*-GGA functionals, which incorporate implicit dependence on the curvature of the density,  $\nabla^2 n$ , through

the non-interacting kinetic energy density,  $\tau(\mathbf{r})$ ,

$$e_{xc} = e_{xc}[n_\alpha, n_\beta, \nabla n_\alpha, \nabla n_\beta, \tau_\alpha, \tau_\beta] . \quad (4.76)$$

The kinetic energy density is calculated as a sum of occupied orbital gradients, so is technically “non-local”, but in practice has similar cost to GGA functionals so is normally called “semilocal”,

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_i^{\text{occ}} |\nabla \psi_i|^2 . \quad (4.77)$$

In TUNA, this sum over occupied orbitals is achieved through the density matrix,

$$\tau(\mathbf{r}) = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \varphi_\mu(\mathbf{r}) \nabla \varphi_\nu(\mathbf{r}) . \quad (4.78)$$

The kinetic energy density can be incorporated through the dimensionless iso-orbital indicator,

$$\alpha(\mathbf{r}) = \frac{\tau(\mathbf{r}) - \tau_W(\mathbf{r})}{\tau_U(\mathbf{r})} , \quad (4.79)$$

which compares the kinetic energy density to the von Weizsäcker kinetic energy density of a one orbital system,  $\tau_W(\mathbf{r})$  and the uniform electron gas kinetic energy density,  $\tau_U(\mathbf{r})$ . This allows *meta*-GGA functionals to identify regions with one-electron character or slowly varying densities and adjust their behaviour accordingly, thereby satisfying more constraints of the exact exchange–correlation functional compared to GGAs.

The exchange–correlation matrix contribution here requires the [basis function](#) gradients, as in GGA functionals, but also requires the derivative of the functional with respect to  $\tau(\mathbf{r})$ .

$$K_{xc}^{\mu\nu} = \int \nabla \varphi_\mu(\mathbf{r}) \frac{\delta f_{xc}}{\delta \tau} \nabla \varphi_\nu(\mathbf{r}) d\mathbf{r} \quad (4.80)$$

[Table 4.7](#) lists the *meta*-GGA exchange–correlation functionals in TUNA, which are all available self-consistently for both restricted and unrestricted references. Because *meta*-GGAs rely on second derivatives of the density, they are particularly sensitive to the precision of the [integration grid](#). It is probably a good idea to use TIGHTGRID for these functionals for high precision work.

TUNA SPE : Li H 1.0 : r2SCAN cc-pVDZ : PLOTHOMO TIGHTGRID

The B97M-V functional is unique in that it has been parameterised [\[73\]](#) with [non-local VV10 dispersion correction](#). This dispersion calculation happens at the end of the SCF cycle.

**Table 4.7** *meta*-Generalised gradient approximation exchange–correlation functionals in TUNA

| Functional               | $E_x[n]$            | $E_c[n]$            |
|--------------------------|---------------------|---------------------|
| TPSS [68]                | TPSS                | TPSS                |
| revTPSS [69]             | revTPSS             | revTPSS             |
| SCAN [70]                | SCAN                | SCAN                |
| rSCAN [71]               | rSCAN               | rSCAN               |
| r <sup>2</sup> SCAN [72] | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN |
| B97M-V [73]              | B97M-V              | B97M-V              |

```
TUNA SPE : C O 1.1 : B97M-V def2-SVP :
```

#### 4.5.4 Hybrid Functionals

The next rung up Jacob’s ladder is “hybrid”, non-local exchange–correlation functionals. A hybrid exchange–correlation functional uses a proportion,  $\beta$ , of the Fock exchange expression evaluated on the Kohn–Sham orbitals

$$E_{xc}[n] = \beta E_x^{\text{HF}} + (1 - \beta) E_x^{\text{KS}}[n] + E_c^{\text{KS}}[n] . \quad (4.81)$$

A number of hybrid functionals are implemented in TUNA, listed in Table 4.8. These include the popular PBE0 and B3LYP functionals. The version of B3LYP used in the Gaussian program, which uses VWN-III LSDA correlation instead of VWN-V, can be requested with B3LYP/G.

The proportion of [Hartree–Fock exchange](#) can be changed with the HFX keyword, with the desired proportion of exchange. This will not change the proportion of exchange from the DFT expression, which can be changed with the DFX keyword. For example, the PBE0 functional can be built as:

```
TUNA SPE : H H 1.0 : PBE 4-31G : HFX 0.25 DFX 0.75
```

Similarly, the B3LYP functional can be specified by:

```
TUNA SPE : H H 1.0 : BLYP def2-TZVPD : HFX 0.50 DFX 0.50
```

### 4.5.5 Double-hybrid Functionals

The next rung up the ladder contains “double-hybrid” functionals, which additionally incorporate a proportion,  $\gamma$ , of [second-order perturbation theory correlation](#).

$$E_{xc}[n] = \beta E_x^{\text{HF}} + (1 - \beta) E_x^{\text{KS}}[n] + \gamma E_c^{\text{MP2}} + (1 - \gamma) E_c^{\text{KS}}[n] \quad (4.82)$$

The [MP2](#) part of the energy is applied only to the converged Kohn–Sham orbitals at the end of the [SCF](#). The [unrelaxed density matrix](#) is also implemented for double-hybrid functionals, allowing calculation of [properties](#) including natural orbitals and [dipole and quadrupole moments](#).

**Table 4.8** Hybrid exchange–correlation functionals in TUNA

| Functional                 | $E_x[n]$            | $E_c[n]$            | HFX  |
|----------------------------|---------------------|---------------------|------|
| PBE0 [65]                  | PBE                 | PBE                 | 0.25 |
| revPBE0 [79]               | revPBE              | PBE                 | 0.25 |
| revPBE38 [79]              | revPBE              | PBE                 | 0.38 |
| B1LYP [81]                 | B88                 | LYP                 | 0.25 |
| BHLYP [82]                 | B88                 | LYP                 | 0.50 |
| B3LYP [83]                 | B88                 | LYP                 | 0.20 |
| B3LYP/G [83]               | B88                 | LYP                 | 0.20 |
| B1P86 [60, 76]             | B88                 | P86                 | 0.25 |
| B3P86 [60, 76]             | B88                 | P86                 | 0.20 |
| PW1PW [65]                 | PW91                | PW91                | 0.25 |
| mPW1PW [65]                | mPW91               | PW91                | 0.25 |
| mPW1LYP [65]               | mPW91               | LYP                 | 0.25 |
| B3PW91 [60, 64]            | B88                 | PW91                | 0.20 |
| TPSSH [84]                 | TPSS                | TPSS                | 0.10 |
| TPSS0 [84]                 | TPSS                | TPSS                | 0.25 |
| SCAN0 [85]                 | SCAN                | SCAN                | 0.25 |
| r <sup>2</sup> SCANh [86]  | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.10 |
| r <sup>2</sup> SCAN0 [86]  | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.25 |
| r <sup>2</sup> SCAN50 [86] | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.50 |
| B97 [66]                   | B97                 | B97                 | 0.19 |



The first double-hybrid functional developed was B2PLYP [87], which incorporates 53% [Hartree–Fock exchange](#) and 27% [MP2 correlation](#). This functional can be requested by either of these:

```
TUNA SPE : Li H 1 : B2PLYP 6-311G :
```

```
TUNA SPE : Li H 1 : BLYP 6-311G : HFX 0.53 DFX 0.47 MPC 0.27 DFC 0.73
```

The MPC keyword defines the proportion of [MP2 correlation](#), and the DFC keyword defines the proportion of DFT correlation. Similarly, the non-empirical PBE0-DH functional can be built by:

```
TUNA SPE : Li H 1 : PBE 6-311G : HFX 0.50 DFX 0.50 MPC 0.125 DFC 0.875
```

A number of double-hybrids are implemented in TUNA, listed in [Table 4.9](#). These include non-empirical double-hybrids like PBE0-DH,  $r^2$ SCAN0-DH and PBE-QIDH, as well as empirical double-hybrids including B2PLYP, mPW2PLYP and DSD-BLYP. The latter functional, as well as  $r^2$ SCAN-based double-hybrids, also incorporates [spin-component scaling](#) of the [MP2 correlation](#) part. The proportion of [same- and opposite-spin scaling](#) can be changed with the SSS and OSS keywords.

The values of HFX, DFX, MPC and DFC are printed at the start of a hybrid or double-hybrid DFT calculation's [SCF](#) cycle. The values of OSS and SSS are printed during the [MP2 calculation](#) for a spin-scaled double-hybrid, or when those keywords are used otherwise.

#### 4.5.6 Efficient Grid Integration

Density functional theory calculations involved numerical integrations over a grid, on which the electron density is expressed. The integral of the density should be the same as the number of electrons in the system, and prints for the guess density before the [self-consistent field cycle](#), and the final converged density. If the guess density is far from the ideal, a warning will be printed, and if it is totally wrong the calculation will crash politely.

The tightness of this grid can be controlled directly with the INTACC keyword, which has a default value of 4.0. The extent of the grid (which may be important for large atoms) is controlled via the general integration grid keywords in [Table 4.10](#).

Integration on this grid is performed by numerical quadrature. The grids are atom centered, and then Becke weighting is applied to build the molecular grid [96]. The atom centered grids use Gauss–Legendre quadrature for the radial part and Lebedev quadrature for the angular part. The order of Lebedev integration is chosen by the closest Lebedev order available to match the desired integral accuracy. The following is a very high density grid, beyond that from EXTREMEGRID:

**Table 4.9** Double-hybrid exchange–correlation functionals implemented within TUNA.

| Functional                    | $E_x[n]$            | $E_c[n]$            | HFX  | MPC  | SSS  | OSS  |
|-------------------------------|---------------------|---------------------|------|------|------|------|
| PBE0-DH [88]                  | PBE                 | PBE                 | 0.50 | 1/8  | 1.00 | 1.00 |
| PBE-QIDH [89]                 | PBE                 | PBE                 | 0.69 | 0.33 | 1.00 | 1.00 |
| PBE0-2 [90]                   | PBE                 | PBE                 | 0.79 | 0.50 | 1.00 | 1.00 |
| B2PLYP [87]                   | B88                 | LYP                 | 0.53 | 0.27 | 1.00 | 1.00 |
| B2K-PLYP [91]                 | B88                 | LYP                 | 0.72 | 0.42 | 1.00 | 1.00 |
| B2T-PLYP [91]                 | B88                 | LYP                 | 0.60 | 0.31 | 1.00 | 1.00 |
| B2G-PLYP [91]                 | B88                 | LYP                 | 0.65 | 0.36 | 1.00 | 1.00 |
| B2NC-PLYP [92]                | B88                 | LYP                 | 0.81 | 0.55 | 1.00 | 1.00 |
| mPW2PLYP [93]                 | mPW91               | LYP                 | 0.55 | 0.25 | 1.00 | 1.00 |
| DSD-BLYP [94]                 | B88                 | LYP                 | 0.75 | 1.00 | 0.60 | 0.46 |
| r <sup>2</sup> SCAN0-DH [95]  | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.50 | 1/8  | 0.00 | 4/3  |
| r <sup>2</sup> SCAN-CIDH [95] | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.55 | 4/3  | 0.00 | 4/3  |
| r <sup>2</sup> SCAN-QIDH [95] | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.69 | 1/3  | 0.00 | 4/3  |
| r <sup>2</sup> SCAN0-2 [95]   | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.79 | 0.50 | 0.00 | 4/3  |
| Pr <sup>2</sup> SCAN50 [95]   | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.50 | 0.25 | 0.00 | 4/3  |
| Pr <sup>2</sup> SCAN69 [95]   | r <sup>2</sup> SCAN | r <sup>2</sup> SCAN | 0.69 | 4/9  | 0.00 | 4/3  |

TUNA SPE : Ar : HFS 6-31G : INTACC 8.0 TIGHTGRID

Becke’s heteroatomic weighting system is used, to distribute integration weights for heteronuclear systems. This relies on a proxy for atomic size, for which the van der Waals radius is used.

Both the density (and density gradient for GGA functionals) need to be expressed on the grid. This is done through the density matrix, by expressing [basis functions](#) and basis function gradients on the grid. The mathematical expressions for the primitive Gaussians, weighted by their coefficients in an atomic orbital, are evaluated on the grid, centered around each atomic basis function centre.

**Table 4.10** Integration grid tightness criteria in TUNA

| Grid        | Integral Accuracy | Grid Range |
|-------------|-------------------|------------|
| LOOSEGRID   | 3.0               | 0.7        |
| MEDIUMGRID  | 4.0               | 0.9        |
| TIGHTGRID   | 5.0               | 1.0        |
| EXTREMEGRID | 7.0               | 1.2        |

## 4.6 Excited States

Excited state calculations, to evaluate excitation energies and absorption intensities, are available in TUNA. As with other electronic structure methods, any excited state method can be used with [optimisations](#), [vibrational frequency calculations](#) and, in principle, [molecular dynamics simulations](#). In reality, there is significant risk to doing this, as unintended crossings between excited state potential energy surfaces may occur and TUNA currently has no root following algorithm.

Excited state calculations are available for both restricted and unrestricted references. The default [SCF convergence criteria](#) is set to TIGHT for all excited state calculations.

Linear response time-dependent [Hartree–Fock](#) and [density functional theory](#) allow for the calculation of excitation energies,  $\omega$ , through the Casida equations,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} = \omega \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix}, \quad (4.83)$$

where the elements of the  $\mathbf{A}$  and  $\mathbf{B}$  matrices are the same as in [stability analysis](#),

$$A_{ia,jb} = (\varepsilon_a - \varepsilon_i)\delta_{ij}\delta_{ab} + 2(ia|jb) - \beta(ij|ab) + (1 - \beta)K_{ijab}^{\text{xc}} \quad (4.84)$$

and

$$B_{ia,jb} = 2(ia|bj) - \beta(ib|aj) + (1 - \beta)K_{ijab}^{\text{xc}}, \quad (4.85)$$

where  $\beta$  is the proportion of Hartree–Fock exchange, which is unity for TDHF. In the Casida equations,  $\mathbf{X}$  and  $\mathbf{Y}$  represent the transition amplitudes for excitation and deexcitation, so an excited many-body wavefunction is

$$|\Psi\rangle = \sum_{ia} X_{ia} |\Phi_i^a\rangle + \sum_{ia} Y_{ia} |\Phi_a^i\rangle. \quad (4.86)$$

For spin-restricted calculations, the orbital Hessian is calculated separately for singlet-to-singlet and singlet-to-triplet excitations. By default, both singlet and triplet excited states are calculated, but can be disabled with the NOSINGLETs and NOTRIPLETS keywords respectively.

In an excited state calculation, the excitation energies from the reference determinant to each state is printed, along with the percentage contribution of each occupied-to-virtual transition. The default threshold for the contributions to be printed is 1%, but this can be changed with the EXTHRESH parameter. The total number of excited states to print — counted from lowest excitation energy — can be controlled with the NSTATES keyword, which has a default value of 10. This will not speed up the calculation, as presently all states are calculated at the same time.

After the percentage contributions are calculated, TUNA determines the transition dipoles between the excited and ground state,  $\mu$ , and uses these to find the oscillator strengths, by

$$f_{\text{osc}} = \frac{2}{3} \omega |\mu|^2. \quad (4.87)$$

These values, along with the excitation energy in hartree, eV, and excitation wavelength in nm, are printed in the absorption spectrum. Finally, the unrelaxed density matrix is calculated.

This is added to the ground state density matrix to give the full unrelaxed one-particle reduced density matrix of a particular state. This state can then be investigated further in [population analysis](#) and [dipole moment](#) calculations with the additional print keyword, P. The state of interest is the lowest energy state by default, but can be changed using the ROOT keyword. This determines for which state the energy and unrelaxed density matrix is calculated.

#### 4.6.1 Time-dependent Hartree–Fock

The time-dependent Hartree–Fock method uses both the **A** and **B** matrices. It can be requested by the TDHF method keyword or through the HF method with the TD keyword.

```
TUNA SPE : H H 1.0 : HF STO-3G : TD
```

The following calculation performs a [population analysis](#) using the unrelaxed density matrix of the second excited state — which is the first *singlet* excited state in this case.

```
TUNA SPE : H H 1.0 : TDHF STO-3G : ROOT 2 P
```

### 4.6.2 Configuration Interaction Singles

The [configuration interaction](#) single excitations (CIS) method is the analogue of Hartree–Fock for excited states [97], and therefore is not very good. In CIS, the wavefunction is expressed as a linear combination of singly-excited determinants,

$$|\Psi_{\text{CIS}}\rangle = \sum_{ia} X_i^a |\Phi_i^a\rangle, \quad (4.88)$$

where only the  $\mathbf{X}$  matrix is used and deexcitations from  $\mathbf{Y}$  are ignored. This is equivalent to solving the much simpler eigenvalue problem

$$\mathbf{A}\mathbf{X} = \omega\mathbf{X}. \quad (4.89)$$

Contributions from the  $\mathbf{B}$  matrix of deexcitations are ignored. This is known as the Tamm–Dancoff approximation. Either the TDA keyword can be used in addition to TD, or the method keyword CIS:

TUNA SPE : H H 1.0 : HF cc-pVDZ : TD TDA

TUNA SPE : H H 1.0 : CIS cc-pVDZ :

### 4.6.3 Perturbative Doubles Correction

Analogous to the [MP2 correlation](#) correction to the [Hartree–Fock ground state](#) energy, CIS(D) is a perturbative doubles correction to an excited state’s energy [98]. This has been shown to improve excitation energies considerably compared to configuration interaction singles.

This method is callable in TUNA with the CIS[D] keyword and the correction will be applied to a chosen ROOT of interest only, as, like [MP2](#), CIS(D) scales with  $\mathcal{O}(N^5)$  with [basis functions](#).

The correction to the excitation energy  $\omega$ , consists of a “direct” contribution from the  $u_{ij}^{ab}$  tensor, which accounts for electron correlation effects of the electron involved in the excitation. The other term is “indirect”, accounting for correlation between electrons not involved in the excitation, calculated by contraction of the  $v_i^a$  tensor, which contains contributions from the [MP2 amplitudes](#),

$$\omega_{\text{CIS(D)}} = \frac{1}{4} \sum_{ijab} \frac{(u_{ij}^{ab})^2}{\mathcal{E}_{ij}^{ab} + \omega} + \sum_{ia} c_i^a v_i^a. \quad (4.90)$$

The state to which the energy correction should be applied is the first excited state by default, but can be changed with the ROOT keyword. For example, a CIS(D) [optimisation](#) of the fourth

excited state of dihydrogen, which is a singlet, is requested by:

```
TUNA OPT : H H 1.0 : CIS[D] 6-311G : ROOT 4
```

The CIS(D) code is spin-adapted so will be fast for restricted references, on either singlet or triplet excited states. It is also available for unrestricted calculations.

The perturbative doubles can also be used with other linear response time-dependent methods with the [D] keyword. For instance, a [single point energy](#) calculation with a perturbative doubles correction on a time-dependent Hartree–Fock state can be requested by:

```
TUNA SPE : H H 1.0 : HF 6-311G : TD [D]
```

#### 4.6.4 Time-dependent Density Functional Theory

The infrastructure for [TDHF](#) can be easily modified to allow TD-DFT calculations, which are requested with the TD keyword with the [exchange–correlation functional](#) of your choice. Currently, TD-DFT calculations are only available with [LSDA](#) functionals, although a proportion of [Hartree–Fock exchange](#) can be incorporated with the HFX keyword.

The TDA keyword activates the Tamm–Dancoff approximation, which is the [CIS](#) analogue for TD-DFT, ignoring contributions from deexcitation. This dramatically reduces the computational time for excited state calculations, halving matrix size, so is probably a good idea.

The TD-DFT method works the same as [TDHF](#), except that the contribution of the exchange–correlation kernel matrix,  $K_{ijab}^{\text{xc}}$ , in [Equation 4.84](#) and [Equation 4.85](#) is not neglected. The matrix elements are integrals over the [standard DFT integration grid](#) in TUNA,

$$K_{ijab}^{\text{xc}} = \sum_{\mu\kappa\nu\lambda} \int \psi_i^\mu(\mathbf{r}) \psi_a^\kappa(\mathbf{r}) f_{\text{xc}}(\mathbf{r}) \psi_j^\nu(\mathbf{r}) \psi_b^\lambda(\mathbf{r}) d\mathbf{r} \quad (4.91)$$

The integrals run over the occupied and virtual Kohn–Sham orbitals,  $\psi(\mathbf{r})$ , and the exchange–correlation kernel,

$$f_{\text{xc}} = \frac{\delta^2 E_{\text{xc}}[n]}{\delta n^2} . \quad (4.92)$$

These analytical second derivatives are hard-coded in TUNA. The  $K_{ijab}^{\text{xc}}$  matrix is also needed for [orbital Hessian calculations for stability analysis](#). A TD-DFT calculation with [LSDA](#) is requested:

```
TUNA SPE : H H 1.0 : LSDA 6-311G : TD
```

The Tamm–Dancoff approximation to speed up calculations can be requested as:

```
TUNA SPE : H H 1.0 : LSDA 6-311G : TD TDA
```

Hybrid and double-hybrid excited state calculations can be requested, but only with [LSDA exchange–correlation functionals](#). The following tells TUNA to use 20% of [Hartree–Fock](#) exchange, and apply a [perturbative doubles](#) correction with 50% (D) correlation and 50% DFT correlation.

```
TUNA SPE : H H 1.0 : LSDA 6-311G : TD HFX 0.2 DFX 0.8 [D] MPC 0.5 DFC 0.5
```

#### 4.6.5 Plotting Absorbance Spectra

The calculated transition peaks can be plotted, as a function of wavelength, with the `ABS PLOT` keyword. This works in the same way as [coordinate scan plots](#) with `SCAN PLOT` in that the `ADD PLOT` and `DEL PLOT` keywords allow multiple spectra to be overlaid on the same axes. Any of the colours or line styles for [coordinate scan plots](#) can also be used with absorbance plots.

```
TUNA SPE : Li H 1.0 : HF cc-pVTZ : TD ABS PLOT
```

The absorbance peaks are plotted with their calculated oscillator strength by Gaussian functions. The width of these can be modified with the `PEAK WIDTH` keyword which is 3.0 by default.

```
TUNA SPE : Li H 1.0 : HF cc-pVTZ : TD ABS PLOT PEAK WIDTH 1.0
```

The following two calculations produced [Figure 4.2](#), showing a change on using a larger basis set. Using the `DEL PLOT` keyword in the first case clears any currently saved plots:

```
TUNA SPE : C O 1.0 : HF cc-pVDZ : ABS PLOT ADD PLOT TD RED DEL PLOT
```

```
TUNA SPE : C O 1.0 : HF cc-pVQZ : ABS PLOT ADD PLOT TD BLUE
```

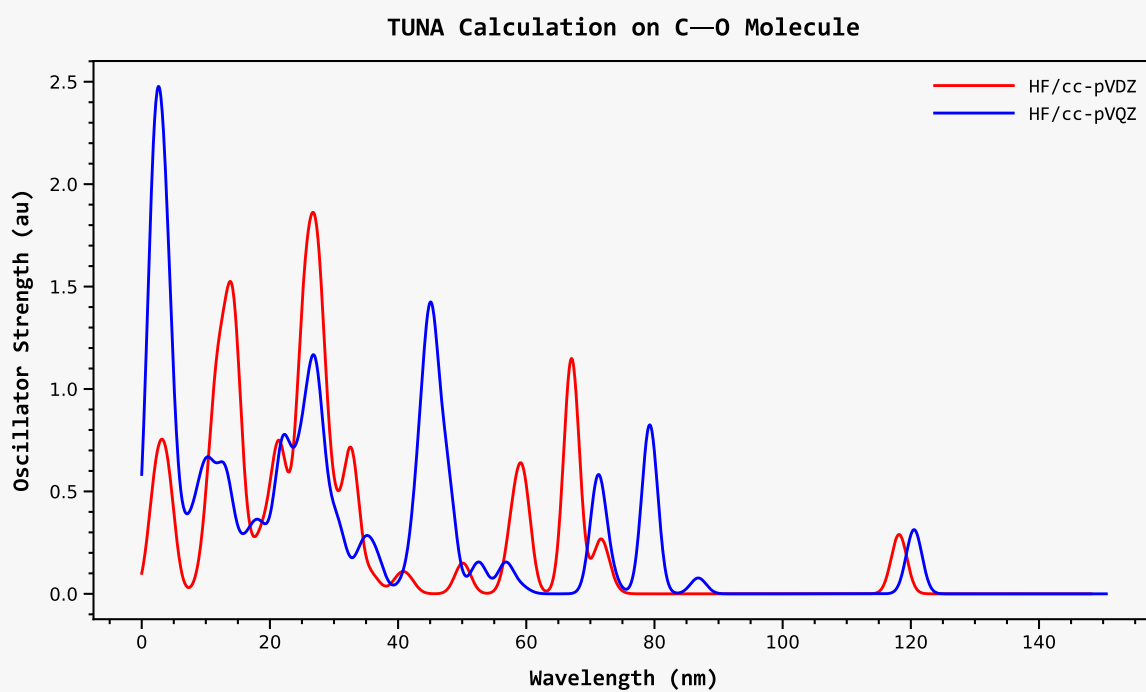


Figure 4.2: Absorbance spectra of carbon monoxide, calculated by TUNA with ABSPLIT



## 5 Basis Sets

Basis sets in TUNA consist of contracted Gaussian functions,

$$\varphi(\mathbf{r}) = \sum_{\gamma} C_{\gamma} g_{\gamma}(\mathbf{r}) \quad (5.1)$$

where  $\{g_{\alpha}(\mathbf{r})\}$  are primitive Gaussians,

$$g_{\gamma}(\mathbf{r}) = N x^l y^m z^n e^{-\gamma(x^2+y^2+z^2)}, \quad (5.2)$$

where  $l$ ,  $m$  and  $n$  are angular momenta and  $N$  is a normalisation constant. Using the additional print keyword, P, will show the detailed basis data at the start of a TUNA calculation, including the Gaussian exponent  $\gamma$  and contraction coefficient  $C_{\gamma}$  for each basis function.

A basis can be decontracted with the DECONTRACT keyword. The following command dedicates the three primitive Gaussians in the hydrogen STO-3G basis to three separate basis functions, rather than a single 1s orbital, giving a lower energy to the hydrogen atom than contracted STO-3G:

```
TUNA SPE : H : HF STO-3G : DECONTRACT
```

### 5.1 Basis Set Extrapolation

Calculations that approach the complete basis set limit, with quadruple- $\zeta$  or higher basis sets, are very computationally expensive. In TUNA, two calculations can be run — for example with double- and triple- $\zeta$  basis sets — and the results from these two energy evaluations extrapolated to give an approximation to the complete basis set limit.

Because in TUNA [derivatives of the energy are numerical](#), a complete basis set extrapolated energy can be carried forwards to give a complete basis set extrapolated [geometry](#), [molecular dynamics simulation](#), [electric response property](#) or [vibrational frequency](#). Extrapolation can be done starting with double-, triple-, quadruple-, and quintuple- $\zeta$  basis sets.

Requesting a lower- $\zeta$  basis set and using the EXTRAPOLATE keyword tells TUNA to run two calculations — first with the lower- $\zeta$  basis, then the higher- $\zeta$  calculation begins:

```
TUNA SPE : H H 1.0 : CISD cc-pVQZ : EXTRAPOLATE
```

The convergence of the [Hartree–Fock energy](#) to the basis set limit [99] is expected to be

$$E_X^{\text{SCF}} = E_\infty^{\text{SCF}} + A \exp(-\alpha \sqrt{X}) , \quad (5.3)$$

where  $A$  and  $\alpha$  are constants that depend on the basis set. This can be solved, with  $X$  being the lower cardinal number and  $Y$  the higher, for an equation for the complete basis set limit,

$$E_\infty^{\text{SCF}} = E_X^{\text{SCF}} + \frac{E_Y^{\text{SCF}} - E_X^{\text{SCF}}}{1 - \exp[\alpha(\sqrt{X} - \sqrt{Y})]} . \quad (5.4)$$

The correlation energy is supposed to converge as

$$E_\infty^{\text{corr}} = \frac{X^\beta E_X^{\text{corr}} - Y^\beta E_Y^{\text{corr}}}{X^\beta - Y^\beta} . \quad (5.5)$$

The values of  $\alpha$  and  $\beta$  are dependent on the basis set. For double- $\zeta$  basis sets,  $\beta$  is always taken as 2.4 in TUNA, and for triple- $\zeta$  and higher bases  $\beta$  is 3.0. The values of  $\alpha$  are listed in [99] for double- and triple- $\zeta$  basis sets. We have internally optimised the  $\alpha$  values for quadruple- $\zeta$  and quintuple- $\zeta$  sets to reproduce the complete basis [Hartree–Fock](#) limit energy of the hydrogen atom.

An example of a [single point energy](#) calculation on a sodium atom using extrapolation with the cc-pVDZ and cc-pVTZ basis sets can be requested by:

```
TUNA SPE : Na : MP2 cc-pVDZ : EXTRAPOLATE
```

An extrapolated [optimisation and frequency](#) run using def2-TZVPPD and def2-QZVPPD is:

```
TUNA OPTFREQ : H H 0.74 : CCSD def2-TZVPPD : EXTRAPOLATE
```

Finally, the [CISD](#) energy of the helium atom with quadruple-to-quintuple- $\zeta$  basis set extrapolation:

```
TUNA SPE : He : CISD aug-cc-pVQZ : EXTRAPOLATE
```

## 5.2 Custom Basis Sets

If you want a basis set which is not implemented natively in TUNA, a custom basis set can be read in. This is done by setting the basis type to CUSTOM and using the BASIS [filepath.tuna] keyword, where filepath.tuna is the path to the custom basis file.

For example, reading in basis data from “basis-data.tuna” can be done by:

```
TUNA SPE : H H 1.0 : HF CUSTOM : BASIS BASIS-DATA.TUNA
```

Note that any file path typed in the terminal will be converted to *lower* case, so you need to have a *fully lower case* basis data file name.

The .tuna file type stores basis set data, and has essentially the same form of the .orca file type that can be downloaded from the Basis Set Exchange [100] — so .orca basis data files can be used directly. In this file, lines to skip begin with “\$”, then elements are spelled out with their full names. Underneath this is the type of subshell (S, P, D, etc.) with the number of primitive Gaussians for that subshell. Next is a list of primitive Gaussians in three columns: the primitive number (1, 2, 3, etc.), the exponent and the coefficient. This process repeats for all subshells.

An example of this file type is shown, for the cc-pVDZ basis set for hydrogen and oxygen:

\$DATA

HYDROGEN

S 4

|   |              |              |
|---|--------------|--------------|
| 1 | 1.301000E+01 | 1.968500E-02 |
| 2 | 1.962000E+00 | 1.379770E-01 |
| 3 | 4.446000E-01 | 4.781480E-01 |
| 4 | 1.220000E-01 | 5.012400E-01 |

S 1

|   |              |              |
|---|--------------|--------------|
| 1 | 1.220000E-01 | 1.000000E+00 |
|---|--------------|--------------|

P 1

|   |              |              |
|---|--------------|--------------|
| 1 | 7.270000E-01 | 1.000000E+00 |
|---|--------------|--------------|

OXYGEN

S 9

|   |              |               |
|---|--------------|---------------|
| 1 | 1.172000E+04 | 7.100000E-04  |
| 2 | 1.759000E+03 | 5.470000E-03  |
| 3 | 4.008000E+02 | 2.783700E-02  |
| 4 | 1.137000E+02 | 1.048000E-01  |
| 5 | 3.703000E+01 | 2.830620E-01  |
| 6 | 1.327000E+01 | 4.487190E-01  |
| 7 | 5.025000E+00 | 2.709520E-01  |
| 8 | 1.013000E+00 | 1.545800E-02  |
| 9 | 3.023000E-01 | -2.585000E-03 |

```

S 9
1 1.172000E+04 -1.600000E-04
2 1.759000E+03 -1.263000E-03
3 4.008000E+02 -6.267000E-03
4 1.137000E+02 -2.571600E-02
5 3.703000E+01 -7.092400E-02
6 1.327000E+01 -1.654110E-01
7 5.025000E+00 -1.169550E-01
8 1.013000E+00 5.573680E-01
9 3.023000E-01 5.727590E-01
S 1
1 3.023000E-01 1.000000E+00
P 4
1 1.770000E+01 4.301800E-02
2 3.854000E+00 2.289130E-01
3 1.046000E+00 5.087280E-01
4 2.753000E-01 4.605310E-01
P 1
1 2.753000E-01 1.000000E+00
D 1
1 1.185000E+00 1.000000E+00

```

\$END

### 5.3 List of Basis Sets

Native basis options in TUNA are listed in [Table 5.1](#), and are taken from the Basis Set Exchange [100]. There are a large variety, including sets with diffuse functions to describe the tails of electron density, core functions to describe near-nuclear electrons and polarisation functions to flexibly describe the molecular correlation energy which results from spontaneous multipole formation.

**Table 5.1** Basis sets implemented in TUNA

| Basis Set | Citation | Description |
|-----------|----------|-------------|
| STO-2G    | [101]    | Minimal     |
| STO-3G    | [101]    | Minimal     |
| STO-4G    | [101]    | Minimal     |
| STO-5G    | [101]    | Minimal     |

| Basis Set      | Citation        | Description                                                                |
|----------------|-----------------|----------------------------------------------------------------------------|
| STO-6G         | [101]           | Minimal                                                                    |
| 3-21G          | [102]           | Double- $\zeta$                                                            |
| 4-31G          | [103]           | Double- $\zeta$                                                            |
| 6-31G          | [103]           | Double- $\zeta$                                                            |
| 6-31+G         | [103]           | Double- $\zeta$ , diffuse functions on heavy atoms                         |
| 6-31++G        | [104]           | Double- $\zeta$ , diffuse functions                                        |
| 6-31G*         | [105]           | Double- $\zeta$ , polarisation functions on heavy atoms                    |
| 6-31G**        | [106]           | Double- $\zeta$ , polarisation functions                                   |
| 6-31+G*        | [107, 108]      | Double- $\zeta$ , diffuse and polarisation functions on heavy atoms        |
| 6-31++G*       | [107, 108]      | Double- $\zeta$ , diffuse functions, polarisation functions on heavy atoms |
| 6-31+G**       | [107, 108]      | Double- $\zeta$ , diffuse functions on heavy atoms, polarisation functions |
| 6-31++G**      | [107, 108]      | Double- $\zeta$ , diffuse and polarisation functions                       |
| 6-31G[2df,p]   | [108–110]       | Double- $\zeta$ , many polarisation functions                              |
| 6-31G[3df,3pd] | [108, 109, 111] | Double- $\zeta$ , many polarisation functions                              |
| 6-311G         | [109]           | Triple- $\zeta$ basis set                                                  |
| 6-311+G        | [104]           | Triple- $\zeta$ basis set with diffuse functions on heavy atoms            |
| 6-311++G       | [104]           | Triple- $\zeta$ basis set with diffuse functions on all atoms              |
| 6-311G*        | [109]           | Triple- $\zeta$ basis set with polarisation functions on heavy atoms       |
| 6-311G**       | [109]           | Triple- $\zeta$ basis set with polarisation functions on all atoms         |
| 6-311+G*       | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions          |
| 6-311++G*      | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions          |
| 6-311+G**      | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions          |

| Basis Set         | Citation        | Description                                                       |
|-------------------|-----------------|-------------------------------------------------------------------|
| 6-311++G**        | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions |
| 6-311G[2df,2pd]   | [109, 110]      | Triple- $\zeta$ basis set with polarisation and diffuse functions |
| 6-311+G[2d,p]     | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions |
| 6-311++G[2d,2p]   | [104, 109]      | Triple- $\zeta$ basis set with polarisation and diffuse functions |
| 6-311++G[3df,3pd] | [104, 109, 110] | Triple- $\zeta$ basis set with polarisation and diffuse functions |
| cc-pVDZ           | [112, 113]      | Polarised double- $\zeta$                                         |
| cc-pVTZ           | [112, 113]      | Polarised triple- $\zeta$                                         |
| cc-pVQZ           | [112, 113]      | Polarised quadruple- $\zeta$                                      |
| cc-pV5Z           | [112, 113]      | Polarised quintuple- $\zeta$                                      |
| cc-pV6Z           | [114]           | Polarised sextuple- $\zeta$                                       |
| aug-cc-pVDZ       | [115]           | Polarised double- $\zeta$ , diffuse functions                     |
| aug-cc-pVTZ       | [115]           | Polarised triple- $\zeta$ , diffuse functions                     |
| aug-cc-pVQZ       | [115]           | Polarised quadruple- $\zeta$ , diffuse functions                  |
| aug-cc-pV5Z       | [115]           | Polarised quintuple- $\zeta$ , diffuse functions                  |
| aug-cc-pV6Z       | [115]           | Polarised sextuple- $\zeta$ , diffuse functions                   |
| d-aug-cc-pVDZ     | [115]           | Polarised double- $\zeta$ , lots of diffuse functions             |
| d-aug-cc-pVTZ     | [115]           | Polarised triple- $\zeta$ , lots of diffuse functions             |
| d-aug-cc-pVQZ     | [115]           | Polarised quadruple- $\zeta$ , lots of diffuse functions          |
| d-aug-cc-pV5Z     | [115]           | Polarised quintuple- $\zeta$ , lots of diffuse functions          |
| d-aug-cc-pV6Z     | [115]           | Polarised sextuple- $\zeta$ , lots of diffuse functions           |
| t-aug-cc-pVDZ     | [115]           | Polarised double- $\zeta$ , tons of diffuse functions             |
| t-aug-cc-pVTZ     | [115]           | Polarised triple- $\zeta$ , tons of diffuse functions             |
| t-aug-cc-pVQZ     | [115]           | Polarised quadruple- $\zeta$ , tons of diffuse functions          |
| t-aug-cc-pV5Z     | [115]           | Polarised quintuple- $\zeta$ , tons of diffuse functions          |
| t-aug-cc-pV6Z     | [115]           | Polarised sextuple- $\zeta$ , tons of diffuse functions           |

| Basis Set     | Citation | Description                                                               |
|---------------|----------|---------------------------------------------------------------------------|
| cc-pCVDZ      | [116]    | Polarised double- $\zeta$ , core functions                                |
| cc-pCVTZ      | [116]    | Polarised triple- $\zeta$ , core functions                                |
| cc-pCVQZ      | [116]    | Polarised quadruple- $\zeta$ , core functions                             |
| cc-pCV5Z      | [116]    | Polarised quintuple- $\zeta$ , core functions                             |
| aug-cc-pCVDZ  | [116]    | Polarised double- $\zeta$ , core and diffuse functions                    |
| aug-cc-pCVTZ  | [116]    | Polarised triple- $\zeta$ , core and diffuse functions                    |
| aug-cc-pCVQZ  | [116]    | Polarised quadruple- $\zeta$ , core and diffuse functions                 |
| aug-cc-pCV5Z  | [116]    | Polarised quintuple- $\zeta$ , core and diffuse functions                 |
| cc-pwCVDZ     | [117]    | Polarised double- $\zeta$ , core–valence functions                        |
| cc-pwCVTZ     | [117]    | Polarised triple- $\zeta$ , core–valence functions                        |
| cc-pwCVQZ     | [117]    | Polarised quadruple- $\zeta$ , core–valence functions                     |
| cc-pwCV5Z     | [117]    | Polarised quintuple- $\zeta$ , core–valence functions                     |
| aug-cc-pwCVDZ | [117]    | Polarised double- $\zeta$ , core–valence and diffuse functions            |
| aug-cc-pwCVTZ | [117]    | Polarised triple- $\zeta$ , core–valence and diffuse functions            |
| aug-cc-pwCVQZ | [117]    | Polarised quadruple- $\zeta$ , core–valence and diffuse functions         |
| aug-cc-pwCV5Z | [117]    | Polarised quintuple- $\zeta$ , core–valence and diffuse functions         |
| ano-pVDZ      | [99]     | Polarised double- $\zeta$ , atomic natural orbitals                       |
| ano-pVTZ      | [99]     | Polarised triple- $\zeta$ , atomic natural orbitals                       |
| ano-pVQZ      | [99]     | Polarised quadruple- $\zeta$ , atomic natural orbitals                    |
| ano-pV5Z      | [99]     | Polarised quintuple- $\zeta$ , atomic natural orbitals                    |
| aug-ano-pVDZ  | [99]     | Polarised double- $\zeta$ , atomic natural orbitals, diffuse functions    |
| aug-ano-pVTZ  | [99]     | Polarised triple- $\zeta$ , atomic natural orbitals, diffuse functions    |
| aug-ano-pVQZ  | [99]     | Polarised quadruple- $\zeta$ , atomic natural orbitals, diffuse functions |
| aug-ano-pV5Z  | [99]     | Polarised quintuple- $\zeta$ , atomic natural orbitals, diffuse functions |

| Basis Set   | Citation   | Description                                                               |
|-------------|------------|---------------------------------------------------------------------------|
| pc-0        | [118, 119] | Double- $\zeta$ , polarisation-consistent                                 |
| pc-1        | [118, 119] | Polarised double- $\zeta$ , polarisation-consistent                       |
| pc-2        | [118, 119] | Polarised triple- $\zeta$ , polarisation-consistent                       |
| pc-3        | [118, 119] | Polarised quadruple- $\zeta$ , polarisation-consistent                    |
| pc-4        | [118, 119] | Polarised quintuple- $\zeta$ , polarisation-consistent                    |
| aug-pc-0    | [118–120]  | Double- $\zeta$ , polarisation-consistent, diffuse functions              |
| aug-pc-1    | [118–120]  | Polarised double- $\zeta$ , polarisation-consistent, diffuse functions    |
| aug-pc-2    | [118–120]  | Polarised triple- $\zeta$ , polarisation-consistent, diffuse functions    |
| aug-pc-3    | [118–120]  | Polarised quadruple- $\zeta$ , polarisation-consistent, diffuse functions |
| aug-pc-4    | [118–120]  | Polarised quintuple- $\zeta$ , polarisation-consistent, diffuse functions |
| pcseg-0     | [121]      | Double- $\zeta$ , polarisation-consistent                                 |
| pcseg-1     | [121]      | Polarised double- $\zeta$ , polarisation-consistent                       |
| pcseg-2     | [121]      | Polarised triple- $\zeta$ , polarisation-consistent                       |
| pcseg-3     | [121]      | Polarised quadruple- $\zeta$ , polarisation-consistent                    |
| pcseg-4     | [121]      | Polarised quintuple- $\zeta$ , polarisation-consistent                    |
| aug-pcseg-0 | [121]      | Double- $\zeta$ , polarisation-consistent, diffuse functions              |
| aug-pcseg-1 | [121]      | Polarised double- $\zeta$ , polarisation-consistent, diffuse functions    |
| aug-pcseg-2 | [121]      | Polarised triple- $\zeta$ , polarisation-consistent, diffuse functions    |
| aug-pcseg-3 | [121]      | Polarised quadruple- $\zeta$ , polarisation-consistent, diffuse functions |
| aug-pcseg-4 | [121]      | Polarised quintuple- $\zeta$ , polarisation-consistent, diffuse functions |
| def2-SVP    | [122]      | Polarised double- $\zeta$                                                 |
| def2-SVPD   | [122, 123] | Polarised double- $\zeta$ , diffuse functions                             |
| def2-TZVP   | [122]      | Polarised triple- $\zeta$                                                 |



| Basis Set   | Citation   | Description                                           |
|-------------|------------|-------------------------------------------------------|
| def2-TZVPD  | [122, 123] | Polarised triple- $\zeta$ , diffuse functions         |
| def2-TZVPP  | [122, 123] | Very polarised triple- $\zeta$                        |
| def2-TZVPPD | [122, 123] | Very polarised triple- $\zeta$ , diffuse functions    |
| def2-QZVP   | [122]      | Polarised quadruple- $\zeta$                          |
| def2-QZVPD  | [122, 123] | Polarised quadruple- $\zeta$ , diffuse functions      |
| def2-QZVPP  | [122, 123] | Very polarised quadruple- $\zeta$                     |
| def2-QZVPPD | [122, 123] | Very polarised quadruple- $\zeta$ , diffuse functions |

## 6 Molecular Integrals

One of the first steps in all TUNA calculations is the calculation of one- and two-electron integrals over [contracted Gaussian basis functions](#), which are necessary to compute expectation values such as the molecular energy. These are based on Cartesian primitive Gaussians of the form

$$g(x, y, z) = Nx^l y^m z^n e^{-\gamma(x^2+y^2+z^2)} \quad (6.1)$$

where  $l$ ,  $m$ , and  $n$  are the angular momenta,  $N$  is a normalisation constant and  $\gamma$  is the primitive Gaussian exponent, defining the “steepness” of the function.

Molecular integrals in TUNA are computed with the McMurchie–Davidson scheme [124]. In this widely used method recurrence relations between Hermite Gaussians simplify the calculation [125].

The molecular integral code is stored in the `tuna_integral` module, which is written in Cython [11]. Cython is a compiled version of Python which can easily interface with the rest of the TUNA program. This dramatically increases the speed of the integrals, at the cost of having to be compiled. Diatomic parity is leveraged in the calculation — for a molecule aligned to the  $z$  axis, several of the two-electron integrals are necessarily zero, so their calculation can be skipped.

The integrals module demands the molecule be aligned along the  $z$ -axis, or located at the origin for an atom. This allows all the symmetry of diatomics to be taken into account, leading to an approximately tenfold increase in speed from prior versions. A diatomic molecule can always be rotated onto the  $z$ -axis, which is the approach taken in the [molecular dynamics](#) module to allow three-dimensional rotation along the trajectory but energy evaluation in one dimension.

The two-electron integral evaluation is often the slowest step of a TUNA calculation. However, each integral is completely independent of the others, so their calculation can be trivially made parallel using OpenMP over the loops [126]. This is done by default with four OpenMP threads, and the number of threads can easily be changed with the `THREADS` keyword by:

```
TUNA SPE : H H 1.0 : HF cc-pV5Z : THREADS 8
```

Note that this only affects the calculation of the molecular integrals, not the rest of the code.

## 6.1 Types of Integrals

Several matrices of integrals over [basis functions](#) are calculated by TUNA, used in the [Hartree–Fock self-consistent field method](#) and in [Kohn–Sham density functional theory](#).

The simplest are overlap integrals,

$$\mathbf{S} = S_{\mu\nu} = \int \varphi_{\mu}(\mathbf{r})\varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} . \quad (6.2)$$

The kinetic matrix is

$$\mathbf{T} = T_{\mu\nu} = -\frac{1}{2} \int \varphi_{\mu}(\mathbf{r})\nabla^2\varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} . \quad (6.3)$$

The nuclear–electron interaction matrix is

$$\mathbf{V} = V_{\mu\nu} = - \sum_{\text{nuclei A}} \int \varphi_{\mu}(\mathbf{r}) \frac{Z_{\text{A}}}{|\mathbf{r} - \mathbf{R}_{\text{A}}|} \varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} . \quad (6.4)$$

A Cartesian component of the dipole matrix is

$$\mathbf{D}^{\alpha} = D_{\mu\nu}^{\alpha} = \int \varphi_{\mu}(\mathbf{r})(\mathbf{r}_{\alpha} - \mathbf{R}_0)\varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} , \quad (6.5)$$

where  $\mathbf{R}_0$  is the dipole origin. A similar component of the quadrupole matrix is

$$\mathbf{Q}^{\alpha} = Q_{\mu\nu}^{\alpha} = \int \varphi_{\mu}(\mathbf{r})(\mathbf{r}_{\alpha} - \mathbf{R}_0)^2\varphi_{\nu}(\mathbf{r}) \, d\mathbf{r} . \quad (6.6)$$

Finally, and with most difficulty, the electron–electron repulsion integrals are calculated as

$$g_{\mu\kappa\nu\lambda} = (\mu\kappa|\nu\lambda) = \iint \varphi_{\mu}(\mathbf{r})\varphi_{\kappa}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \varphi_{\nu}(\mathbf{r}')\varphi_{\lambda}(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' . \quad (6.7)$$

## 6.2 Spherical and Cartesian Harmonics

After the two-electron integrals are calculated by TUNA, they are transformed from the redundant Cartesian harmonics to spherical harmonics by

$$(\mu\kappa|\nu\lambda) = \sum_{\bar{\mu}\bar{\kappa}\bar{\nu}\bar{\lambda}} U_{\mu\bar{\mu}}U_{\kappa\bar{\kappa}}U_{\nu\bar{\nu}}U_{\lambda\bar{\lambda}}(\bar{\mu}\bar{\kappa}|\bar{\nu}\bar{\lambda})_{\text{cart}} . \quad (6.8)$$

This is a formally  $\mathcal{O}(N^5)$  scaling process, but is made dramatically faster in TUNA by the use of sparse matrix multiplication, since the transformation matrices  $U$  are mostly zeroes.

This transformation reduces the size of the basis set, and makes basis functions “proper” atomic orbitals — there are now 5 d orbitals rather than 6. Although it requires additional transformations, this approach reduces the basis set size for future parts of the calculation and reduces the linear dependence of the basis set. Regardless, it can be disabled with the CARTHARM keyword:

```
TUNA SPE : H H 1.0 : HF cc-pVTZ : CARTHARM
```

## 7 Performance of TUNA

### 7.1 Speed and Efficiency

Python is extremely slow. However, TUNA escapes this fate by having very few loops in pure Python — the computationally intensive elements of calculations are array manipulations, handled efficiently with NumPy's [6] vectorised operations. This makes TUNA surprisingly quick. An optimisation and harmonic frequency calculation with B3LYP/cc-pVTZ on a dihydrogen molecule takes about 13 seconds in TUNA and 3 minutes in ORCA, on a single core of the same device.

However, moving to larger basis sets, the story changes — ORCA becomes much *faster* than TUNA, like in real life. Table 7.1 shows the relative speeds of ORCA and TUNA on a CCSD optimisation. For small basis sets, TUNA is the winner by far, but once over a hundred basis functions ORCA dominates. Improvements in version 0.11.0 have made a significant difference to the speed. This is largely down to more efficient molecular integral evaluation and performing calculations in spherical harmonics.

To overcome their slowness, the molecular integrals are written in Cython [11] (a compiled version of Python) which alleviates the problem for low angular momentum basis functions. However integrals over d, f and higher functions, which use recursion in the McMurchie–Davidson scheme, are still extremely slow. At the moment, TUNA is therefore very well suited to performing correlated calculations with small-to-medium sized basis sets. The routine use of larger basis sets will require a substantially rewritten molecular integrals module, which I do not want to do soon. It is possible to vectorise the integrals, to some extent, as has been done in ORCA's SHARK package [125].

**Table 7.1** Calculation times for a full CI optimisation and frequency calculation on dihydrogen

| Basis Set | ORCA    | TUNA 0.10.0 | TUNA 0.11.0 |
|-----------|---------|-------------|-------------|
| cc-pVDZ   | 97.2 s  | 0.4 s       | 0.2 s       |
| cc-pVTZ   | 108.1 s | 1.6 s       | 0.6 s       |
| cc-pVQZ   | 129.3 s | 30.0 s      | 6.3 s       |
| cc-pV5Z   | ~6 min  | > 1 hr      | ~3 min      |

The use of Numpy's "einsum" function is ubiquitous throughout the TUNA code. This makes the implementation of tensorial equations with lots of indices very straightforward — increasing readability of the code, and dramatically increasing speed and reducing memory use in some cases.

For example, the [transformation](#) of the [two-electron integrals](#) into the molecular orbital basis is

$$(pr|qs) = \sum_{\mu\kappa\nu\lambda} M_{\mu}^p M_{\kappa}^r (\mu\kappa|\nu\lambda) M_{\nu}^q M_{\lambda}^s. \quad (7.1)$$

The implementation of this in pure Python would require at least four loops — one for each summation index. However, with einsum the code practically writes itself:

```
ERI_MO = np.einsum("mp,kr,mkn1,nq,ls->prqs", M, M, ERI_AO, M, M, optimize = True)
```

This reduces the number of lines of code to one, but *increases* code readability!

The "optimize = True" statement tells Numpy to find the most efficient "path" to perform this contraction — it turns out that this is four  $\mathcal{O}(N^5)$  steps, contracting one molecular orbital with the atomic orbital basis [two-electron integrals](#) at a time, and saving the resulting temporary array. Complicated tensorial equations — as are found within [coupled cluster theory](#) — can therefore be efficiently and easily be implemented with einsum.

## 7.2 Memory Requirements

Calculations in TUNA are fast due to the program avoiding writing anything to disk, and storing everything in memory. Since there are a maximum of two atoms, this is normally completely fine.

However, for extremely large basis sets memory will become a problem. The most memory-intensive part of the calculation is the storage of the [four-centre two-electron integrals](#) at the start of a calculation. If TUNA predicts that these can not be stored in memory, the calculation aborts.

Using the additional print keyword P will show you the memory footprint of these integrals. On dihydrogen, storing  $(\mu\kappa|\nu\lambda)$  requires around 3 GB of RAM for cc-pV5Z — which is fine — and around 32 GB for cc-pV6Z — which is not fine! For these extreme calculations, a laptop will not suffice. However, they will do very well on a high memory node of a high-performance computing cluster. The highest memory possible calculation uses triply-augmented cc-pV6Z, and will require around a terabyte of RAM. These memory problems, as well as the slowness of huge basis set calculations, can be avoided by using [basis set extrapolation](#) with the EXTRAPOLATE keyword.

**Table 7.2** Numerical derivative step length,  $h$ , for different energy perturbations

| Perturbation                | Property                        | Value of $h$ |
|-----------------------------|---------------------------------|--------------|
| $\partial E/\partial R$     | Geometry optimisation           | 0.00005      |
| $\partial^2 E/\partial R^2$ | Harmonic frequency              | 0.01         |
| $\partial^3 E/\partial R^3$ | Vibrational perturbation theory | 0.025        |
| $\partial^4 E/\partial R^4$ | Vibrational perturbation theory | 0.025        |
| $\partial E/\partial F$     | Dipole moment                   | 0.00001      |
| $\partial^2 E/\partial F^2$ | Polarisability                  | 0.001        |
| $\partial^3 E/\partial F^3$ | Hyperpolarisability             | 0.0015       |

### 7.3 Accuracy and Precision

All derivatives of the energy in TUNA are calculated numerically, allowing total transferability of electronic structure methods between calculation types and properties. This approach means that once a method is implemented it automatically becomes available throughout TUNA for any property or calculation type. The numerical derivative equations have been carefully optimised to maximise their precision, and the step lengths used are listed in [Table 7.2](#) for geometric perturbations based on the bond length,  $R$ , and electric perturbations based on the electric field,  $F$ .

The [energy convergence criteria](#) have also been tested to make sure response properties are stable — conservative default choices have been made here, including TIGHT convergence for first derivative quantities, and EXTREME convergence for higher derivatives. This may sometimes cause a problem (particularly for [CCSDT](#) and [CCSDTQ](#) which are more difficult to converge) and if it does, the convergence criteria can be safely reduced to TIGHT for second derivative properties — third and fourth derivatives might be riskier but still will most likely be okay.

In terms of accuracy, new features implemented in TUNA are always — assuming they exist there — benchmarked against ORCA [\[127\]](#). Naturally, there will be some bugs which we don't know about yet, but a wide range of systems is used to check that newly implemented features behave.

## 8 Acknowledgements

There are a variety of codes and tutorials that have served as valuable inspiration for TUNA over the years. First and foremost is the [ORCA quantum chemistry package](#), which puts clear emphasis on the user experience with the “simple input line” approach [4, 5].

The [Crawford group programming projects](#) for computational chemistry which contain examples for wavefunction methods such as coupled cluster theory and MP2 were very useful for getting started [128]. The “[Psi4Numpy](#)” collection of codes and tutorials has a diverse group of methods which were extremely useful [2].

The collection of Python codes of wavefunction methods based on Hartree–Fock theory, [HarPy](#) [129], was particularly useful for checking the [CCSD](#) and [CCSDT](#) code in TUNA. Finally, [PyDFT](#), a simple educational DFT code that shows how basic exchange–correlation functionals can be implemented, was an extremely useful reference, as was its accompanying documentation [130].

I am very grateful to Hannah Whittome for, among other things, her brutal criticism of an early version of the TUNA logo — this led to the fish being put on an extreme weight loss regime. Thank you also to Chris Field for suggesting features such as custom scan plot colours.



## 9 List of Keywords

Table 9.1 lists all of the mandatory and optional keywords in TUNA, grouped by their functions.

**Table 9.1** Parameter keywords in TUNA

| Parameter                      | Description                                     |
|--------------------------------|-------------------------------------------------|
| P                              | Print more information                          |
| T                              | Print less information                          |
| DEBUG                          | Print extreme information                       |
| PRINTLEVEL [int]               | Print levels of information                     |
| CH [int] or CHARGE [int]       | Molecular charge                                |
| ML [int] or MULTIPLICITY [int] | Molecular multiplicity                          |
| BASIS [string]                 | Custom basis filepath                           |
| EXTRAPOLATE                    | Extrapolate double-zeta basis                   |
| DECONTRACT                     | Decontract basis set                            |
| CARTHARM                       | Cartesian harmonics Gaussian basis              |
| ROTATE [float]                 | Rotate orbitals for initial guess (by degrees)  |
| NOROTATE                       | Don't rotate orbitals for initial guess         |
| STHRESH [float]                | Threshold for overlap matrix eigenvalues        |
| PRINTMOS [int]                 | Number of molecular orbitals to print           |
| M1 [float]                     | Custom mass for first atom                      |
| M2 [float]                     | Custom mass for second atom                     |
| THREADS [int]                  | Number of OpenMP threads for integrals          |
| COREGUESS                      | Core Hamiltonian guess density                  |
| SADGUESS                       | Superposition of atomic densities guess density |
| SCFGUESS                       | Minimal self-consistent guess density           |

| Parameter             | Description                                       |
|-----------------------|---------------------------------------------------|
| LOOSE or LOOSESCF     | Loose SCF convergence                             |
| MEDIUM or MEDIUMSCF   | Normal SCF convergence                            |
| TIGHT or TIGHTSCF     | Tight SCF convergence                             |
| EXTREME or EXTREMESCF | Extreme SCF convergence                           |
| DAMP [float]          | Static damping for SCF convergence                |
| NODAMP                | No damping for SCF convergence                    |
| MAXDAMP [float]       | Maximum damping factor                            |
| DIIS [int]            | Fock matrix extrapolation, and number of matrices |
| NODIIS                | No Fock matrix extrapolation                      |
| SLOWCONV              | High static damping (50%) for SCF convergence     |
| VERYSLOWCONV          | Extreme static damping (85%) for SCF convergence  |
| MAXITER [int]         | Maximum number of SCF iterations                  |
| ECONV [float]         | Iterative energy convergence criteria             |
| MAXDP [float]         | Maximum change in density matrix                  |
| RMSDP [float]         | Root-mean-square change in density matrix         |
| DIISERR [float]       | Root-mean-square commutator                       |
| STAB                  | Do SCF stability analysis                         |
| LOOSEOPT              | Loose geometry convergence                        |
| MEDIUMOPT             | Normal geometry convergence                       |
| TIGHTOPT              | Tight geometry convergence                        |
| EXTREMEOPT            | Extreme geometry convergence                      |
| CALCHESS              | Calculate exact Hessian for optimisation          |
| DEFAULTHESS [float]   | Default Hessian for optimisation                  |
| D2                    | Semi-empirical dispersion correction with D2      |
| MAXSTEP [float]       | Maximum step for optimisation                     |
| MOREAD                | Use density from previous step                    |

| Parameter          | Description                                        |
|--------------------|----------------------------------------------------|
| NOMOREAD           | Recalculate density from scratch each step         |
| OPTMAX             | Optimise to a local maximum, rather than a minimum |
| TRAJ [string]      | Write trajectory                                   |
| NOTRAJ             | Do not write trajectory                            |
| MAXGEOMITER [int]  | Maximum number of optimisation steps               |
| ANHARMCONV [float] | Convergence criteria for anharmonic frequencies    |
| NELEC [int]        | Number of electrons for IP or EA calculation       |
| VERTICAL           | Vertical calculation for IP or EA                  |
| ZPE                | Calculate zero-point energy for BDE                |
| NOCP               | Do not use counterpoise correction in BDE          |
| VPT1               | First-order vibrational perturbation theory        |
| VPT2               | Second-order vibrational perturbation theory       |
| STEP [float]       | Distance increment for coordinate scan             |
| NUM [int]          | Number of distance increments for coordinate scan  |
| SCANPLOT           | Plot potential energy surface from coordinate scan |
| SAVEPLOT [string]  | Save plot from coordinate scan                     |
| DELPLOT            | Delete temporary plot                              |
| ADDPLOT            | Add another plot to the axes                       |
| DOT                | Use dotted lines on the coordinate scan plot       |
| DASH               | Use dashed lines on the coordinate scan plot       |
| BLACK              | Use black lines on the coordinate scan plot        |
| BLUE               | Use blue lines on the coordinate scan plot         |
| GREEN              | Use green lines on the coordinate scan plot        |
| RED                | Use red lines on the coordinate scan plot          |
| YELLOW             | Use yellow lines on the coordinate scan plot       |
| MAGENTA            | Use magenta lines on the coordinate scan plot      |
| CYAN               | Use cyan lines on the coordinate scan plot         |
| WHITE              | Use white lines on the coordinate scan plot        |
| COLOUR [str]       | Change colour of coordinate scan plot              |

| Parameter                           | Description                                         |
|-------------------------------------|-----------------------------------------------------|
| STEP                                | Timestep for MD simulation                          |
| NUM                                 | Number of timesteps for MD simulation               |
| TEMP [float] or TEMPERATURE [float] | Sets temperature for thermochemistry and MD         |
| PRES [float] or PRESSURE [float]    | Sets pressure for thermochemistry calculations      |
| FREEZECORE [int]                    | Freeze core orbitals                                |
| SSS [float]                         | Same-spin scaling for SCS-MP2                       |
| OSS [float]                         | Opposite-spin scaling for SCS-MP2                   |
| MP3S [float]                        | Scaling for SCS-MP3                                 |
| MPGRID [int]                        | Grid size for Laplace transform MP2                 |
| AMPCONV [float]                     | Convergence criteria for coupled cluster amplitudes |
| CORRMAXITER [int]                   | Maximum iterations for correlation                  |
| CORRDAMP [float]                    | Use damping in correlation, parameter               |
| PRINTAMPS [int]                     | Print this number of largest $t$ -amplitudes        |
| NATORBS                             | Calculate natural orbitals                          |
| RELAXED                             | Calculate relaxed density                           |
| TD                                  | Do a time-dependent calculation                     |
| TDA                                 | Use the Tamm–Dancoff approximation                  |
| EXTHRESH [float]                    | Threshold for printing excited state contributions  |
| ROOT [int]                          | Choice of excited state                             |
| NSTATES [int]                       | Number of states to print                           |
| [D]                                 | Calculate perturbative doubles                      |
| LOOSEGRID                           | Loose DFT integration grid                          |
| MEDIUMGRID                          | Normal DFT integration grid                         |
| TIGHTGRID                           | Tight DFT integration grid                          |
| EXTREMEGRID                         | Extreme DFT integration grid                        |
| INTACC [float]                      | Integral accuracy for DFT                           |

| Parameter         | Description                                      |
|-------------------|--------------------------------------------------|
| DFX [float]       | Density functional theory exchange percentage    |
| HFX [float]       | Hartree–Fock theory exchange percentage          |
| DFC [float]       | Density functional theory correlation percentage |
| MPC [float]       | Perturbation theory correlation percentage       |
| NOX               | Turn off DFT exchange                            |
| NOC               | Turn off DFT correlation                         |
| XA [float]        | Value of $\alpha$ in $X\alpha$ LDA exchange      |
| NL                | Calculate non-local VV10 dispersion              |
| DENSLOT           | Plot the electron density                        |
| SPINDENSLOT       | Plot the spin density                            |
| DIFFDENSLOT       | Plot the difference electron density             |
| DIFFSPINDENSLOT   | Plot the difference spin density                 |
| PLOTMO [int]      | Plot a molecular orbital                         |
| PLOTNO [int]      | Plot a natural orbital                           |
| PLOTHOMO          | Plot the highest occupied molecular orbital      |
| PLOTLUMO          | Plot the lowest unoccupied molecular orbital     |
| VIBLOT            | Plot nuclear wavefunctions                       |
| ABSPLOT           | Plot absorbance spectrum                         |
| PEAKWIDTH [float] | Gaussian peak width                              |
| EX [float]        | Electric field in $x$ direction                  |
| EY [float]        | Electric field in $y$ direction                  |
| EZ [float]        | Electric field in $z$ direction                  |
| EGX [float]       | Electric field gradient in $x$ direction         |
| EGY [float]       | Electric field gradient in $y$ direction         |
| EGZ [float]       | Electric field gradient in $z$ direction         |
| DIPOLE            | Calculate numerical dipole moment                |
| QUADRUPOLE        | Calculate numerical quadrupole moment            |

| Parameter | Description                             |
|-----------|-----------------------------------------|
| POLAR     | Calculate numerical polarisability      |
| HYPER     | Calculate numerical hyperpolarisability |

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